

The Effect of Public Sector Relocations on Regional Development in Germany

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Can governments create private jobs in lagging regions by relocating their own employees? I study a program in Germany's second most populous state that is moving about 5,200 public-sector jobs from thriving cities to lagging towns. Using self-collected relocation data and linked employer–employee social security records, I compare treated municipalities with municipalities outside the policy state matched on the indicators that determine program eligibility. Within five years, private employment in municipalities that gain public jobs rises by 3–4% of its 2014 level, with no offsetting decline in those that lose them. Following individual workers, I find that jobs move, but people largely do not. A quantitative spatial model explains these effects through local public-to-private productivity spillovers and shows that the relocations leave national output and welfare nearly unchanged. Choosing destinations where public jobs raise private output most would cut the small output loss by about two thirds.

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1 Introduction

Can governments use public employment relocation to create private-sector jobs in lagging regions, and can they do so without harming the cities that lose those jobs? Within-country disparities in economic opportunity are widening, and governments have responded with place-based tax incentives, hiring subsidies, special economic zones, and infrastructure investment. These instruments can be difficult to target and expensive to finance. Tax credits and hiring subsidies require eligibility rules to which firms can respond strategically. Competition among jurisdictions can also bid up incentives and transfer part of the resulting surplus to mobile firms rather than local residents (Slattery 2024). Governments are therefore increasingly experimenting with an instrument over which they retain more direct control: the location of their own activity. Germany, Denmark, France, Canada, South Korea, the United Kingdom, and Sweden have all used public employment relocation to direct state activity toward economically lagging regions.¹

In this paper, I ask how relocating public employment affects private employment in the places that gain and lose public jobs, how the release and absorption of public office space shape these effects, and what the relocations imply for aggregate output and welfare. I answer these questions by studying an ongoing public employment relocation program in Germany.

The German constitution anchors the objective of equivalent living conditions across regions as a state goal (*GG Art. 72(2)* 1949). Politically, this is interpreted as a mandate to reduce disparities between urban and rural areas and across regions (German Federal Ministry of Agriculture, Food and Regional Identity 2025). The program I study, the *Heimatstrategie*, is one German state’s response to this mandate; I refer to this state as the policy state. Introduced in 2015, the program envisages 5,196 public-sector jobs and 1,170 paid training positions across its two phases. The jobs alone are equivalent to roughly 1.7% of the pre-policy direct state workforce.² The program also changes the use of scarce urban space: the government estimates that phase I freed around 37,000 square meters of office space, mainly in Munich.

Three features make the program well suited to studying public employment relocation. First, it involves many sending and receiving municipalities.³ Second, it assigns destinations by explicit rules. The policy state identifies eligible destinations using a quantitative index of socioeconomic disadvantage at the municipal or district level, and the government favored officially designated service centers, including *Mittelzentren*, reflecting spatial-planning requirements. Third, it moves jobs out of booming urban centers, so I can study the places that lose public jobs as well as those that gain them.

¹See Freitas (2025) for an overview of past and current public employment relocation programs.

²These figures describe intended final staffing; phase II is still being implemented (Bavarian Ministry of Finance 2026).

³Related evidence largely comes from large, concentrated “big push” episodes, including capital-city relocations such as the Berlin–Bonn case (Becker et al. 2021; Faggio, Schlüter, et al. 2025) and large single-agency moves from capital cities to inland regions, such as the BBC’s partial relocation from London to Salford (Nathan et al. 2024).

The setting raises three empirical challenges: selection into treatment, staggered treatment timing, and economy-wide adjustments that a comparison of treated and untreated places cannot capture. To address selection, I match each treated municipality one-to-one to a municipality outside the policy state, using Mahalanobis matching on pre-policy levels and trends in the economic and demographic indicators that underlie the eligibility criteria. Drawing controls from outside the policy state limits contamination by spillovers within the state, and I additionally exclude the counties with the strongest commuting links to the policy state. Identification requires parallel trends within matched pairs in the absence of treatment. To address staggered timing, I use the estimator of Sun and Abraham (2021) as the main event-study specification, alongside long differences with outcomes observed through 2024. To capture economy-wide adjustments, I embed the policy in a two-sector quantitative spatial model.

I find that the program increased private employment in receiving municipalities. Five years after the first relocation, private workplace employment is higher by about 4.3% of its 2014 level and private employment among residents by about 3.3%. The long differences show a gradually rising effect on resident private employment, from about 1% in 2015–2016 to 1.3–1.9% in 2017–2021 and 2.8–3.5% in 2022–2024.

Sending municipalities show no offsetting losses. Their private-employment estimates are negative on some margins in the first years but turn positive in the later long-difference endpoints, although the workplace estimates are imprecise. Excluding Munich, the largest origin, leaves the sending-location estimates positive. Receiving private-employment estimates also remain positive when I date treatment from the last rather than the first relocation, and I compare the main estimates with those from the estimator of Callaway and Sant’Anna (2021).

Worker histories show that jobs move, but people largely do not. Of workers employed at candidate sending establishments in 2014, only 1.8% work at a candidate destination in 2024 after changing residence, compared with 10.3% who do so without moving. About 41% remain employed elsewhere without changing residence, and most of these stayers still work in public administration. Meanwhile, 99% of employees at candidate destinations in 2024 were outside the original sending cohort; among those with a recorded prior sector, 62% already worked in administration in 2014. These patterns motivate the model’s distinction between where public jobs are located and where public workers live.

The reduced-form evidence leaves open how places that neither send nor receive public jobs adjust, and what the relocations imply for aggregate output and welfare. To answer these questions, I build a quantitative spatial model of Germany with a private sector and a state public sector. Public jobs provide administrative services that are not tied to local users, so the model holds their national number fixed and lets the government choose only their location. Private workers choose where to live and work, while public workers choose where to live given their assigned workplace. Public employment raises the productivity of nearby private firms, public offices compete with private firms and households for floor space, and a labor-income tax finances public payroll and office rents. Because private productivity is highest in urban origins such as Munich, moving public jobs to lagging places lowers their opportunity cost in scarce urban labor and space, but it also moves the productivity spillovers they generate.

I estimate the public-to-private spillover elasticity by indirect inference, matching the model’s county-level private-employment responses to their reduced-form counterparts. The estimated elasticity is 0.36. The model reproduces the receiving-county response, its prediction for sending counties lies well within the reduced-form confidence interval, and it implies negligible effects on the matched control counties outside the policy state, which supports the empirical design. Simulating the relocations completed by 2024, the model predicts that private workplace employment rises by 2.0% in receiving counties and falls by 0.8% in sending counties, in line with an imprecise county-level reduced-form estimate close to zero. National private output falls by 0.035%, its spatial concentration declines, and private workers’ expected utility rises by 0.017%. Almost the entire response runs through the public spillover; at the county level, the reallocation of office space matters little. Where the jobs go matters: sending the same positions to the counties where they generate the most private output would reduce the national output loss by about two thirds.

My paper contributes to three strands of literature. First, it contributes to urban economics through the literature on place-based policies, which studies whether spatially targeted interventions can improve outcomes in lagging regions (Moretti 2012; Kline and Moretti 2014; Ehrlich and Seidel 2018; Duranton and Venables 2018; Ehrlich and Overman 2020). I study public employment relocation as a distinct place-based policy instrument. Unlike subsidies, tax incentives, or infrastructure programs, this policy directly changes the spatial allocation of public-sector employment. Because the policy uses explicit eligibility criteria, I can use their economic and demographic dimensions to construct a matched control group and reduce concerns about selection into treatment. I also estimate effects in the places that lose public jobs, which the literature has rarely examined.

Second, the paper contributes to labor economics through the literature on public employment, local labor markets, and worker adjustment to employer shocks. Existing work asks whether public jobs crowd in or crowd out private employment, with evidence from public-sector expansions, contractions, and relocations (Faggio and Overman 2014; Senfleben-König 2014; Auricchio et al. 2020; Faggio 2019; Jofre-Monseny et al. 2020; Lee et al. 2024; Becker et al. 2021; Faggio, Schlüter, et al. 2025). I add evidence from a multi-site program with explicit allocation rules, and I document how the workforce adjusts by distinguishing movers, commuters, stayers, and new hires at destination offices. To do so, I construct a yearly municipal employment panel for Germany from the IAB Integrated Employment Biographies⁴ and combine it with a self-collected relocation dataset and a newly assembled panel of the policy state’s public-agency register.

Third, the paper contributes to public economics by studying government employment as spatial redistribution: the reallocation of a fixed stock of public jobs rather than a net fiscal expansion. It connects this question to the literature that incorporates public employment into spatial equilibrium models (Becker et al. 2021; Guillouzouic et al. 2021; Guillouzouic et al. 2024). I estimate how strongly public employment raises nearby

⁴I thank Wolfgang Dauth, Nina Gläser, and the IAB REGIO group for their support in accessing the IAB data.

private productivity and model the competition between public offices and private users for floor space. This allows me to quantify the trade-off between the lower opportunity cost of public jobs in lagging places and the loss of their spillovers in productive cities, and to evaluate alternative allocations of the same jobs.

The paper proceeds as follows. Section 2 describes the institutional setting and the program. Section 3 presents the data, including the relocation dataset and the administrative sources. Section 4 describes the matching procedure and the empirical methodology, and Section 5 presents the employment effects in receiving and sending locations. Section 6 examines worker adjustment. Section 7 introduces the quantitative spatial model, its quantification, and the policy counterfactuals. Section 8 concludes.

2 Institutional Setting

The public sector is a major employer in Germany and accounts for a substantial share of government expenditure.⁵ By choosing where to locate public jobs, governments can direct employment and recurring wage income toward disadvantaged places. Bavaria used this instrument to address spatial inequality within the state.

Bavaria lies in southeastern Germany, bordering Austria and the Czech Republic. Its 13.25 million inhabitants in 2024 made it Germany’s second most populous state, with a population comparable to Pennsylvania’s (Bavarian Statistical Office 2025a; U.S. Census Bureau 2025). It is also Germany’s largest state by area (Bavarian Statistical Office 2025b).

Economic opportunities differ sharply across Bavaria. Wealthy municipalities around Munich and in the south contrast with struggling areas in Franconia and along the Czech border.⁶ The state thus faces a familiar regional-policy problem: how to bring jobs to places that have fallen behind its economic centers. In 2015, Bavaria began reallocating public employment from its major agglomerations to these places as part of its *Heimatstrategie*, a regional development strategy.

To identify eligible destinations, the government benchmarks municipalities across Bavaria against the state average. Municipalities that fall behind on economic and demographic indicators form the eligible set. The government summarizes these indicators in an index covering population prospects, unemployment, employment density, income, and migration among young adults. A municipality qualifies through either its own index or its district’s index, with a threshold of 90 relative to a state average of 100 (Bavarian Ministry of Finance 2015). Panel (a) of Figure 1 shows the municipalities meeting these eligibility criteria. Appendix A reports the index components and their construction.

⁵On the OECD’s comparable general-government measure, employment shares in 2021 were approximately 11% in Germany, 15% in the United States, 21% in France, and 17% in the United Kingdom (OECD 2023, Figure 12.1). In the same country order, the World Bank reports wage bills of 17.4%, 25.8%, 22.1%, and 23.4% of public expenditure in 2019 (World Bank 2024).

⁶Before the program, average annual wages in high-wage municipalities, concentrated around Munich and Starnberg, were more than twice those in low-wage municipalities, concentrated in Upper Franconia and Lower Bavaria (Bavarian Statistical Office 2015).

Within the eligible set, the government favored towns designated as service centers for surrounding communities and places that had not already received substantial support through higher-education initiatives (Bavarian Ministry of Finance 2025). Ministries assessed whether potential destinations offered suitable premises and infrastructure for the agencies they proposed to relocate. They selected sites in coordination with the finance ministry; the cabinet approved the program, and the responsible ministries implemented it (Kaltenhauser 2023). Panel (b) adds the service-center designations, while panel (c) shows the selected destinations and origins. Some selected destinations lie outside the index-defined eligible set.

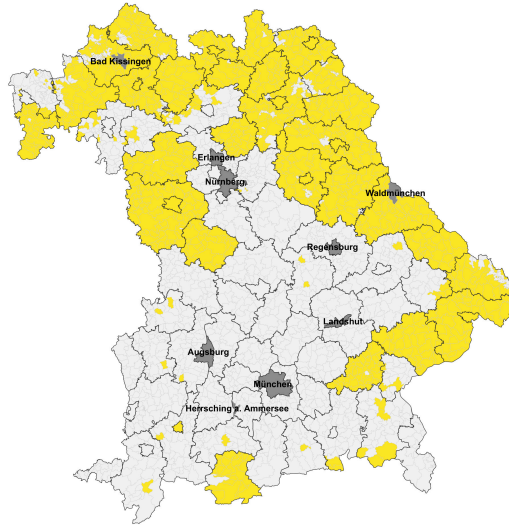
The agencies selected for relocation perform functions that are not tied to local public good provision. Tax processing, information technology, and other specialized administrative functions can serve the same users from a different municipality. Ordinary schools and local police stations are therefore excluded from the relocation program (*ibid.*). The program includes whole and partial agency moves, as well as the establishment or expansion of offices. For agency moves, Munich is an important origin: relocating an office shifts public jobs away from the state’s largest employment center while adding them to a receiving town.

I study both phases of the public employment reallocation program. Together, they envisage 5,196 jobs and 1,170 paid training positions, equivalent to about 1.7% of Bavaria’s pre-policy direct state workforce when counting jobs alone.⁷ The training positions include public-service apprenticeships organized as degree programs: participants receive remuneration during their training, so these positions also bring wage income to receiving places (Freistaat Bayern 2026, Art. 75). I assembled a relocation dataset that records agencies, destinations, origins, staffing, and implementation dates. Appendix A documents the program’s administrative components and intended final staffing.

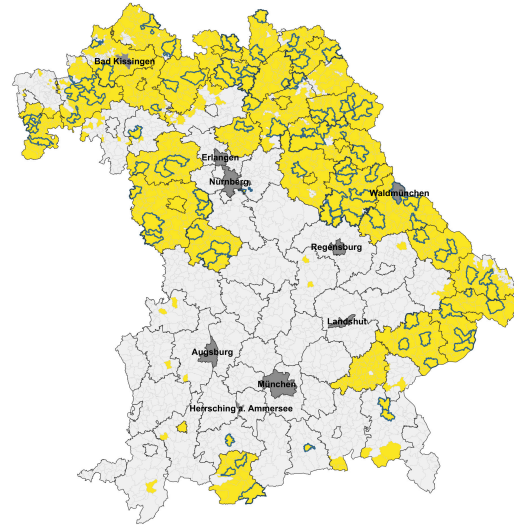
Most workers in Bavaria’s direct state administration are civil servants. Before the program, civil servants and judges accounted for 68.4% of this workforce; the remaining workers held employment contracts (Statistical Offices 2014; Bavarian Statistical Office 2014). Civil servants have a distinct legal status and are absent from the social-security employment records used in the empirical analysis. Municipal civil-service employment data from the statistical offices will allow me to extend the analysis to both groups of public employees.

Public base pay does not vary across Bavarian municipalities for a given position and experience level. Contractual employees are paid under the collective agreement for state-government employees, known as TV-L, which Bavaria negotiates jointly with the other participating German states. Wage setting is thus coordinated nationally rather than determined by individual agencies or municipalities. Civil-service pay is set by Bavarian law. Both systems use schedules in which the job determines the pay grade and experience determines progression within it, much as grades and steps organize pay in the US federal General Schedule (TdL 2025; Freistaat Bayern 2026; OPM 2026).

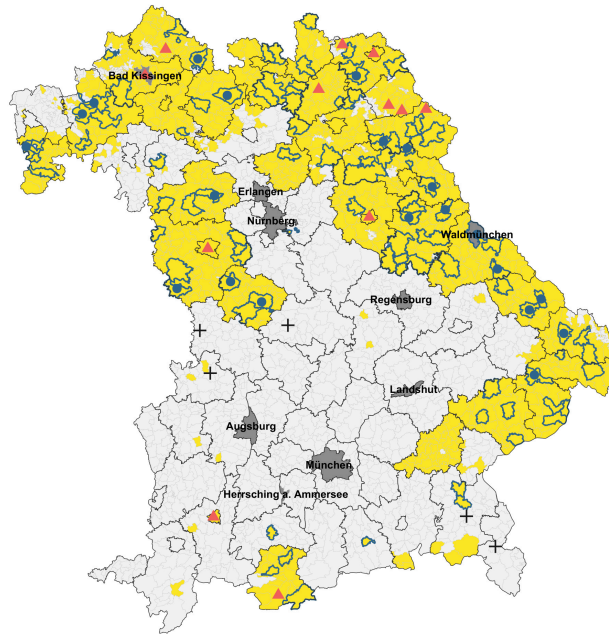
⁷The government’s March 2026 statement reports intended final staffing of 2,511 jobs and 730 training positions in phase I, and 2,685 jobs and 440 training positions in phase II. Phase II remains under way, with implementation extending to 2030 (Bavarian Ministry of Finance 2026). The employment benchmark uses the direct state workforce before the program; Appendix A gives the calculation.



(a) Municipal eligibility



(b) Service-center designations



(c) Receiving and sending locations

Figure 1. Geographical targeting and site selection. Panel (a) marks municipalities meeting the index-based eligibility criteria in yellow. Panel (b) outlines eligible service centers in teal. Panel (c) adds receiving municipalities: blue circles mark service centers, red triangles other municipalities, and black crosses municipalities outside the index-defined eligible set. Sending locations appear in gray throughout.

The schedules provide a benchmark for earnings in the relocated jobs. Under the February 2025 schedule, qualified administrative work accessible without a university degree, illustrated by grade E9a, starts at about €3,520 gross per month. A position requiring a master’s-level qualification, illustrated by E13, starts at about €4,630 (LfF 2025b). Appendix Table 8 gives the corresponding civil-service grades and pay ranges, distinguishing career level from experience.⁸

Agencies build up employment at their destinations gradually. An office may open with a small team, recruit additional staff, and move into permanent premises only later. Positions can shift as workers leave or retire at the origin, allowing employment at the destination to grow over several years (Bavarian Ministry of Finance 2026). I date treatment to the government press release announcing an office’s opening. This convention captures responses that occur while staffing is still being built up, and the event studies allow me to check for responses before the announced opening.

Employees do not have to move when their agency relocates. The personnel framework rules out compulsory permanent transfers, relocation-related dismissal, and involuntary downgrading. Workers can seek reassignment, while those who follow an agency can receive training and support for moving or commuting, including a conditional €3,000 mobility payment. Telework is available where duties permit (Bavarian Ministry of Finance 2021); in practice, two work-from-home days per week are common. Incumbents can therefore stay, commute, or move, while agencies can also recruit at the destination. Public jobs therefore need not bring an equal number of new residents. Section 6 analyzes these worker responses.

Relocations also change the use of office space. Receiving agencies rent, adapt, or construct premises, often before reaching full staffing. At the origins, the government estimates that phase I freed about 37,000 square meters of office space, mainly in Munich (Bavarian Ministry of Finance 2026). Vacated premises create scope for other uses, including private-sector offices. Sending locations thus lose public jobs while potentially releasing space for other employers. Section 5 examines employment responses at both ends of the relocations.

Providing new premises can be a substantial part of relocation costs. For three completed tax-office relocations with matched cost and staffing figures, construction spending averaged about €180,000 per public job (Kaltenhauser 2023). Firm-specific investment incentives provide a benchmark: approved direct grants for Northvolt’s Heide battery plant amounted to about €230,000 per planned job, while proposed support for Intel’s Magdeburg semiconductor plant amounted to €3.3 million per planned job.⁹

⁸A public-sector wage premium also depends on the private-sector alternative. Using German survey data, Bonaccolto-Töpfer et al. (2022) find a wage advantage for women but a pay penalty for men after accounting for worker characteristics and selection. These estimates refer to Germany as a whole.

⁹The relocation ratio covers construction spending per public job accommodated. The Northvolt and Intel ratios use announced support and planned direct employment; Intel cancelled its project in July 2025 (Intel 2025). These figures illustrate spending magnitudes, not relative cost-effectiveness. Appendix A documents the calculations and other reported relocation costs.

3 Data

Studying public employment relocation requires knowing where agencies move and what happens to the people and firms around them. I assemble a record of the Bavarian program, use successive editions of the Bavarian public-agency register to document agency locations, and measure employment with linked employer–employee data from the Institute for Employment Research (IAB). The empirical analysis follows municipalities over time. I also construct county-level measures for the quantitative model, which I quantify at the county level.

3.1 Data on the Relocation Program

I construct the relocation dataset by combining official program records. Starting from the BayernAtlas records, I collect the agency name, responsible ministry, origin and destination, intended staffing, program phase, and implementation status (State Office for Digitization, Broadband and Geodetic Measurement 2024). I then follow each project through the annual *Heimatstrategie* reports, government announcements, agency websites, and local newspaper coverage. These sources turn a list of planned relocations into a record of where the policy was implemented and when employment could begin to change.

Implementation often unfolds over several years. An agency may open a branch before moving into its permanent building, while staffing grows toward the announced target. I therefore distinguish the start of operations from construction milestones and intended final staffing. Where sources disagree or a date remains unresolved, I follow the official state government source. The program totals in the preceding section describe the intended scale of program rollout phases I and II.

I complement these project histories with a new source for studying the geography of public employment: the Bavarian public-agency register (Bavarian State Statistical Office 2026). Its records identify agencies and their addresses, administrative affiliations, municipality codes, and headquarters or branch status. I reconcile the January 2014, March 2023, and June 2026 editions to follow agencies across changes in names, identifiers, and addresses. Comparing these records with the policy histories helps distinguish a move to another municipality from an address correction or an organizational change. It also reveals why a partial relocation may leave the parent agency at its original address. The register and IAB data serve separate purposes: the former documents agency locations, while the latter measures employment at the establishment level. Appendix B describes the registry reconciliation.

3.2 Employment Data

The main employment source is the IAB’s Integrated Employment Biographies (IEB), from which I use linked worker and establishment records for 2007–2024. The IEB combine employers’ social-security notifications with administrative records on benefit receipt, job search, and labor-market programs (IAB FDZ 2026; IAB 2017). This source offers a combination essential for the analysis: broad coverage of insured employment,

detailed locations, and worker histories that continue across employers. It allows me to study small local labor markets and to connect changes in local employment with the movements of individual workers.

The records distinguish the municipality where a person works from the municipality where that person lives. I use this distinction to construct annual workplace and residence employment measures and to observe commuting across municipal boundaries. The same person and establishment identifiers link employment spells over time; the records also report industry, earnings, education, and demographic characteristics. These dimensions make it possible to ask whether employment grows around a receiving agency, whether local residents benefit, and whether the adjustment involves changes in residence or commuting. County-level aggregates provide the corresponding measures for the model comparison.

Table 1 shows that the receiving and sending municipalities differ sharply in size. In the matched policy sample, the 34 receiving municipalities had an average of 8,857 workplace employees in 2014, compared with 228,704 in the six sending municipalities. By 2024, these averages were 9,388 and 262,824. The policy therefore connects small local labor markets with much larger employment centers. The table also reports employment by residence, the administration share, and establishments in each group.

	Receiving municipalities		Sending municipalities	
	2014	2024	2014	2024
Workplace employment	8,857	9,388	228,704	262,824
Administration workplace employment	450	540	9,474	11,635
Other-industry workplace employment	8,407	8,849	219,230	251,190
Employed residents	6,182	6,826	167,362	193,753
Administration share of workplace employment (%)	4.68	5.27	4.24	4.58
Establishments	715	659	14,742	14,213
Municipalities	34	34	6	6

Table 1. Employment and establishments in the matched policy sample. Unweighted means per treated municipality, except the final row. The one-to-one matching uses pre-treatment levels and trends and draws controls from outside Bavaria. The sample covers projects classified as *Regionalisierung von Verwaltung* with a treatment date by 2024. Employment counts are persons; the administration share is the mean municipal percentage. Administration denotes WZ 2008 division 84. Components may not sum to totals because the source reports rounded means.

Industry and employment status answer different questions. Industry codes identify public administration, defense, and social security, but do not identify every publicly owned employer. Public schools and hospitals, for example, fall in other industries. Civil servants are also outside the IEB employment records because their jobs are exempt from social-security contributions, so relocated civil-service positions are not observed in the employment data. Self-employment is not covered either.

The worker histories also provide evidence on how agencies adjust their workforce. I identify candidate establishments in two steps. First, I use the workplace postcodes recorded for program origins and destinations to locate public-administration estab-

lishments in the IAB data. Second, I look for establishment openings, closures, and unusually large employment changes between 2014 and 2024. At destinations, I retain establishments that are absent in 2014 but present in 2024, or that expand by at least two baseline standard deviations. At origins, I retain establishments that disappear by 2024 or contract by at least one baseline standard deviation. The baseline standard deviation is calculated across establishments' employment changes from 2010 to 2013, separately for origin and destination postcode areas. This procedure captures both whole-establishment changes and substantial expansions or contractions within continuing establishments.

I then use worker identifiers to follow employees of candidate sending establishments in 2014 through 2024. Among those subsequently employed at candidate receiving establishments, changes in recorded residence distinguish movers from workers who keep their residence and commute. I also identify employees at receiving establishments who were not in the original sending cohort. Section 6 presents this evidence. Appendix B gives the sample definitions, including the treatment of postcodes that occur at both ends of the program.

The model uses employment from the 2011 German Census. Its reference date, May 9, 2011, precedes the program, and its employment coverage includes civil servants as well as workers outside social-security employment (Statistical Offices' Research Data Centres 2026). The Census provides workplace and residence counts by industry at the county level. It supplies the baseline cross-section for the model, while the annual IAB panel supplies the empirical changes.

Table 2 summarizes the published county Census counts. Bavaria contains 96 of the 412 counties in Germany in 2011. Its average county has about 68,400 people employed at local workplaces, of whom about 4,200 work in public administration. The large dispersion reflects the contrast between small counties and major employment centers. Workplace and residence counts describe different sides of the local labor market; their spatial distribution is central to a model in which workers can live and work in different places. As in the IAB descriptive table, the administration category is based on WZ 2008 division 84.

3.3 Controls Used in Mahalanobis Matching

For matching, I use pre-treatment population, employment, unemployment, income per taxpayer, and the population aged 18–30 from the German Regional Database (German Statistical Offices 2026). These variables capture the economic and demographic dimensions that Bavaria uses to benchmark municipalities against the state average when determining eligibility. I match on pre-treatment levels and trends to compare receiving places with municipalities facing similar disadvantages before the policy. The next section specifies the comparison pools and sample restrictions for receiving and sending locations.

3.4 Travel Costs

Commuting connects the local effects of the policy across space. I use Federal Employment Agency origin–destination flows to measure these connections and road travel times to

	Germany		Bavaria	
	Mean	SD	Mean	SD
Workplace employment	96,533	128,790	68,400	100,740
Public-administration workplace employment	6,794	11,332	4,238	8,102
Employed residents	97,524	109,720	68,522	78,107
Public-administration employed residents	6,821	8,330	4,213	4,597
Administration share of workplace employment (%)	6.9	3.0	6.0	2.8
Administration share of employed residents (%)	7.2	2.0	6.3	1.4
Counties	412		96	

Table 2. County employment in the 2011 Census. Unweighted means and sample standard deviations across counties in the published source files, including independent cities. Employment is in persons; sector shares are percentages calculated within each county. Public administration comprises administration, defense, and compulsory social security (WZ 2008 section O). Source: 2011 Census tables 2000X-2041 and 2000X-2045; Appendix B describes the extraction.

characterize the cost of reaching a workplace. For the county model, the travel-time matrix describes journeys between county locations. I compute road travel times with OpenStreetMap routing through OSRM.

3.5 Rents, Housing Prices, and Commercial Floor Space

To measure the price of residential space, I use the micro-geographic rent indices of Ahlfeldt, Heblich, and Seidel (2023). I take the municipal rent measures for 2011 and aggregate them to counties using population weights for the same year. These inputs describe the housing costs households face in the model and allow location choices to depend on more than wages and commuting.

Office rents provide a separate measure of the cost of space used by firms. I use the municipal office-rent series associated with Ahlfeldt, Heblich, Seidel, and Yin (2026) and construct corresponding population-weighted county measures for 2011.

4 Empirical Methodology

I estimate how local labor markets adjust when public jobs arrive and when they leave. For each receiving municipality, I construct a comparison with a similar untreated municipality; I repeat the procedure separately for sending locations. Long differences measure the change since the pre-policy year, while event studies trace adjustment around the first relocation. The same matches underpin the analyses using IAB employment records and the other municipal data.

4.1 Long Differences

The policy directs public jobs toward disadvantaged places, whose employment might have evolved differently even without a relocation. I therefore match municipalities on both their pre-policy economic conditions and their recent trajectories. The matching variables are population, residence employment, unemployment, income per taxpayer, and the population aged 18–30. For each variable, I use its 2014 level and pre-policy log change. These changes cover 2009–2014, except income, for which they cover 2010–2014. The variables capture the economic and demographic dimensions used in Bavaria’s assessment of regional disadvantage. Within the comparison pool, I select one nearest neighbor by Mahalanobis distance, without replacement, separately for receiving and sending municipalities.

I choose comparison municipalities outside Bavaria to limit contamination by policy spillovers within the state. Even municipalities that neither receive nor send an agency may be affected as workers, firms, and expenditure adjust across nearby locations. As an additional precaution, I exclude municipalities in the 20 non-Bavarian counties with the strongest commuting links to Bavaria.¹⁰ I also use comparison municipalities within Bavaria as a robustness check. This alternative helps assess whether the estimates reflect changes specific to Bavaria, although those controls may themselves be exposed to policy spillovers.

Matching brings receiving municipalities close to their controls across the selected characteristics. Figure 30 in Appendix C reports the absolute standardized differences before and after matching. For receiving locations, all one-to-one post-matching differences are below 0.12. Sending locations are closely matched on population and residence-employment levels, but larger differences remain in several pre-policy trends, particularly the change in the population aged 18–30. Appendix C reports the full balance statistics.

For each endpoint year, I compare the change in the treated municipality with the change in its matched control. The long-difference equation is

$$\Delta Y_{i,t} = a_{p(i),t} + \beta_t D_i + \varepsilon_{i,t}, \quad (1)$$

where the change is measured from 2014 to year t , D_i identifies the treated municipality, and $a_{p(i),t}$ is a fixed effect for matched pair $p(i)$. I estimate the equation separately for receiving and sending locations. For employment counts, the dependent variable is the log change; employment shares enter as level differences. The matching variables determine the pairs and do not enter the regression again as controls. I report heteroskedasticity-robust standard errors with the HC1 correction.

The IAB employment analysis follows outcomes through 2024. At each endpoint, I retain pairs whose treated municipality has already experienced its first relocation. Later-treated places therefore enter the long-difference sample as the program proceeds. Changes across endpoint estimates can reflect both a longer adjustment period and the arrival of new treatment cohorts. The event studies below instead align municipalities by time since treatment.

¹⁰The exclusion uses the counties identified by their shares of commuters to Bavaria in 2013. Appendix C shows their geography and documents the matching procedure.

A causal interpretation requires that, without the relocation, treated municipalities would have followed the same outcome trends as their matched controls. Matching on pre-policy levels and changes addresses observed differences relevant to this comparison. The design also requires that anticipation does not change outcomes during the comparison’s pre-treatment period. Agencies may prepare their premises and employees may adjust before operations begin, so I examine pre-treatment coefficients and alternative timing definitions.

4.2 Staggered Difference-in-Differences

I use the estimator of Sun and Abraham (2021) as the main staggered difference-in-differences specification. It traces employment before and after treatment while allowing the response to differ across cohorts of municipalities first treated in different years. I estimate receiving and sending effects separately, using their matched never-treated municipalities as the comparison groups.

Treatment begins in the year of the first recorded relocation into a receiving municipality or out of a sending municipality. I use the official treatment year where it is available and the recorded completion year otherwise. Some municipalities experience further relocations, so first treatment marks the start of exposure rather than the completion of every project. Figure 2 shows the timing of individual projects by municipality and program; Table 16 reports first- and last-treatment cohorts in the matching roster. The distinction is especially relevant for places receiving several agencies over time.

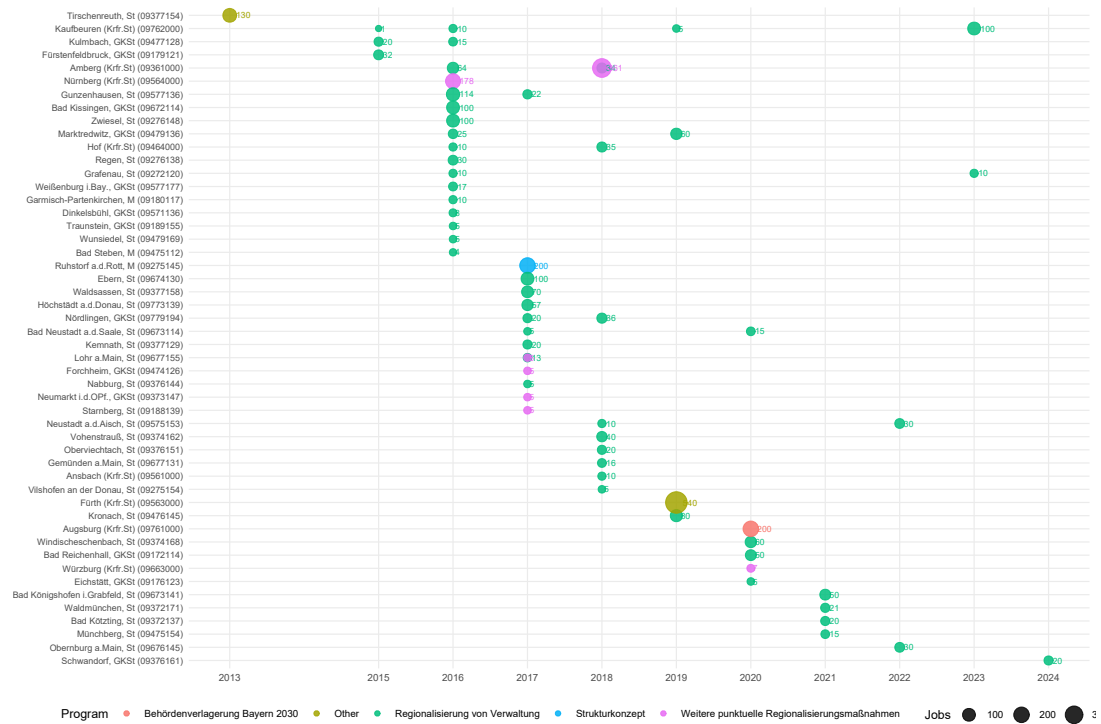
Let G_i denote municipality i ’s first treatment year. I estimate cohort-specific event-time effects using

$$Y_{it} = \alpha_i + \delta_{p(i),t} + \sum_{g \in \mathcal{G}} \sum_{\ell \neq -1} \theta_{g\ell} \mathbf{1}\{G_i = g\} \mathbf{1}\{t - g = \ell\} + u_{it}, \quad (2)$$

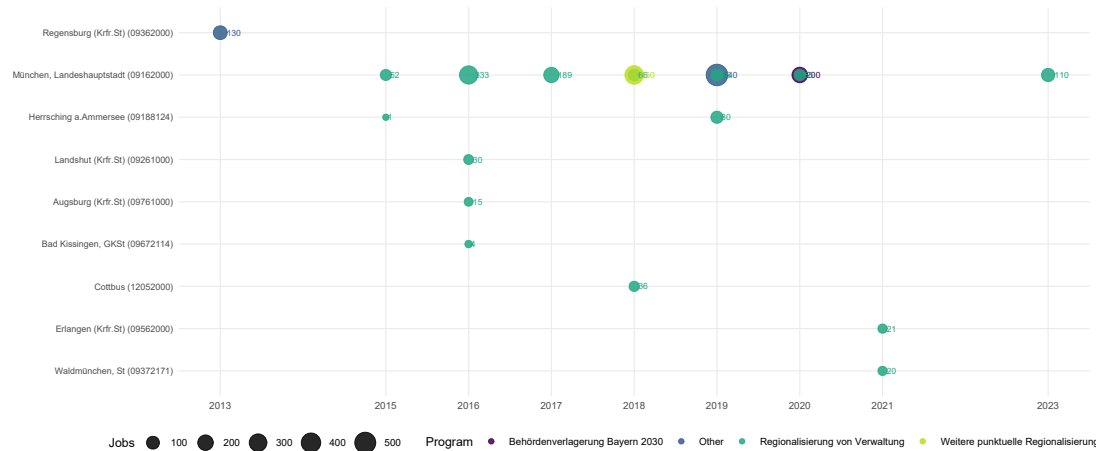
where α_i absorbs time-invariant municipal characteristics and $\delta_{p(i),t}$ absorbs shocks common to a matched pair in a given year. The year before first treatment is the reference period. I aggregate the cohort-specific coefficients at each event time to obtain the reported event-study effects. The main employment-count event studies use levels, so their coefficients measure employees; the long-difference estimates instead measure changes in log employment. Standard errors are clustered by municipality.

As a timing robustness check, I use the last recorded treatment year. This specification compares outcomes before the first relocation with outcomes from the last relocation onward. For municipalities treated more than once, I omit the intervening years rather than classifying them as untreated. The check asks whether the findings depend on dating exposure from its onset while additional relocations are still taking place. Appendix C gives the corresponding difference-in-differences specification.

I also estimate the group-time treatment effects of Callaway and Sant’Anna (2021) as a complementary specification. These estimates use never-treated controls and a common pre-treatment reference year within each cohort, with bootstrap confidence bands for the dynamic effects. Appendix C documents the aggregation and inference. Together with the alternative comparison pools and the sending-location estimates excluding Munich,



(a) Receiving locations



(b) Sending locations

Figure 2. Timing of relocation projects by municipality and program. The figures show the broader program inventory; the matched-sample first- and last-treatment cohorts are reported in Table 16.

these checks address sensitivity to timing, controls, and the influence of the largest origin. The next section presents the employment responses at both ends of the relocations.

5 Results

The relocations gradually expand administration employment at receiving workplaces. Employment in other industries also rises, both at local workplaces and among local residents. I first trace these changes around the start of a relocation, then compare outcomes over a common calendar period. Throughout, I examine sending locations separately: the arrival of jobs in one municipality need not imply an equal measured loss in another. I discuss employment effects as percentages of their 2014 levels, while the event-study figures report employee counts.¹¹

5.1 First Stage: Public Employment Relocation

The first stage asks how administration employment changes when a municipality receives or sends an agency. I measure this response directly in the IAB records, using public-administration employment measured at the workplace. These outcomes capture covered public-administration employment across the municipality. Changes in this employment need not equal the number of posts listed in the relocation program, which can include civil-service positions outside the IAB records.¹²

Administration employment at receiving workplaces grows gradually over several years. Panel (a) of Figure 3 shows an estimated increase equivalent to 2.5% of 2014 administration workplace employment one year after the first relocation, rising to 11.6% after five years, with a standard error of 3.8 percentage points. This gradual expansion accords with the staged implementation described in the institutional setting. The estimates alone do not identify which workers commute, move, or enter these jobs.

Sending municipalities do not show an equally clear contraction in administration employment. Panel (b) of Figure 3 shows estimates close to zero at shorter horizons and negative estimates at longer horizons, with wide confidence intervals. These estimates capture net changes in the municipal administration workforce, which need not equal the number of posts relocated.

¹¹For this comparison, I divide each count estimate and its standard error by the mean 2014 level of the same outcome in the corresponding treated descriptive sample. The receiving baselines are 450 administration workplace employees, 8,407 private workplace employees, and 5,920 private employees at residence; the sending private-employment baselines are 219,230 at workplaces and 161,704 at residence. This rescales the count estimates for interpretation; it does not estimate an average municipal percentage effect.

¹²As described in Section 3, public administration denotes WZ 2008 division 84. I use “private employment” below as shorthand for covered employment in the remaining industries. This category includes publicly owned employers outside administration; it is not a classification by ownership. Private employment is observed directly and does not require subtracting recorded program jobs from total employment.

5.2 Event-Study Results

The main event studies trace private employment around the first relocation using the Sun–Abraham specification in Section 4. Figure 4 follows the same panel order and event-time convention as the first stage. Workplace employment measures activity located in the municipality; residence employment measures employed people living there, including those working elsewhere. Both outcomes are employee counts, relative to the year before treatment and the matched control. Longer horizons cover fewer treatment cohorts, so changes along a plotted path can reflect changes in the contributing municipalities as well as adjustment over time.

5.2.1 Receiving Locations

Private employment grows as receiving municipalities absorb the public jobs. The workplace estimate is close to zero in the treatment year and rises to 2.3% of 2014 private workplace employment after three years and 4.3% after five years. The five-year estimate has a standard error of 2.0 percentage points. Private employment among residents also increases, reaching 3.3% of its 2014 level after five years, with a standard error of 1.2 percentage points. The evidence therefore points to an expansion extending beyond the administration jobs themselves, with gains visible on both sides of the local labor market.

The workplace and residence measures answer different questions about that expansion. More jobs at local workplaces indicate greater local employment demand; more employed residents show that the employment response is also associated with the population living in the receiving municipality. However, the latter group can include newcomers as well as established residents, and its members can work outside the municipality. Subtracting the two estimates would not identify how many incumbent residents find work or how many employees begin commuting. Those worker-level adjustments are examined in Section 6.

The receiving estimates become less precise at longer horizons, especially for workplace employment. At eight years, the workplace estimate is 5.6% of its 2014 level, with a standard error of 3.2 percentage points, while the residence estimate is 8.1%, with a standard error of 2.8 percentage points. These horizons draw only on the early-treated municipalities. Before treatment, the private-employment estimates are small and their confidence intervals include zero, which is consistent with parallel pre-treatment trends.

5.2.2 Sending Locations

Sending municipalities show positive private-workplace employment estimates throughout the post-treatment window, although the longer-horizon intervals are wide. Five years after treatment, the workplace estimate is about 3.5% of 2014 private workplace employment, with a standard error of about 2.9 percentage points. The corresponding residence estimate is about 0.9% of its 2014 level, with a standard error of about 1.2 percentage points. Residence employment turns negative at later horizons and becomes very imprecise. Relative to their initial size, the gains at origins are smaller than at

destinations. The sending evidence therefore points to the absence of losses rather than to a sustained gain on every margin.

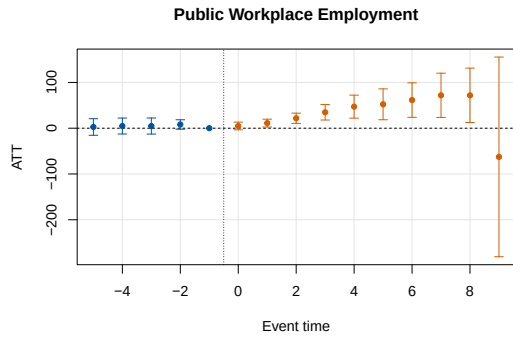
The sending comparison also warrants more caution before treatment. Private workplace and residence employment estimates are negative in the displayed pre-treatment years, indicating that the later rise begins from a relative shortfall. These pre-treatment differences and the small number of origins limit how confidently the post-treatment movement can be attributed to the relocation. The long-difference results below provide a complementary proportional comparison over the full observation period.

5.3 Long-Difference Results

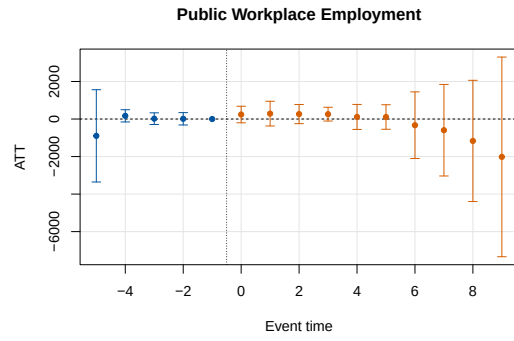
The long differences show how outcomes diverge from their pre-policy levels as the program unfolds. Each point in Figure 5 compares the change from 2014 to the indicated year with the change in the matched control. I report every available post-policy endpoint for each outcome. Unlike the event studies, these comparisons align municipalities by calendar year rather than time since their first relocation. Later-treated municipalities enter as they become exposed, so the paths reflect both a longer adjustment period and the addition of treatment cohorts. Appendix G reports the year-by-year estimates, standard errors, and p values.

5.3.1 Receiving Locations

Private employment among residents grows progressively more in receiving municipalities than in their matched controls. The estimated difference is around 0.8–1.0% at the first two endpoints, 1.3–1.9% during 2017–2021, and 2.8–3.5% during 2022–2024. Workplace employment shows a weaker pattern. Its early estimates alternate around zero; from 2018 onward they are positive, ranging from about 1.1% to 2.9%, but remain imprecise. The long differences therefore give clearer evidence of a sustained increase in employed residents than of proportional growth in local jobs.



(a) Receiving: public workplace employment



(b) Sending: public workplace employment

Figure 3. Public-administration workplace employment around the first relocation. Sun–Abraham event-study estimates in employee counts, with year -1 as the reference. Panels (a) and (b) report receiving and sending municipalities, respectively. Panels have different vertical scales. Bars show pointwise 95% confidence intervals based on standard errors clustered by municipality.

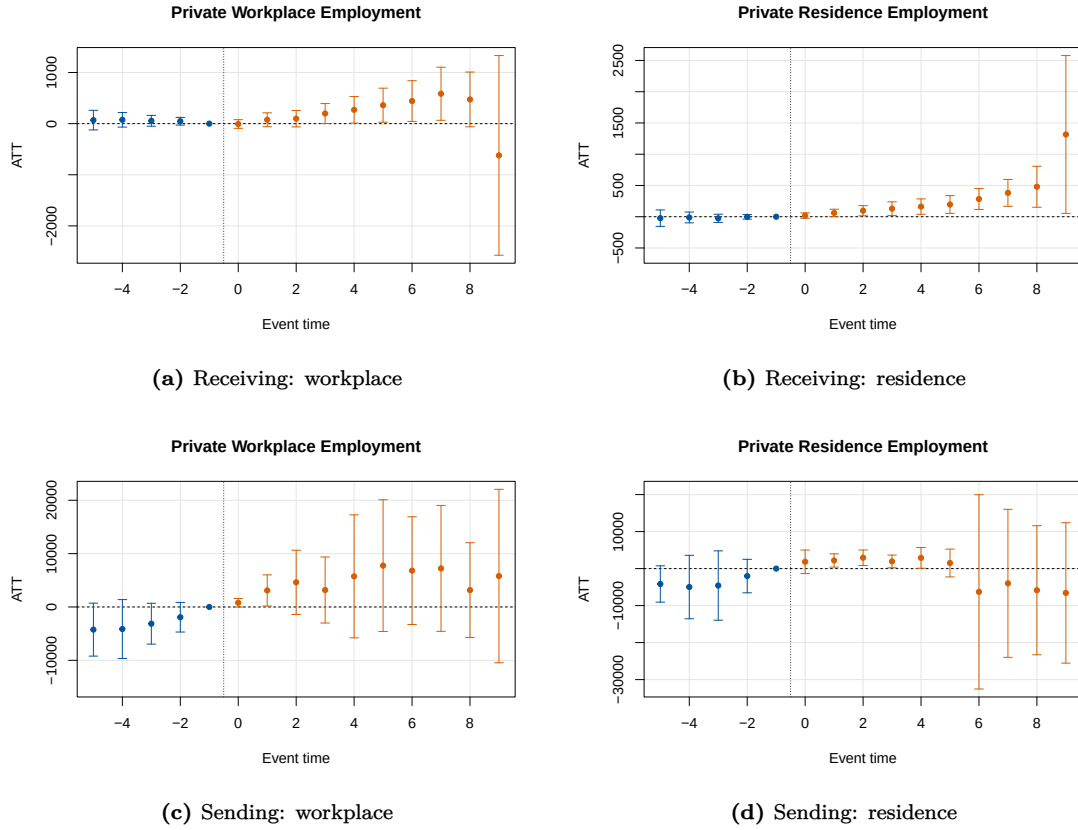


Figure 4. Private employment around the first relocation. Sun–Abraham event-study estimates in employee counts, with year -1 as the reference. Receiving and sending municipalities are in the top and bottom rows; workplace and residence outcomes occupy the left and right columns. Panels have different vertical scales. Bars show pointwise 95% confidence intervals based on standard errors clustered by municipality.

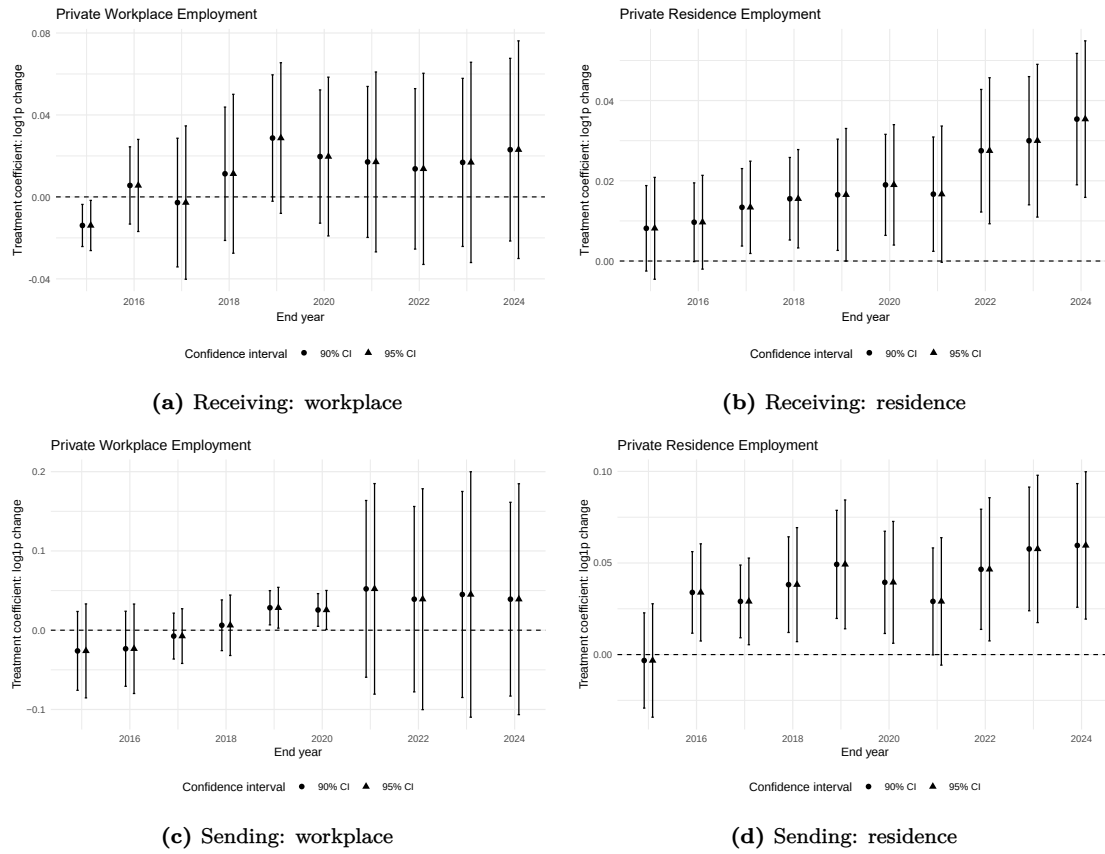


Figure 5. Private-employment long differences across all available endpoints. Each point compares the change since 2014 with the matched control. Receiving and sending municipalities occupy the top and bottom rows, with workplace and residence employment in the left and right columns. The figures show 90% and 95% confidence intervals.

5.3.2 Sending Locations

The sending long differences show a gradual shift toward higher private employment. Workplace estimates are negative at the first three endpoints, turn positive in 2018, and remain positive through 2024. The later estimates are roughly 1–5% relative to 2014 employment. Residence estimates are slightly negative in 2015 and positive at all later endpoints, generally around 3–6%. Neither path is monotonic, and the workplace estimates are substantially less precise. These calendar-year comparisons complement the event studies, whose longer horizons draw on fewer early-treated municipalities.

5.3.3 Robustness

The receiving-location sensitivity checks in Figure 38 compare outcomes using Bavarian controls over all available endpoints. This change reduces the residence estimates to within about 0.7% of zero throughout 2015–2024; workplace estimates are mostly negative and imprecise. The alternative comparison cannot distinguish shared Bavarian shocks from exposure of Bavarian controls to policy spillovers, but its smaller residence estimates show why the comparison pool matters. I use outside-Bavaria controls for the main design for the reasons set out in Section 4; the quantitative model in Section 7 implies that spillovers onto these controls are negligible. Excluding Munich, the largest origin, the 2014–2024 long difference for private employment among residents of sending municipalities is 6.6% (standard error 1.7 percentage points), and the workplace estimate remains positive but imprecise (Table 24).

The alternative timing comparison addresses a different concern: agencies may continue expanding after their first relocation. In the simple difference-in-differences check in Table 17, dating the post-period from the last relocation raises the receiving private-residence estimate from 3.8% to 4.2% of its 2014 level, with standard errors of about 1.6 percentage points in both cases. The sending comparison in Figure 31 is also positive but imprecise under both dates. Under the last-treatment definition, these comparisons omit the transition years between the first and the last relocation.

The place-level evidence leaves a central adjustment question open. Receiving municipalities gain administration jobs and private employment; sending municipalities show positive residence-employment long differences but less conclusive employment dynamics. To understand how these changes arise, I next follow workers: who moves with an agency, who keeps a residence and commutes, and who joins at the destination.

6 Worker Adjustment

Public jobs can change location without the workers who hold them changing residence. To study how the workforce adjusts, I follow two groups: workers employed at candidate origin establishments in 2014, and workers employed at candidate destination establishments in 2024. As described in Section 3 and Appendix B, I identify candidate establishments among public-administration establishments in the workplace postcodes of program origins and destinations, using openings, closures, and large employment changes. These

candidates are not matched one-to-one to agencies in the relocation register, and the procedure does not pair a specific origin with a specific destination. The evidence in this section is therefore descriptive: it characterizes adjustment around relocating offices rather than decomposing the municipal treatment effects.

Few incumbents relocate with their jobs. Of the 2,352 workers employed at candidate origin establishments in 2014, 284 work at a candidate destination in 2024 (Table 3). Of these, 242 (10.3% of the origin cohort) have the same recorded residence as in 2014 and therefore commute, while only 42 (1.8%) have changed residence. A further 957 workers (40.7%) are in regular employment elsewhere without having changed residence; I refer to them as stayers. Another 941 workers (40.0%) are not observed in regular employment in 2024. This category combines retirement, unemployment, self-employment, and moves into civil-service status, which the records do not cover. The remaining 170 workers (7.2%) fall into none of these categories.¹³

Table 3. Origin-worker classifications and destination employment

Category	Workers	Share (%)
A. Origin cohort: 2,352 workers employed in 2014		
At a candidate destination in 2024, changed residence (movers)	42	1.8
At a candidate destination in 2024, unchanged residence (commuters)	242	10.3
In regular employment elsewhere, unchanged residence (stayers)	957	40.7
Outside observed regular employment in 2024	941	40.0
Not classified in the categories above	170	7.2
Origin cohort total	2,352	100.0
B. Destination cohort: 31,401 workers employed in 2024		
Outside the 2014 origin cohort	31,117	99.1
In the 2014 origin cohort	284	0.9
Destination cohort total	31,401	100.0

Notes: Each panel has its own denominator. Candidate establishments are selected within program workplace postcodes and are not verified matches to individual agencies. Movers and commuters work at a candidate destination in 2024 with changed and unchanged recorded residence, respectively. Stayers are in regular employment elsewhere with unchanged residence; they need not retain their employer or workplace. Outside observed regular employment includes missing records and employment states other than regular employment; it does not identify unemployment. Panel B is an employment stock, not a count of hires caused by the policy. Percentages are rounded to one decimal.

Most stayers remain in public administration. Of the 957 stayers, 746 (78.0%) work in public administration, defense, and compulsory social security in 2024, and the remaining 211 work in other industries (Table 4 and Figure 6). Because a stayer need not keep the same employer or workplace, these figures are consistent with reassignment within the public sector, which the personnel framework explicitly allows, but they do not identify it.

¹³The classification compares the 2014 and 2024 endpoints and therefore does not capture moves followed by returns in between.

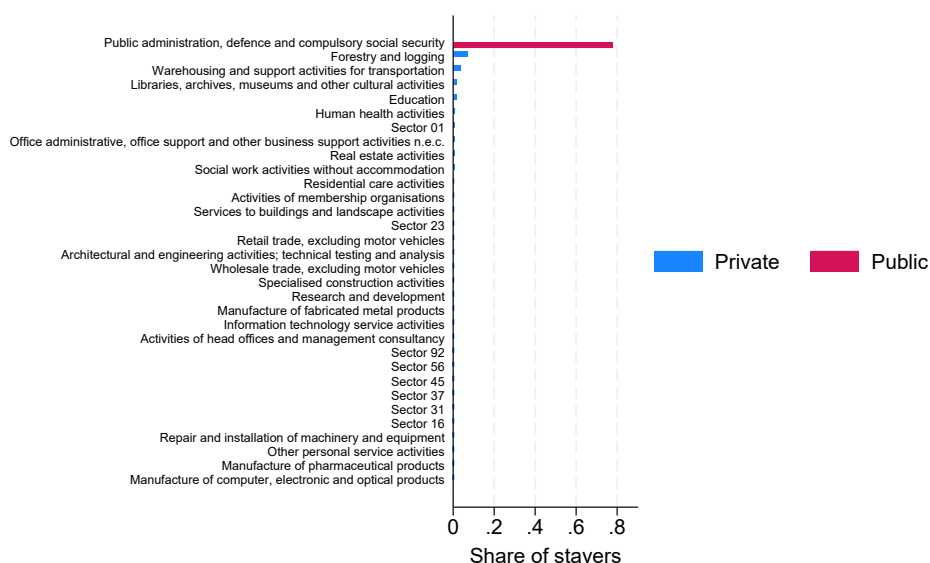


Figure 6. Sectors of origin-cohort stayers in 2024. Shares are calculated across the 957 workers classified as stayers. Public denotes public administration, defense, and compulsory social security; other industries are grouped as private in the figure.

Destination offices are staffed mostly by workers from outside the original workforce. Of the 31,401 workers employed at candidate destination establishments in 2024, 31,117 (99.1%) were not employed at the candidate origin establishments in 2014. This share describes the composition of destination employment rather than hiring caused by the relocations: the candidate set includes expanding establishments, whose existing employees are also outside the origin cohort.

Many of these workers come from public administration. Among the 20,217 outside-cohort employees with a recorded sector in 2014, 12,606 (62.4%) worked in administration and 7,611 in other industries (Figure 7). The remaining 10,900 workers (35.0%) have no sector record for 2014. They may have been in education, self-employment, civil service, or out of employment in 2014, and the records do not distinguish these states.

Average gross daily wages rise between 2014 and 2024 in both groups (Figure 8): by 10.1% for destination employees outside the origin cohort and by 8.3% for stayers. These are changes in group means. For the destination employees, they also reflect changes in the composition of the group, which has 20,217 wage observations in 2014 and 31,117 in 2024; the stayer comparison contains 957 observations in each year.

Taken together, the worker evidence shows that relocating public jobs reallocates workplaces far more than it reallocates workers. Incumbents mostly commute or remain in public employment without moving, while destination offices fill their positions with workers from outside the original workforce, many of whom already worked in administration. This helps explain why workplace and resident employment can respond differently

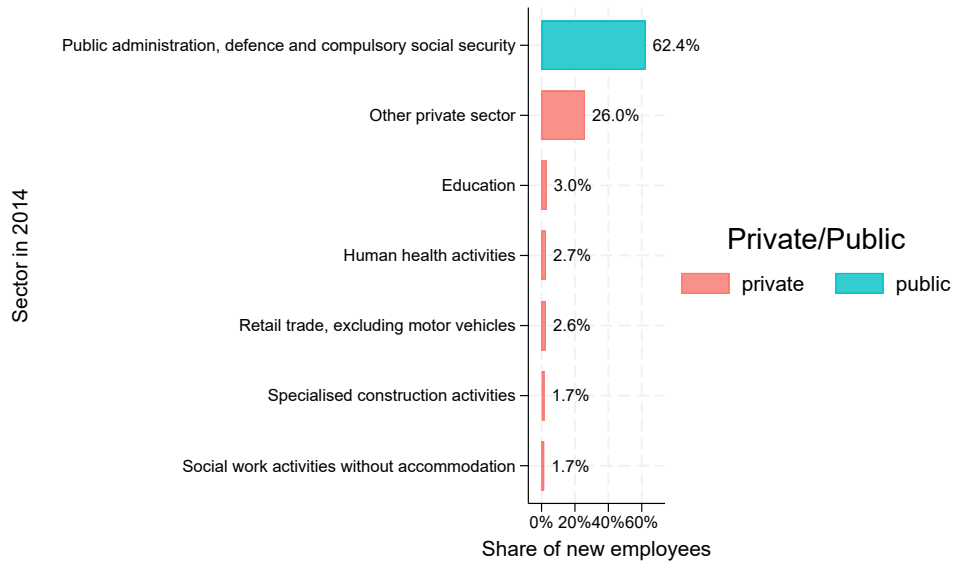


Figure 7. Sectors in 2014 of destination employees outside the origin cohort. The figure covers the 20,217 workers with a recorded sector in 2014. New employees denotes destination employees outside the 2014 origin cohort.

Table 4. Sector composition and coverage of the worker samples

Category	Workers	Share (%)
A. Stayers' industry in 2024: denominator 957		
Administration (WZ 84)	746	78.0
Other industries	211	22.0
Total	957	100.0
B. Outside-cohort employees' industry in 2014		
<i>Denominator: 20,217 with a recorded sector in 2014</i>		
Administration (WZ 84)	12,606	62.4
Other industries	7,611	37.6
Total with a recorded sector	20,217	100.0
C. Sector coverage: denominator 31,117		
Recorded sector in 2014	20,217	65.0
No recorded sector in 2014	10,900	35.0
Total outside-cohort destination employees	31,117	100.0

Notes: Panel A describes origin-cohort stayers, as defined in Table 3. Panels B and C describe 2024 employees of candidate destination establishments who are outside the 2014 origin cohort. WZ 2008 division 84 comprises public administration, defense, and compulsory social security; other industries can include publicly owned employers. Panel B measures industry in 2014, not immediately before hiring. Percentages are rounded to one decimal.

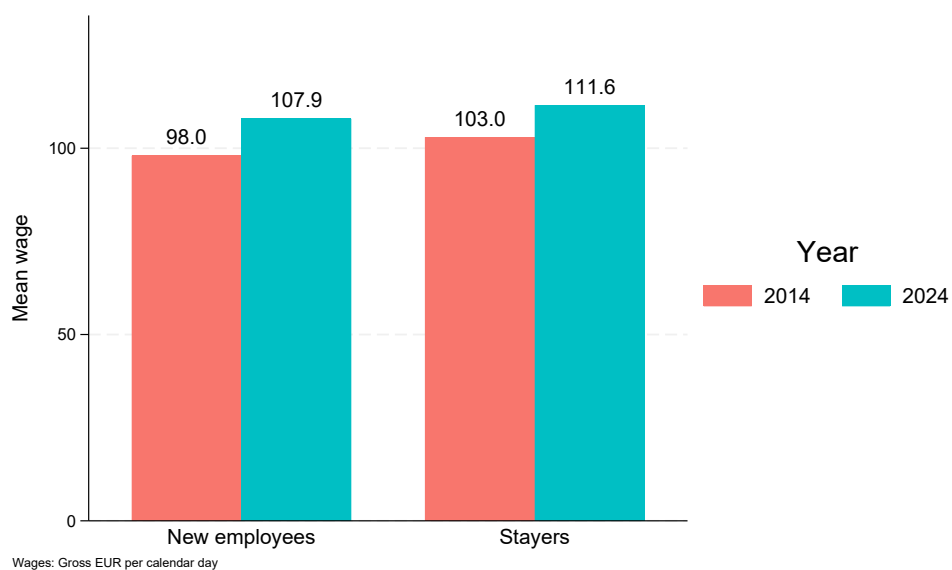


Figure 8. Mean gross wages in euros per calendar day, by worker group and year. New employees denotes destination employees outside the 2014 origin cohort.

in the place-level estimates, and it motivates the distinction in the model between the location of public jobs and the residential choices of public workers.

7 Two-Sector Quantitative Spatial Model

Local employment responses do not by themselves reveal whether relocating public workplaces raises private production or improves workers' living standards. A receiving place can gain employment by drawing workers from elsewhere, and changes in wages, housing costs, and commuting affect the people who hold the new jobs. I therefore use a quantitative spatial model to connect the place-level evidence to private output, its spatial distribution, and worker welfare throughout Germany. The model distinguishes workplaces from residences, consistent with the worker evidence: moving an office changes where its jobs are located without requiring its workers to live there.

Two mechanisms link public jobs to private activity. First, public and private employment enter private productivity separately, so public employment can generate spillovers for nearby private firms that differ from those of private employment. Second, public offices compete with private firms and households for floor space. Moving offices out of an origin releases part of its stock, while accommodating them at the destination reduces the space available to other uses. A common labor-income tax finances public payroll and office costs, linking these adjustments to disposable income. The model contains one freely traded private good and does not specify public-service production, so it is silent on whether service quality changes when agencies move.

7.1 Model Setup

I extend the spatial framework of Ahlfeldt, Redding, et al. (2015) to distinguish assigned public workplaces from private workplace choice. Locations cover all of Germany, with i indexing residences and j workplaces. Each location has land area $K_i > 0$ and a fixed effective floor-space stock $L_i > 0$. I quantify the model at the county level. Solving it for the whole country allows locations outside the policy state to respond as well.

The economy contains fixed national masses $\bar{H}^P$ and $\bar{H}^G$ of private and public workers. Each worker supplies one unit of labor, and sector membership is predetermined. The model therefore addresses spatial reallocation conditional on employment; it does not explain unemployment, participation, or switches between sectors. Both sectors pay the same endogenous gross wage at workplace j , $w_j^P = w_j^G = w_j$, which private labor demand determines. This is a simplifying assumption: actual public pay follows the schedules described in Section 2.

Public sector. The government assigns $\bar{H}_{Mj}^G$ public jobs to workplace j , with $\sum_j \bar{H}_{Mj}^G = \bar{H}^G$. A public worker chooses a residence given that assignment, so public resident employment, H_{Ri}^G , is endogenous. Consistent with the worker evidence, assignment fixes where public jobs are located, not where their holders live. Public jobs are paid the wage at their current workplace, including after relocation.

Private sector. Private workers choose both residence and workplace, generating private workplace employment H_{Mj}^P and private resident employment H_{Ri}^P . Private firms hire

competitively. Total workplace and resident employment satisfy

$$H_{Mj} = H_{Mj}^P + \bar{H}_{Mj}^G, \quad H_{Ri} = H_{Ri}^P + H_{Ri}^G.$$

Here, residents refers to employed residents. Each sector's workplace and residence totals equal its national mass.

7.1.1 Workers

Workers value consumption, residential space, amenities, and access to work. They spend share β of disposable labor income on the freely traded good and share $1 - \beta$ on housing. Let Q_i^R denote residential rent, t the common proportional labor-income tax, and z_{ijo}^m worker o 's preference draw for an admissible alternative in sector m . Indirect utility is

$$u_{ijo}^m = \frac{z_{ijo}^m B_i (1 - t) w_j}{(Q_i^R)^{1-\beta} d_{ij}}, \quad d_{ij} = \exp(\kappa \tau_{ij}), \quad (3)$$

where B_i is the amenity level of residence i and τ_{ij} is road travel time in minutes. Commuting reduces utility without subtracting cash from the household budget. The preference draws are independent Fréchet with common shape $\epsilon > 1$ and scales $T_i^P E_j^P$ for private alternatives and T_i^G for public residences given an assignment. These scales separate tastes from physical amenities and wages.

Private workers compare all residence–workplace pairs. Define $D_{ij} = \exp(-\epsilon \kappa \tau_{ij})$ and $Z_i^m = T_i^m B_i^\epsilon (Q_i^R)^{-\epsilon(1-\beta)}$. Their choice probabilities and worker flows are

$$\phi_{ij}^P = Z_i^P D_{ij} E_j^P [(1 - t) w_j]^\epsilon, \quad \Phi^P = \sum_{r,s} \phi_{rs}^P, \quad \pi_{ij}^P = \frac{\phi_{ij}^P}{\Phi^P}, \quad N_{ij}^P = \bar{H}^P \pi_{ij}^P. \quad (4)$$

Summing N_{ij}^P across residences yields private workplace employment; summing across workplaces yields private resident employment. Higher wages make a workplace more attractive, while housing costs and commuting make residence–workplace pairs less attractive.

Public workers compare residences for their assigned workplace. Writing $\Psi_j^G = \sum_r Z_r^G D_{rj}$, their conditional probabilities and flows are

$$\pi_{ij}^G = \frac{Z_i^G D_{ij}}{\Psi_j^G}, \quad N_{ij}^G = \bar{H}_{Mj}^G \pi_{ij}^G, \quad H_{Ri}^G = \sum_j N_{ij}^G. \quad (5)$$

The assigned wage and the tax cancel from this conditional comparison because they are the same for every residence available to that job. They still enter income, housing demand, and utility. A relocation changes the commuting opportunities attached to a public job, but it does not mechanically move its holder's residence.

The two choice sets imply distinct expected utilities. With $\Gamma_\epsilon = \Gamma(1 - 1/\epsilon)$,

$$\bar{U}^P = \Gamma_\epsilon (\Phi^P)^{1/\epsilon}, \quad \bar{U}^{G|j} = \Gamma_\epsilon (1 - t) w_j (\Psi_j^G)^{1/\epsilon}. \quad (6)$$

Private expected utility values access to the full private choice set. Public expected utility is conditional on a particular workplace assignment. Appendix F derives these objects and the corresponding employment margins.

7.1.2 Private Firms

Private firms produce a freely traded numeraire good using labor and private commercial floor space L_j^P :

$$Y_j^P = A_j^P (H_{Mj}^P)^\alpha (L_j^P)^{1-\alpha}. \quad (7)$$

They take productivity A_j^P , wages, and commercial rent q_j as given. Competition and constant returns imply

$$w_j H_{Mj}^P = \alpha Y_j^P, \quad q_j L_j^P = (1 - \alpha) Y_j^P. \quad (8)$$

Government offices are not part of L_j^P . Released offices can therefore expand the space available to private production or housing, with the allocation determined by rents and demand.

7.1.3 Within- and Between-Sector Productivity Spillovers

Private productivity depends separately on nearby private and public workplace density:

$$A_j^P = a_j (\Upsilon_j^P)^{\lambda_{PP}} (\Upsilon_j^G)^{\lambda_{PG}}, \quad (9)$$

$$\Upsilon_j^m = \sum_s \exp(-\delta_A \tau_{js}) \frac{H_{Ms}^m}{K_s}, \quad m \in \{P, G\}, \quad H_{Ms}^G = \bar{H}_{Ms}^G. \quad (10)$$

The fundamental a_j is fixed across policy scenarios. The elasticities λ_{PP} and λ_{PG} measure private-to-private and public-to-private spillovers, and δ_A governs their decay with travel time. Each source location contributes employment per unit of its own land area.

Moving public workplaces changes the public density term directly. The responses of private firms and workers then change the private density term, reinforcing or offsetting the direct effect. I maintain the private elasticity of Ahlfeldt, Redding, et al. (2015) and estimate the public elasticity from the county employment responses to the relocations.

7.1.4 Amenities, Floor Space, and Market Clearing

Both worker types contribute to amenities through surrounding resident density:

$$B_i = b_i \Omega_i^\eta, \quad \Omega_i = \sum_s \exp(-\delta_B \tau_{is}) \frac{H_{Rs}^P + H_{Rs}^G}{K_s}. \quad (11)$$

The amenity fundamental b_i is fixed, so amenities rise when weighted resident density rises.

Public offices occupy $\bar{L}_i^G$, leaving private firms and households to share the remaining floor space:

$$L_i = L_i^P + \bar{L}_i^G + L_i^R, \quad F_i = L_i - \bar{L}_i^G > 0. \quad (12)$$

The government does not choose its office space. At baseline, a public job uses a fixed fraction γ_G of the floor space used per private worker at the same location,

$$\ell_i^{G,0} = \gamma_G \frac{1 - \alpha}{\alpha} \frac{w_i^0}{q_i^0}, \quad \bar{L}_i^{G,0} = \ell_i^{G,0} \bar{H}_{Mi}^{G,0}. \quad (13)$$

Income earned at workplaces finances housing at residences. Gross resident income is $I_i = \sum_j w_j (N_{ij}^P + N_{ij}^G)$, and disposable income is $v_i = (1 - t)I_i$. Housing demand satisfies $Q_i^R L_i^R = (1 - \beta)v_i$. Every county is in mixed use, and a fixed land-use wedge ξ_i relates the two rents by $q_i = \xi_i Q_i^R$. Floor-space market clearing then gives

$$q_i = \frac{(1 - \alpha)Y_i^P + \xi_i(1 - \beta)v_i}{L_i - \bar{L}_i^G}, \quad Q_i^R = \frac{q_i}{\xi_i}. \quad (14)$$

Public occupancy reduces the floor space available to private uses. Its effect on rents also depends on the responses of private production and resident income.

The government finances public payroll and office rents with a common tax on labor income. Define $W^P = \sum_j w_j H_{Mj}^P$, $W^G = \sum_j w_j \bar{H}_{Mj}^G$, and $C^G = \sum_j q_j \bar{L}_j^G$. The budget implies

$$t(W^P + W^G) = W^G + C^G, \quad t = \frac{W^G + C^G}{W^P + W^G}. \quad (15)$$

Public wages are taxed on the same basis as private wages. Payroll and office costs change with equilibrium prices even though the number of public workers is fixed. The government pays the commercial rental equivalent for all offices, including state-owned premises, and the budget covers only the modeled administration sector.

An equilibrium makes workers' choices, private factor demands, assigned public employment, floor-space use, spillovers, and the government budget mutually consistent. Private labor supplied to each workplace equals firms' demand, public assignments are filled, and both national worker masses are conserved. Floor-space use must be feasible and $0 \leq t < 1$, which requires $C^G < W^P$. Expected utility adjusts endogenously because national worker masses are fixed.

7.2 Quantification

I quantify the model for the 412 German counties in their 2011 boundaries. The baseline combines Census employment by sector at workplaces and residences for 2011, county residential and office rents for 2011, commuting flows from the Federal Employment Agency for 2013, and road travel times in minutes (Section 3). The Census industry public administration, defense, and compulsory social security defines public employment, and all other industries define private employment.

Parameters. Table 5 lists the parameters. I take the production and consumption shares, the private-to-private spillover elasticity, the amenity elasticity, and the decay rates of productivity and amenity spillovers from Ahlfeldt, Redding, et al. (2015). I estimate the commuting decay $\nu = \epsilon\kappa$ from a gravity equation for county commuting flows, estimated by Poisson pseudo-maximum likelihood with origin and destination fixed effects and a control function for travel time. I choose the Fréchet shape ϵ so that the variance of log model wages across counties equals the variance of log observed gross wages in 2011, which gives $\epsilon = 2.49$ and a commuting cost of $\kappa = \nu/\epsilon = 0.0172$ per minute. The land-use wedge ξ_i is the observed ratio of commercial to residential rents in

each county. Unlike Ahlfeldt, Redding, et al. (2015), I do not restrict ξ_i to be at least one; the ratio is below one in 69 of the 412 counties.¹⁴ I set $\gamma_G = 0.25$, the ratio of the roughly 30 square meters per job in newly built state office buildings to the model's floor space per private worker in Munich, about 117 square meters. The tax rate that balances the government budget at baseline is 7.54%.

Table 5. Model parameters

Symbol	Description	Value	Source
<i>Estimated</i>			
λ_{PG}	Public-to-private productivity spillover	0.364	Indirect inference, county employment responses
ϵ	Fréchet shape	2.486	Variance of log county wages, 2011
$\nu = \epsilon\kappa$	Commuting decay (per minute)	0.0429	Gravity equation, 2013 county flows
<i>Calibrated or derived</i>			
κ	Commuting cost (per minute)	0.0172	ν/ϵ
ξ_i	Commercial-to-residential rent ratio	1.145	Observed 2011 county rents; median, range 0.76–2.05
γ_G	Office space per public job relative to private	0.25	Floor space per job in new state offices
t	Labor-income tax rate	0.0754	Balanced government budget
<i>Maintained</i>			
λ_{PP}	Private-to-private productivity spillover	0.071	Ahlfeldt, Redding, et al. (2015)
δ_A	Decay of productivity spillovers (per minute)	0.3617	Ahlfeldt, Redding, et al. (2015)
η	Amenity spillover elasticity	0.1553	Ahlfeldt, Redding, et al. (2015)
δ_B	Decay of amenity spillovers (per minute)	0.7595	Ahlfeldt, Redding, et al. (2015)
α	Labor share in private production	0.80	Ahlfeldt, Redding, et al. (2015)
β	Consumption share	0.75	Ahlfeldt, Redding, et al. (2015)

Given the parameters, I invert the model to recover the unobserved fundamentals that rationalize the baseline: private productivity, amenities, and the taste scales of both worker types. Appendix F describes the inversion. The baseline matches observed private workplace and resident employment by construction. Public resident employment, wages, and own-county commuting shares are not targeted. The model fits public residences closely and captures the cross-county wage gradient, but it understates the share of

¹⁴In Ahlfeldt, Redding, et al. (2015), $\xi_i \geq 1$ is one plus the tax equivalent of land-use restrictions on commercial use. A ratio below one can reflect an implicit subsidy to commercial use or measurement error in county rent ratios.

residents who work in their own county (Appendix Figure 34).

Estimating the public-to-private spillover. I estimate λ_{PG} by indirect inference. For each candidate value, I recover the fundamentals that rationalize the baseline, solve the relocation counterfactual described below, and compute the model-implied log change in private workplace employment, $g_j(\lambda_{PG}) = \log H_{Mj}^{P,cf} - \log H_{Mj}^{P,0}$. I then run the same matched-pair regression on the model outcomes that I run on the data. In each receiving or sending sample $\mathcal{S}_a$ of matched county pairs,

$$g_j(\lambda_{PG}) = \mu_{p(j),a} + \beta_a^M(\lambda_{PG})\text{Treat}_{j,a} + e_{j,a}, \quad j \in \mathcal{S}_a, \quad (16)$$

where $\mu_{p(j),a}$ is a matched-pair fixed effect. Control counties receive their model-implied outcomes, which need not be zero. The targets are the corresponding reduced-form coefficients for the 2014–2024 log change in county private workplace employment: 0.0238 (standard error 0.0140) for receiving counties and -0.0069 (0.0335) for sending counties, with standard errors clustered by matched pair. I choose

$$\hat{\lambda}_{PG} = \arg \min_{\lambda_{PG}} J(\lambda_{PG}), \quad J(\lambda_{PG}) = \sum_{a \in \{\text{receiving, sending}\}} \left[\frac{\beta_a^M(\lambda_{PG}) - \hat{\beta}_a^{RF}}{s_a^{RF}} \right]^2, \quad (17)$$

searching over $\lambda_{PG} \in [-0.1, 1]$.

The estimate is $\hat{\lambda}_{PG} = 0.364$, with an objective value of $J = 0.014$ at an interior minimum (Figure 9). The relocation moments identify only a combination of the two productivity elasticities, because private agglomeration amplifies the private response to a public shock. I therefore hold λ_{PP} at its value in Ahlfeldt, Redding, et al. (2015). The two elasticities apply to densities of very different size, since public employment is a small share of local employment, so their magnitudes are not directly comparable.

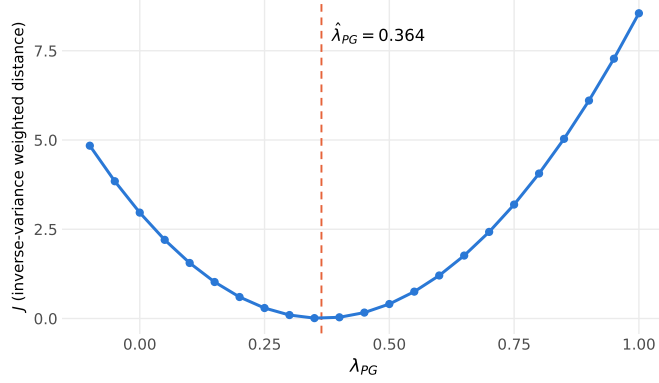


Figure 9. Indirect-inference objective. The figure plots the inverse-variance weighted distance J between model-implied and reduced-form coefficients against the public-to-private spillover elasticity λ_{PG} , holding $\lambda_{PP} = 0.071$ fixed. The dashed line marks the estimate, $\hat{\lambda}_{PG} = 0.364$.

Model fit and validation. Figure 10 compares the model and reduced-form coefficients. The model matches the receiving coefficient (0.0234 against 0.0238) and predicts a sending coefficient of -0.0108 against -0.0069 in the data. Among the untargeted moments, which cover private resident employment, wages, and residential and commercial rents in receiving and sending counties, all model coefficients lie within two standard errors of their reduced-form counterparts except one. In sending counties, the model predicts a 0.7% decline in commercial rents as offices are vacated, whereas commercial rents rose by 6.7% relative to the matched controls. As a holdout test, I re-estimate λ_{PG} from the receiving moment alone and obtain 0.369. At this value, the model predicts a sending coefficient of -0.0110 , close to the reduced-form estimate, although the sending moment is too imprecise for a sharp test (Appendix Figure 35).

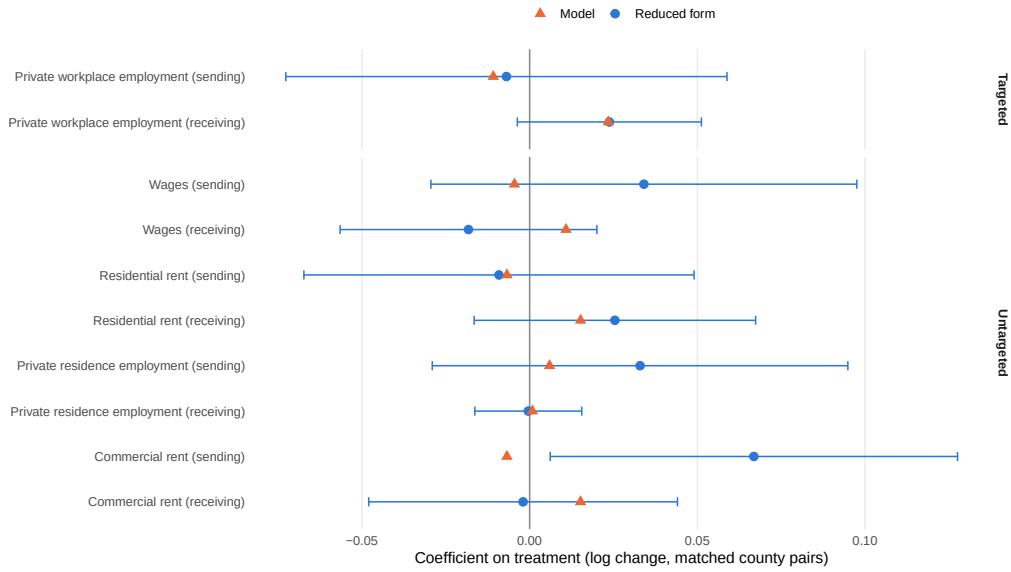


Figure 10. Model and county reduced-form coefficients at $\hat{\lambda}_{PG} = 0.364$. Each row reports the coefficient on treatment in a matched-pair regression of county log changes. Triangles show model coefficients and circles reduced-form coefficients with 95% confidence intervals based on standard errors clustered by matched pair. The top panel shows the two targeted moments; the bottom panel shows untargeted outcomes. Employment changes cover 2014–2024, and wage and rent changes cover 2014–2023.

The model also provides a check on the empirical design. Because it is solved for all of Germany, it predicts how the relocations affect the matched control counties outside the policy state. These effects are small: private workplace employment in the controls changes by -0.045% in the receiving design and by -0.041% in the sending design. Effects on the controls therefore shift the matched-pair coefficient by 2.0% and -3.6% of the effect on the treated counties, respectively (Appendix Figure 36). Contamination of the outside-state comparison group by policy spillovers is negligible.

7.3 Policy Counterfactual

I simulate the relocations completed between 2015 and 2024 as a single shock to the baseline economy. The shock comprises 2,076 positions, counting jobs and paid training positions, in 29 net-receiving counties. Of these, 1,409 are moves from an identified origin county. The remaining 667 positions have no identifiable origin county, for example because they belong to newly established offices; I draw them from the national public workforce in proportion to baseline public employment. Five counties are net senders, led by Munich with 1,051 positions. I classify counties that both send and receive by their net position. Let $n_{rs} \geq 0$ denote positions transferred from origin s to destination r . The net change in public workplace employment is $\Delta \bar{H}_{Mj}^G = \sum_s n_{js} - \sum_r n_{rj}$, so that

$$\bar{H}_{Mj}^{G,cf} = \bar{H}_{Mj}^{G,0} + \Delta \bar{H}_{Mj}^G, \quad (18)$$

$$\sum_j \Delta \bar{H}_{Mj}^G = 0. \quad (19)$$

National public and private employment are unchanged.

Each transferred position carries its origin's baseline office requirement, and positions drawn from the national pool carry the national average. Public office space changes by

$$\bar{L}_j^{G,cf} = \bar{L}_j^{G,0} + \sum_s n_{js} \ell_s^{G,0} - \sum_r n_{rj} \ell_j^{G,0}, \quad (20)$$

so national public office space is conserved while its location changes. Offices vacated at the origins return to private and residential use. Most arriving offices are rented from the existing stock. For relocations into new state-built offices completed by 2024, which cover 388 positions, I instead add the corresponding floor space to the receiving county's stock. A transferred job keeps its origin's office requirement, but its pay follows the destination's equilibrium wage, so public payroll is an equilibrium outcome. The tax rate adjusts to balance the budget.

I re-solve for wages, residential and commercial rents, private workplace and residence choices, public residence choices, floor-space use, productivity, amenities, and the tax rate. Productivity and amenity fundamentals, taste scales, land-use wedges, land, travel times, and national worker masses remain at their baseline values. Solving the model with a zero shock reproduces the baseline to numerical precision. The comparison is static: it describes the difference between two equilibria rather than the path of annual adjustment.

7.4 Counterfactual Results

Table 6 summarizes the main results. Figures 11 to 14 report changes for four groups of counties: receiving counties, sending counties, other counties in the policy state, and the rest of Germany. All changes compare the relocation equilibrium with the baseline.

Table 6. Counterfactual effects of the 2015–2024 relocations

Outcome	Change
<i>Private workplace employment (%)</i>	
Receiving counties	2.01
Sending counties	−0.80
Other counties in the policy state	−0.05
Rest of Germany	−0.04
<i>National aggregates</i>	
Private output, Germany (%)	−0.035
Private output, policy state (%)	0.031
Concentration of private output across counties, HHI (%)	−0.35
Private workers' expected utility (%)	0.017
<i>Public budget</i>	
Public payroll (%)	−0.06
Public office costs (%)	−0.11
Tax rate (percentage points)	−0.002

Notes: Changes from the baseline equilibrium to the equilibrium with the relocations completed by 2024, at $\hat{\lambda}_{PG} = 0.364$. County-group changes are percentage changes in group totals. HHI denotes the Herfindahl–Hirschman index of county shares in national private output.

7.4.1 Employment

Private workplace employment rises by 2.0% in receiving counties and falls by 0.8% in sending counties (Figure 11). Because the national private workforce is fixed, these changes reallocate private employment across space: counties elsewhere in the policy state and in the rest of Germany lose 0.05% and 0.04% of their private workplace employment. The prediction for sending counties is consistent with the county-level reduced form, which is close to zero and imprecise. It does not reproduce the positive municipal estimates for sending locations, which the office channel at the county level is too small to generate.

Residence responds much less than workplace employment because workers commute. Private resident employment rises slightly in all three groups of counties in the policy state, by 0.10% in receiving, 0.09% in sending, and 0.12% in other counties, and falls by 0.02% in the rest of Germany. Public residents follow the relocated jobs only partly: their number rises by 0.22% in receiving counties and falls by 0.29% in sending counties, as many public workers keep their residence and commute.

7.4.2 Productivity, Output, and Amenities

Private productivity rises by 0.97% in receiving counties and falls by 0.50% in sending counties (Figure 12). Holding fundamentals fixed, the change in productivity decomposes into

$$\Delta \log A_j^P = \lambda_{PP} \Delta \log \Upsilon_j^P + \lambda_{PG} \Delta \log \Upsilon_j^G, \quad (21)$$

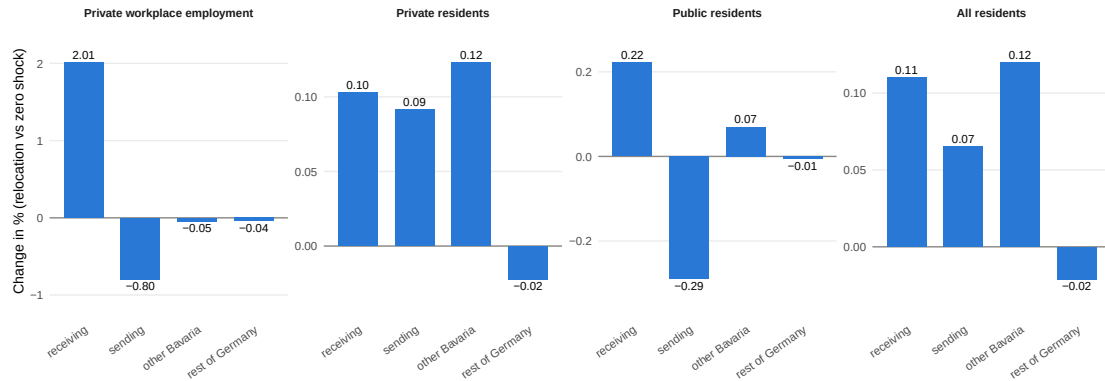


Figure 11. Counterfactual changes in employment, by county group. Percentage changes in group totals of private workplace employment, private resident employment, public resident employment, and all resident employment.

which combines the direct effect of public employment density with the induced response of private density. Together with the employment response, the productivity change raises private output by 2.8% in receiving counties and lowers it by 1.2% in sending counties. Amenities barely change, by at most 0.02%, because resident density changes little.

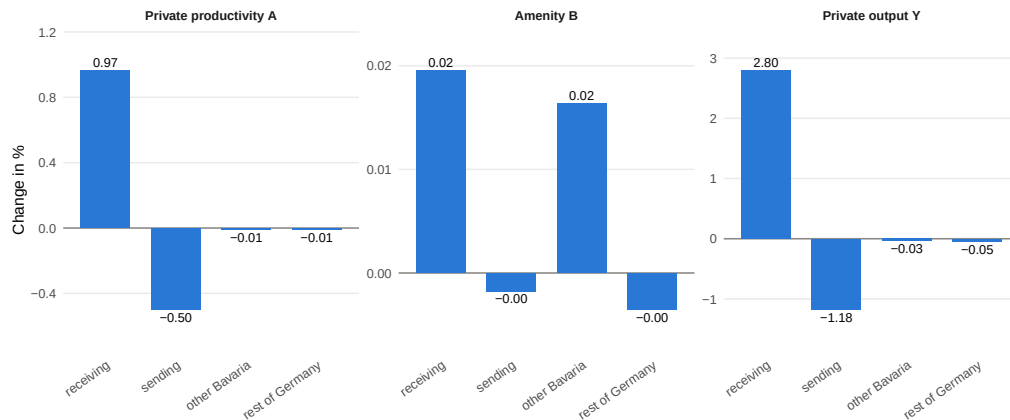


Figure 12. Counterfactual changes in private productivity, amenities, and private output, by county group. Productivity and amenities are group means; output is the group total.

7.4.3 Wages and Rents

Wages rise by 0.88% in receiving counties and fall by 0.40% in sending counties (Figure 13). Commercial and residential rents move in the same direction, by 1.27% and 1.24% in receiving counties and by -0.89% and -0.91% in sending counties. Higher wages in receiving counties therefore come with higher housing costs. Because public pay follows

the local wage, the wage increase also raises the pay of all public jobs located there, including those that did not move.

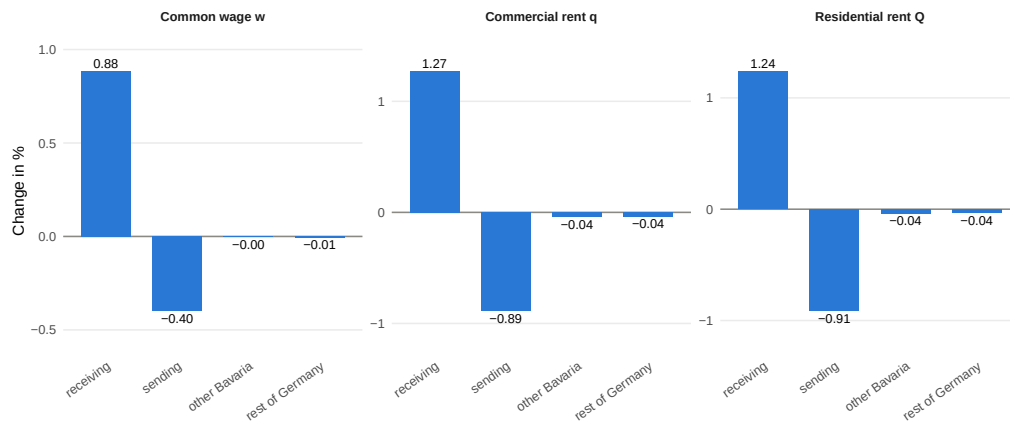


Figure 13. Counterfactual changes in wages, commercial rents, and residential rents, by county group. Group means.

7.4.4 Floor Space

Public office space rises by 2.6% in receiving counties and falls by 1.1% in sending counties (Figure 14). In receiving counties, private floor space expands by 1.5% along with private employment, while residential floor space contracts by 1.0%; the commercial share of floor space rises by 0.6 percentage points. In sending counties, the space released by public offices and contracting private firms shifts toward housing, which expands by 0.5%.

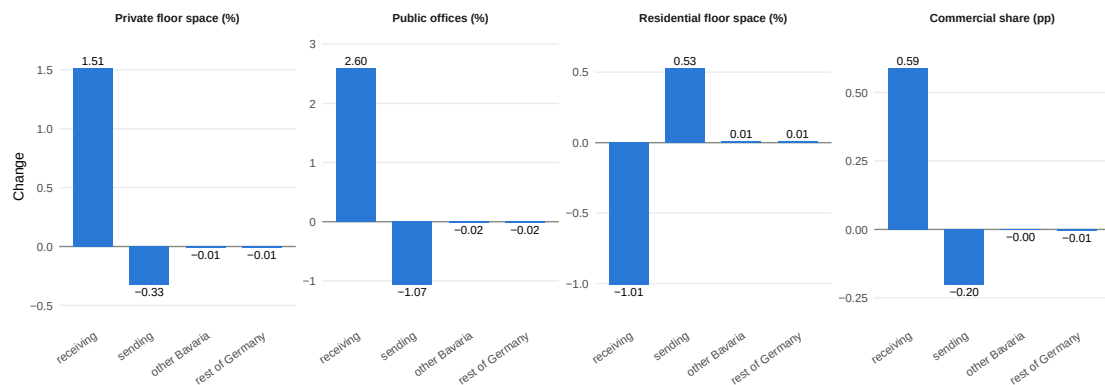


Figure 14. Counterfactual changes in floor-space use, by county group. Percentage changes in private commercial floor space, public office space, and residential floor space, and percentage-point changes in the commercial share of floor space.

7.4.5 National Aggregates and the Public Budget

National aggregates change little (Table 6). Private output falls by 0.035% in Germany but rises by 0.031% in the policy state. The concentration of private output across counties, measured by the Herfindahl–Hirschman index, falls by 0.35%, or 0.50 points on a scale from 0 to 10,000. Private workers’ expected utility rises by 0.017%. Public payroll falls by €79 million per year: €65 million because jobs move to counties with lower baseline wages, and €14 million because wages adjust. Together with lower office costs, this reduces the budget-balancing tax rate by 0.002 percentage points. I do not report a welfare measure for the relocated public workers. Their assignment-conditional utility compares different workplaces before and after the move and mainly reflects the baseline wage and access gap between Munich and the destinations rather than an effect of the policy.

7.4.6 Channel Decomposition

To identify the mechanisms behind these results, I switch off one channel at a time: the public spillover ($\lambda_{PG} = 0$), private agglomeration ($\lambda_{PP} = 0$), the relocation of office space (offices remain at their origin), state-built floor space, and the tax adjustment (Figure 15). The public spillover accounts for almost the entire effect. Without it, private employment in receiving counties changes by -0.02% instead of rising by 2.0% , and national private output rises by 0.0006% instead of falling by 0.035% . Private agglomeration amplifies the response: without it, receiving private employment rises by 1.7% . The office channels are small at the county level. Relocating offices lowers private employment in receiving counties by 0.007 percentage points and raises it in sending counties by the same amount, and state-built floor space adds 0.002 percentage points in receiving counties. Holding the tax rate at its baseline value would leave a budget surplus of €38 million per year and reduce the gain in private welfare by 0.002 percentage points.

7.4.7 Allocation Rule

Finally, I ask whether the program chose destinations where public jobs generate the most private output. I move 100 positions from Munich to each county in the policy state in turn. National private output falls in every case, by between €9 million and €101 million per year depending on the destination (Figure 16). The losses are smallest for destinations close to Munich, such as the surrounding district of Munich and Rosenheim, and largest for Schwabach and the district of Fürth. Actual receiving counties are neither systematically higher- nor lower-value destinations. In a linear probability model of receiving status, the coefficient on a county’s standardized marginal value is -0.001 (standard error 0.045), and 0.10 (0.09) conditional on the baseline wage (Figure 17).

Where the jobs go matters for the aggregate cost of the program. Sending the same positions from the same origins to the 29 counties with the highest marginal value would reduce the national output loss from 0.035% to 0.012% , reduce output concentration by more (0.60 instead of 0.50 HHI points), and raise private welfare by more (0.024%

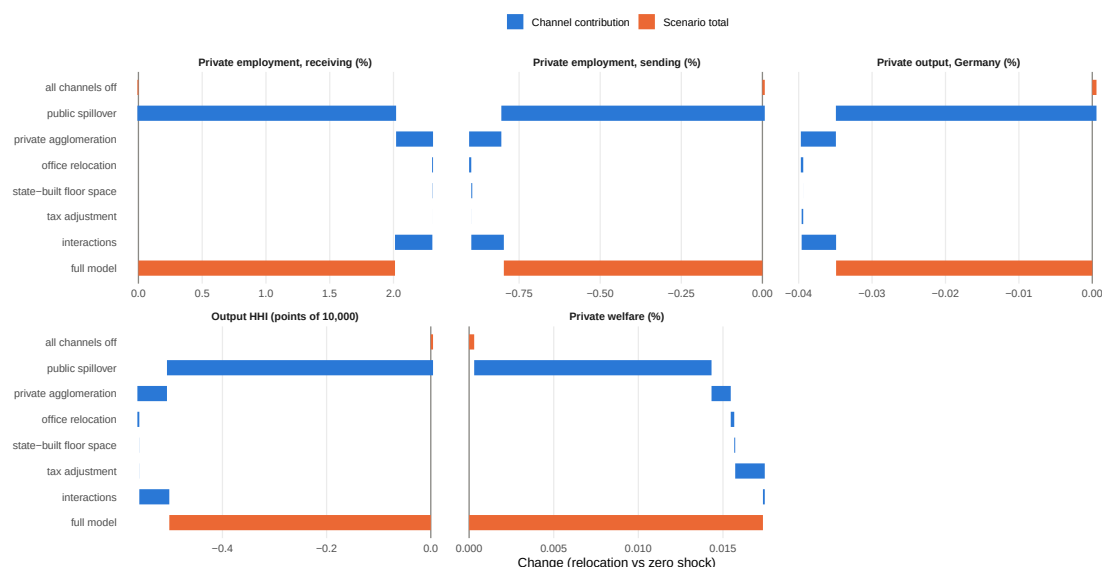


Figure 15. Channel contributions to the counterfactual effects. Each blue bar shows the contribution of one channel, measured as the full-model change minus the change with that channel switched off. Orange bars show the full-model change and the change with all channels switched off. Interactions are the residual. Appendix Figure 37 reports each scenario separately.

instead of 0.017%). Sending them instead to the 29 lowest-wage counties would raise the output loss to 0.046% (Figure 18).

Overall, the model attributes the gains in receiving counties to productivity spillovers from public employment rather than to the reallocation of office space. These gains come largely at the expense of sending counties. The program thus shifts private activity toward lagging places at a small aggregate output cost while slightly raising private workers' welfare, and a different choice of destinations could achieve similar redistribution at a lower cost. At the county level, vacated offices are small relative to the floor-space stock; quantifying the model at the municipal level will allow a sharper assessment of the office channel.

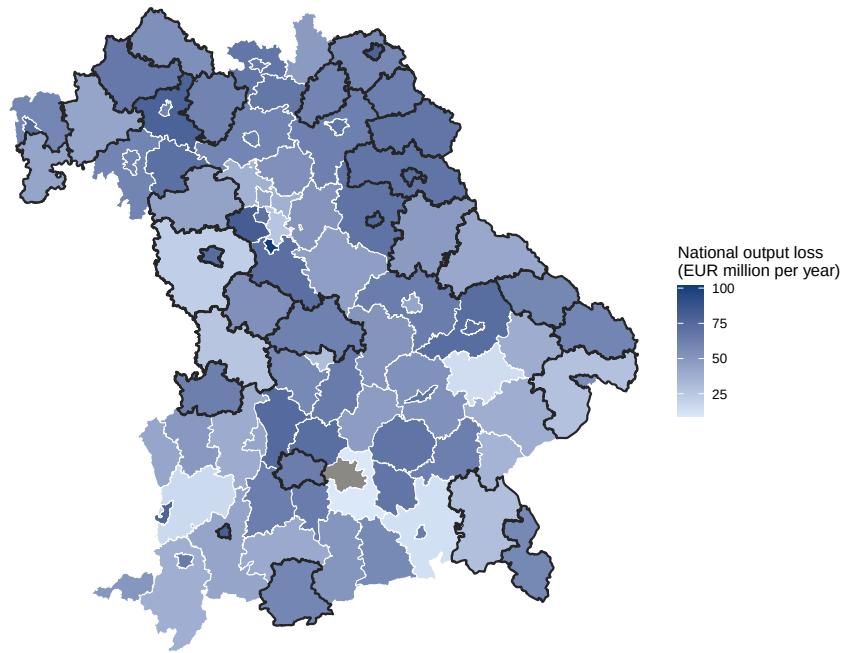


Figure 16. National private output loss from moving 100 public positions from Munich to each county in the policy state, in millions of euros per year. Darker shading indicates larger losses. Thick outlines mark actual net-receiving counties; Munich is shown in gray.

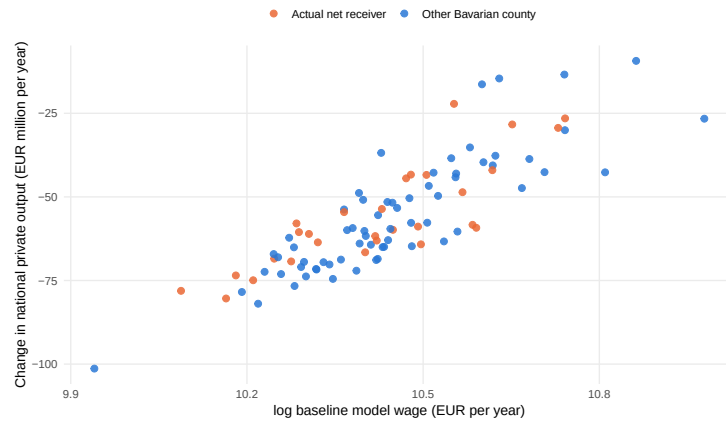


Figure 17. Marginal value of destinations and actual receiving status. Each point is a county in the policy state. The vertical axis shows the change in national private output from moving 100 positions from Munich to that county; the horizontal axis shows its log baseline model wage. Orange points mark actual net-receiving counties.

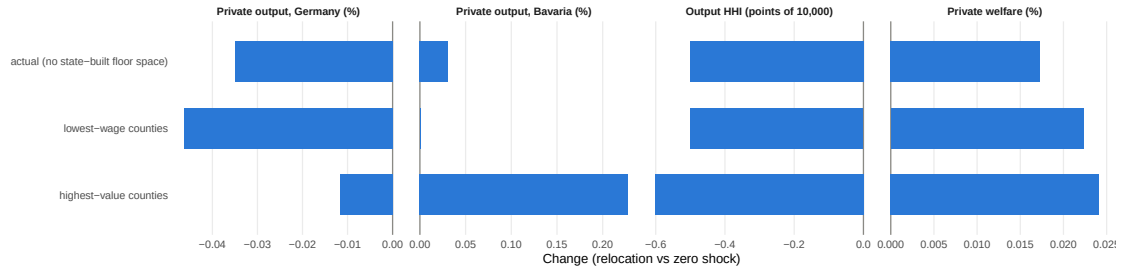


Figure 18. Alternative allocations of the same positions. Changes in private output in Germany and in the policy state, output concentration (HHI points on a scale from 0 to 10,000), and private welfare when the positions moved by 2024 go to the actual destinations, to the 29 lowest-wage counties, or to the 29 counties with the highest marginal value. All three scenarios exclude state-built floor space.

8 Conclusion

This paper studies whether relocating public-sector employment can operate as an effective place-based policy. I examine the *Heimatstrategie*, a policy in Germany that reallocates public-sector jobs and training positions from urban sending locations, especially Munich, toward smaller and economically lagging municipalities within the policy state. The policy's explicit eligibility criteria help construct a comparison group, while its implementation across many municipalities allows me to study effects at both ends of the relocation.

The empirical results show that public employment relocation increases private employment in receiving municipalities. The gains grow over the post-policy period and extend beyond the public jobs that move. Matching municipalities on pre-policy levels and trends and using comparisons outside the policy state, I find positive effects in both the long-difference estimates and the staggered event studies.

Sending municipalities show no offsetting losses in private employment. This finding is difficult to reconcile with a simple local-demand mechanism in which removing public jobs mechanically reduces private activity at the origin. One explanation is that origins can accommodate more private activity as the policy releases scarce urban space; at the county level, however, the quantitative model assigns only a small role to this channel.

The worker evidence shows that jobs move, but people largely do not. Origin-cohort workers employed at destination offices mostly commute rather than relocate, most workers who remain employed elsewhere continue to work in administration, and the destination workforce consists largely of people outside the original cohort. Moving public workplaces is therefore distinct from moving public workers, a distinction that matters for assessing who benefits from the policy.

To quantify the aggregate implications, I build a two-sector quantitative spatial model in which public employment raises nearby private productivity and public offices compete with private uses for floor space. Matching county-level employment responses, I estimate a public-to-private spillover elasticity of 0.36. Through this spillover, the relocations raise private employment and output in receiving counties at the expense of sending counties. Nationally, private output falls slightly, its spatial concentration declines, and

private workers' welfare rises slightly. The choice of destinations matters: sending the same jobs to the counties with the highest marginal value would cost about one third as much national output.

The paper contributes to labor, public, and urban economics by studying public employment relocation as a place-based policy and estimating its effects on both receiving and sending locations. A self-collected relocation dataset and the agency register document the policy, while linked employer–employee records measure local labor-market outcomes and describe worker adjustment. The spatial model connects these local responses to the allocation of private production and to the costs and benefits of redistributing public employment.

Two extensions will sharpen these results. Civil-service employment data will complete the measurement of public employment, and quantifying the model at the municipal level will align its geography with the empirical analysis and allow a closer assessment of the office-space channel.

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Appendix

Guide to the appendices

Topic	Appendix
Program scope, staffing, and costs	Appendix A: Institutional Setting
Coverage, agency records, and worker definitions	Appendix B: Data
Matching, spillover exclusions, cohorts, and timing	Appendix C: Empirical Methodology
Long differences with Bavarian controls	Appendix D
Event studies with Bavarian controls	Appendix E
Model derivations, inversion, and fit	Appendix F: Quantitative Model
Annual long-difference estimates	Appendix G: Year-by-Year Long-Difference Results
Matching ratio, Munich exclusion, and estimators	Appendix H: Additional Robustness Checks

Table 7. Overview of the appendices.

Appendix A: Institutional Setting

Public Employment and Pay

Germany’s June 2024 public-service workforce comprised approximately 527,000 federal employees, 2.67 million state employees, 1.80 million municipal employees, and 381,000 employees of social insurance institutions (Destatis 2025; Destatis 2026). In the World Bank series, the public wage bill remained close to one-sixth of total public expenditure from 2000 to 2019. Its share declined in 2020 as total spending rose during the pandemic (World Bank 2024).

Bavaria’s direct state administration employed 311,580 people in 2014, including 213,090 civil servants and judges. The count includes full-time and part-time staff and excludes municipal employers and legally separate public institutions (Statistical Offices 2014; Bavarian Statistical Office 2014). I use this population to benchmark the policy’s intended size.

Bavarian budget law pairs E9 with A9 and E13 with A13 for the purpose of comparing budgeted posts; this is not a rule assigning employees to grades (Freistaat Bayern 2024, Art. 6a). E9a covers administrative duties accessible without a university degree. In the non-graduate civil-service career, workers normally enter at A6 or A7 and reach A9 through promotion. A9 can also be an entry grade for the degree-based career track. E13 and A13 illustrate master’s-level work rather than all university-qualified employment (TdL 2025; Freistaat Bayern 2026).

A non-graduate who reaches A9 through promotion will already have accumulated service. The first published A9 step therefore does not provide an equal-experience comparison with an E9a entrant. Civil servants and contractual employees also face

Grade (published steps)	Monthly gross base pay (€)	
	First step	Last step
E9a (1, 6)	3,520.10	4,490.04
A9 (2, 10)	3,294.99	4,108.12
E13 (1, 6)	4,629.74	6,580.44
A13 (5, 11)	5,247.58	6,168.25

Table 8. Contractual and civil-service pay schedules, effective February 1, 2025. Amounts are full-time monthly base pay at each grade’s first and last published steps. The endpoints do not represent equal experience across employment statuses. Annual special payments and other compensation are excluded. Source: LfF (2025b) and LfF (2025a).

different pension and insurance arrangements, so gross base pay alone does not determine their relative take-home pay (Freistaat Bayern 1992). A net-pay comparison would require a specified career history, tax situation, and health-insurance arrangement. For the policy, the common feature is that a given grade and experience step carries the same scheduled base pay across Bavarian municipalities.

Program Boundaries and Position Counts

Phase	Administrative designation	Coverage in this paper
I	<i>Regionalisierung von Verwaltung</i>	Included
I	<i>Strukturkonzept – Chancen im ganzen Land</i>	Included
II	<i>Behördenverlagerung Bayern 2030</i>	Included; implementation under way
Other measures	Individual projects outside phases I and II	Not included

Table 9. The two phases of the relocation program. The empirical sample covers projects with an observed opening within the outcome-data window. Source: Bavarian Ministry of Finance (2025) and Kaltenhauser (2023).

Phase	Intended jobs	Paid training positions
I	2,511	730
II	2,685	440
Total	5,196	1,170

Table 10. Intended final staffing across both phases. Source: Bavarian Ministry of Finance (2026, pp. 6, 16). These are the government’s final staffing figures, not current employment or the number of incumbents who move. Training positions include remunerated public-service degree programs.

The intended jobs equal 1.668% of the pre-policy direct state workforce (5,196/311,580). Including paid training positions gives 2.043% (6,366/311,580). Section 2 reports the

jobs-only ratio to keep the two categories distinct.

Geographical Targeting

Weight	Indicator	District reference period
30%	Projected population change	2010–2030
30%	Unemployment rate	2007–2011
10%	Employment density	2007–2011
20%	Disposable household income per resident	2005–2009
10%	Net migration of ages 18–29 per 1,000 in that age group	2006–2010

Table 11. Composition of the structural index. Weights and periods refer to the district index underlying the 2014 expansion of eligibility. The Bavarian mean is 100 and the expanded threshold is 90. Municipal qualification uses a comparable index with municipal proxies, including taxable income. Source: Bavarian Ministry of Finance (2015) and StMWi (2014).

The local reconstruction identifies 829 municipalities through the district index and 392 through the municipal index. Their union contains 885 of Bavaria’s 2,056 municipalities, with overlap counted once. The reconstruction uses an index value of 90 or below; the government describes eligibility as below 90, and published scores are rounded to one decimal place. Service-center designations and ministerial site selection operate in addition to these index-based eligibility criteria. Figure 1 illustrates these geographical selection layers.

Implementation Costs

Reported category	Amount (€)
Budgeted temporary staffing costs, 2016–2022	16,623,849.00
Construction expenditure, four sites	74,923,320.64
Renovation expenditure	3,708,003.37
Standardized moving-cost estimates, 2015–2021	450,713.00
Standardized rental-cost estimates, 2015–2022	18,188,015.00
Sum of reported categories	113,893,901.01

Table 12. Reported cost categories for completed relocation projects. Source: Kaltenhauser (2023, pp. 13–14), response dated May 23, 2023. The sum combines actual construction and renovation spending, budgeted costs of temporary posts, and standardized estimates. Categories cover different periods. It is neither a measure of total cash expenditure nor the full program’s lifetime net cost.

For three tax-office relocations, the same report provides construction spending and the number of jobs established by completion. Höchstädt an der Donau accounts for €9,712,273.52 and 53 jobs, Waldmünchen for €4,358,424.71 and 21 jobs, and Bad Kötzing for €2,691,643.64 and 20 jobs (Kaltenhauser 2023, pp. 8, 13). Summing

spending and jobs across these sites gives €16,762,341.87/94, or €178,323 per public job, rounded to €180,000 in the main text. This construction-only ratio excludes other transition costs, savings at origins, and the value of the resulting assets. It does not measure the fiscal cost per additional private job.

Regional investment subsidies under the German GRW program provide a further benchmark. Using regional variation in subsidy rates, Grunau et al. (2025) estimate a cost of about €25,000 per additional job. Their causal employment denominator differs from the public positions accommodated by the three relocation projects.

For Northvolt's planned battery plant in Heide, the European Commission approved €700 million in direct grants and €202 million in guarantees in January 2024 (European Commission 2024). The government announced 3,000 direct jobs (German Federal Government 2024). Dividing the grants by planned direct employment gives €233,333 per job, rounded to €230,000 in the main text. This calculation excludes guarantees, whose face value is not a grant expenditure, and projected indirect jobs.

For Intel's planned semiconductor plant in Magdeburg, the federal government proposed €9.9 billion in support in June 2023, subject to European Commission approval, alongside about 3,000 direct jobs (German Federal Government 2023). This implies €3.3 million per planned direct job, excluding temporary construction and projected indirect employment. Intel cancelled the project in July 2025 (Intel 2025). This ratio therefore describes the original subsidy offer, not a realized cost per job.

Both firm-specific benchmarks measure announced support per planned direct job, not a causal cost per additional job. All amounts are in nominal euros at their respective source dates. A cost-effectiveness comparison would require net policy costs and additional employment measured over comparable horizons.

Structured News Search

I conducted a structured German-language news search in Nexis Uni using the search string

**Behördenverlagerung OR Behördenumzug OR “Umzug von Behörden”
OR “Umzug vom Amt” OR “Amtsumzug”.**

I restricted the search period to articles published from 1991 onward and limited the sample to news articles, excluding private-company publications, individual webpages, and legal texts.

The query yielded 1,531 results. I identified news reports on 22 federal and 87 state agencies between 2005 and 2032. In total, the reports cover 13,395 relocated jobs. The largest number of relocations occurs in 2019, with 5,843 jobs, followed by 2030, with 2,300 jobs, and 2021, with 1,070 jobs. Table 13 reports the number of relocated jobs and study places by destination state, and Table 14 lists all sending and destination cities.

The search identifies both single-agency relocations and broader relocation programs. The three main programs are the federal program for coal regions, the Bavarian *Heimatstrategie*, and the Saxon *Lausitzprogramm 2038*. I use this search to document the broader institutional context of public employment reallocation in Germany.

State	N of Relocated Jobs	N of Relocated Study Places
Bavaria	9,681	1,730
Brandenburg	510	
Hesse	100	
Saxony	2,214	
Saxony-Anhalt	850	
Thuringia	40	

Table 13. Number of public-sector jobs and study places relocated to each state according to the Nexis Uni news search.

Sending Cities	Destination Cities
Ammersee	Amberg, Ansbach, Augsburg
Berlin, Bonn	Bad Kissingen, Bad Königshofen im Grabfeld,
Bad Kissingen	Bad Kötzing, Bad Neustadt an der Saale, Bad Reichenhall,
	Bad Steben, Bautzen, Berlin, Bischofswerda, Borna
	Cottbus
Dresden	Dessau, Dinkelsbühl, Dresden
Erlangen, Eschborn, Essen	Ebern, Eichstätt
	Freital, Freyung, Fürstenfeldbruck, Fürth
Großenhain	Garmisch-Partenkirchen, Gemünden am Main, Gera,
	Görlitz, Grafenau, Gunzenhausen
Herrsching a. Ammersee	Halle (Saale), Höchstädt a.d.Donau, Hof
	Ingolstadt
	Kaufbeuren, Kemnath, Kitzingen, Kronach, Kulmbach
	Lauterbach, Leipzig, Lohr am Main
München	Marktredwitz, Meißen, Merseburg, Mittenwald
	Mühlendorf a. Inn, Münchberg
Nürnberg	Nabburg, Neustadt an der Aisch, Nördlingen, Nürnberg
	Obernburg am Main, Oberviechtach
	Pegnitz
Regensburg	Regen, Rosenheim
Starnberg	Schwandorf, Schweinfurt
	Tirschenreuth, Traunstein
	Vilshofen an der Donau, Vohenstrauß
Würzburg	Waldkirchen, Waldmünchen, Waldsassen, Weiden in der Oberpfalz
	Weißenburg i. Bay., Weißwasser, Windischeschenbach, Wunsiedel
	Zittau, Zwiesel

Table 14. Sending and destination cities identified in the Nexis Uni news search, listed alphabetically. Some cities, such as Berlin and Dresden, appear both as sending and destination cities.

Appendix B: Data

Relocation records and the agency register

The project records distinguish intended staffing, reported implementation dates, and construction milestones. Source links document the evidence behind each entry. A program entry may represent an entire agency, a branch, or part of an existing organization. Neither an agency's continued presence at its origin nor the absence of a separately named destination in the register establishes that no employment was relocated.

I reconcile the 2014, 2023, and 2026 register editions first by agency identifier. Among unmatched records, I use normalized agency names that occur only once in each edition and then normalized addresses that occur only once in each edition. Normalization removes differences in capitalization, punctuation, and spacing. I review location changes against agency and program sources, distinguishing changes within a municipality from moves between municipalities. Shared addresses and renamed units require particular care. The register records administrative units and locations; it is not an employment census. Its coverage includes state agencies and schools, as well as other institutions with state personnel. These cross-year links remain separate from the establishment identifiers in the IAB data.

IAB descriptive samples and candidate establishments

Table 1 reports the treated municipalities in the one-to-one matched samples used for the municipal analysis. Matching uses pre-treatment levels and trends, with controls outside Bavaria. Treatment dates come from the official year where available and otherwise the recorded completion year. The sample retains *Regionalisierung von Verwaltung* projects with a treatment date by 2024 and excludes projects that are still being planned. The receiving sample contains 34 municipalities and the sending sample six. The table reports municipal means, rather than pooled worker totals.

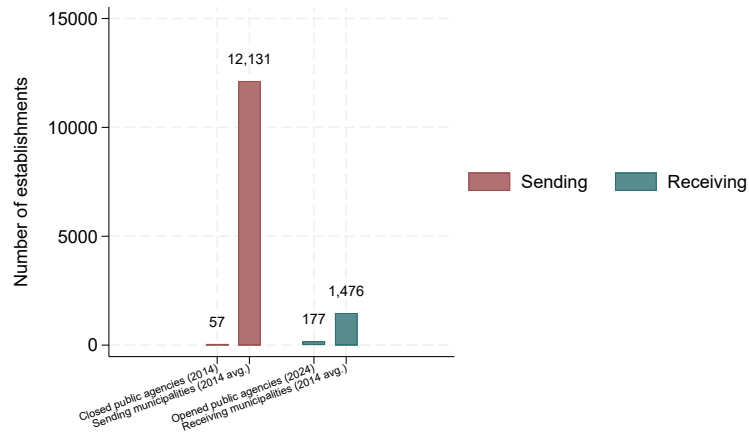
For the worker analysis, workplace postcodes identify public-administration establishments near program origins and destinations. An establishment is classified by its identifier and postcode. A postcode can appear among both origins and destinations and is retained in both sets. Establishments observed in both 2010 and 2013 provide the baseline distribution of employment changes over that interval. I calculate its standard deviation separately for the origin and destination groups. Among continuing establishments, the change from 2014 to 2024 must be at least twice the destination-group standard deviation to qualify as a candidate expansion, or at most minus the origin-group standard deviation to qualify as a candidate contraction. Establishments observed only in 2024 qualify as candidate openings, and those observed only in 2014 as candidate closures. These rules identify potential program-related establishments; they do not pair a particular origin with a particular destination.

The source cohort consists of workers employed at candidate sending establishments in 2014. Residential movers and commuters are distinguished among those employed at candidate receiving establishments in 2024 using changes in recorded residence. Employees at candidate receiving establishments who were not in the source cohort are additional

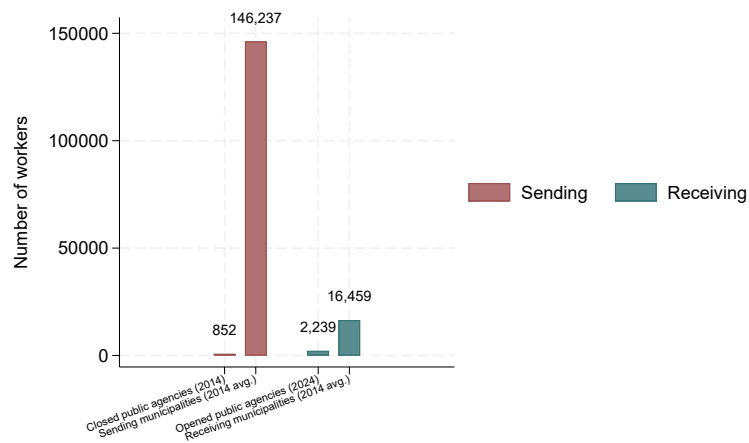
employees relative to that cohort, not necessarily new labor-market entrants. A worker absent from the employed sample in 2024 may have entered unemployment, retirement, civil-service employment, self-employment, or another unobserved state. This absence alone cannot distinguish these transitions.

Comparing closing and opening establishments

Figures 19–24 compare establishments classified as closing or opening with their sending and receiving municipal benchmarks. The closing group is measured in 2014 and the opening group in 2024; municipal benchmarks refer to 2014. The comparisons describe establishment size, workforce composition, and wages. Figures 25–27 additionally include downsizing establishments among the sending candidates. These comparisons characterize the candidate groups; they do not establish direct matches between agency-register records and IAB establishment identifiers.

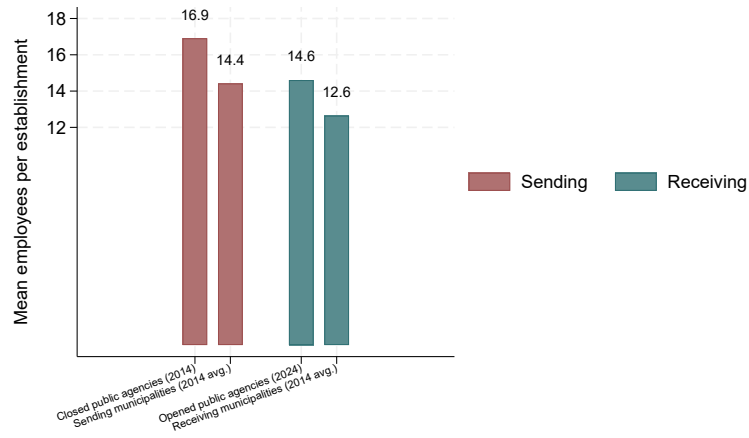


(a) Establishments

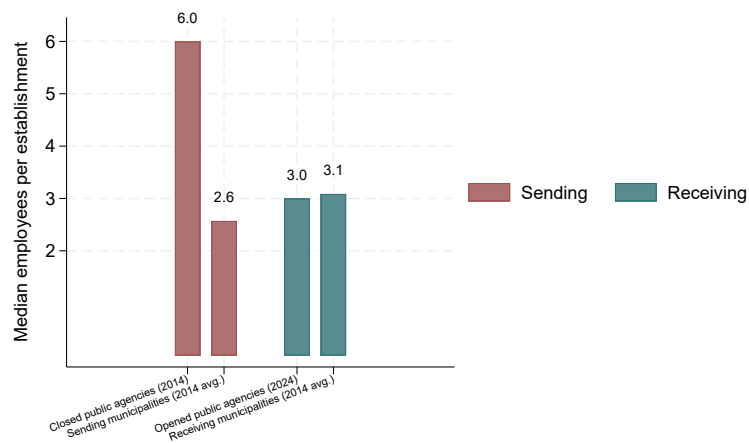


(b) Workers

Figure 19. Numbers of establishments and workers in the closing and opening groups and their municipal benchmarks. Closing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.

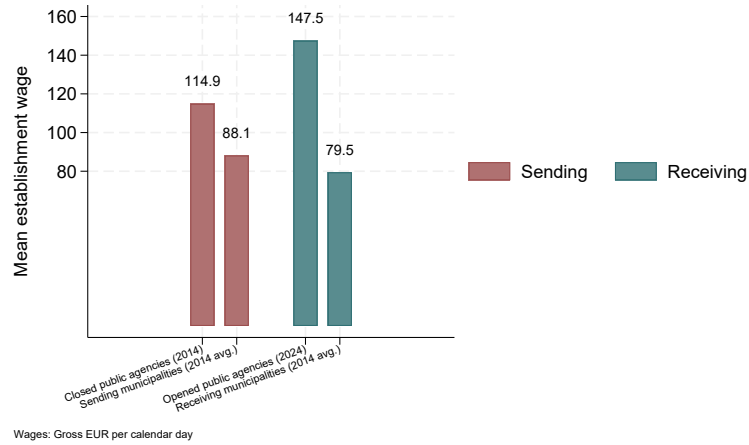


(a) Mean employment

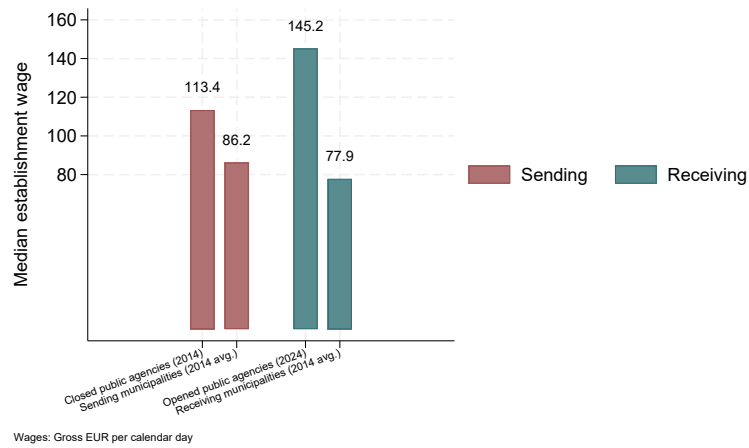


(b) Median employment

Figure 20. Mean and median employment per establishment in the closing and opening groups and their municipal benchmarks. Closing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.

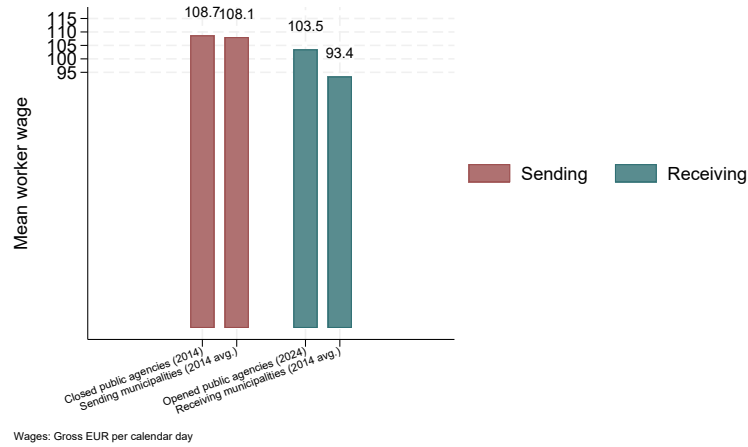


(a) Mean establishment wage

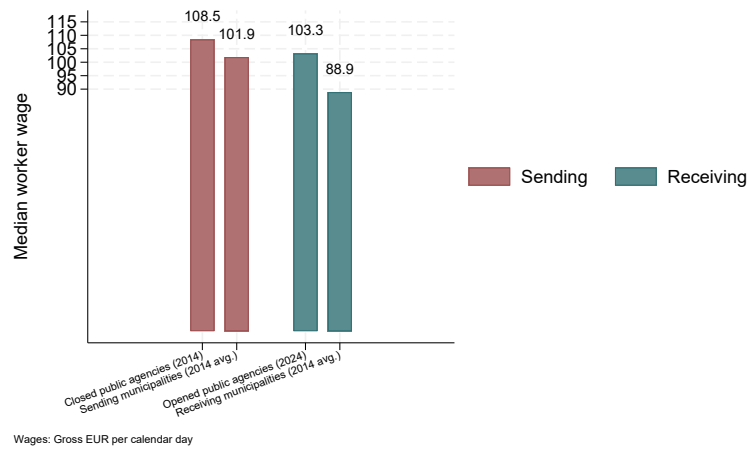


(b) Median establishment wage

Figure 21. Establishment wage comparisons for the closing and opening groups and their municipal benchmarks. Closing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.

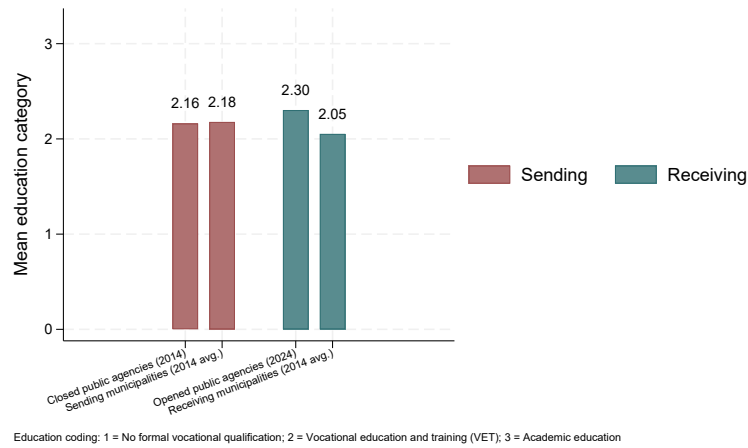


(a) Mean worker wage

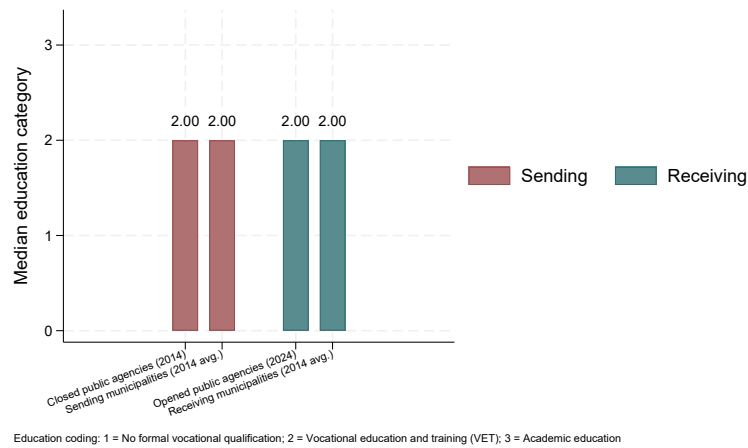


(b) Median worker wage

Figure 22. Worker wage comparisons for the closing and opening groups and their municipal benchmarks. Closing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.



(a) Mean education category



(b) Median education category

Figure 23. Education in the closing and opening groups and their municipal benchmarks. Categories are 1 for no formal vocational qualification, 2 for vocational training, and 3 for academic education; category means are descriptive indices. Closing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.

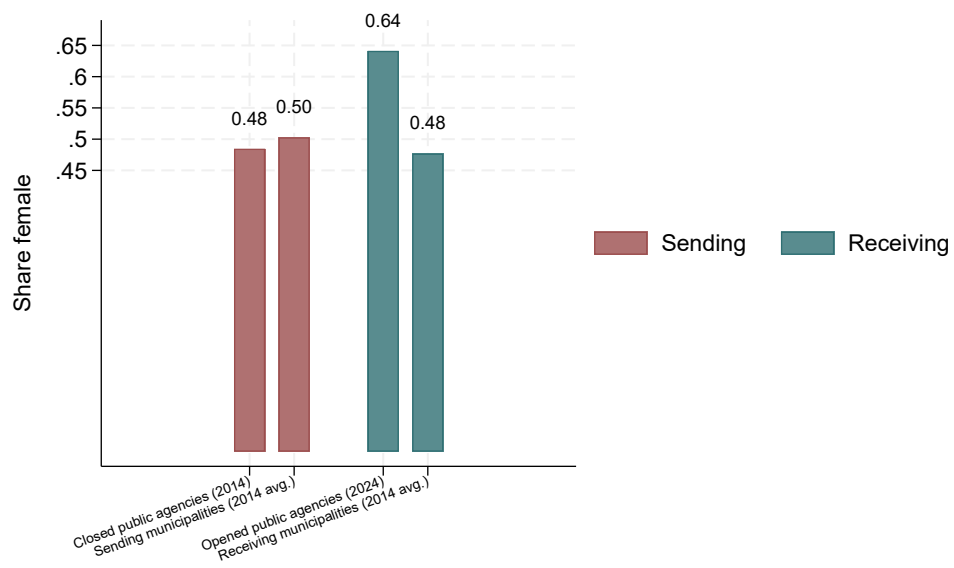
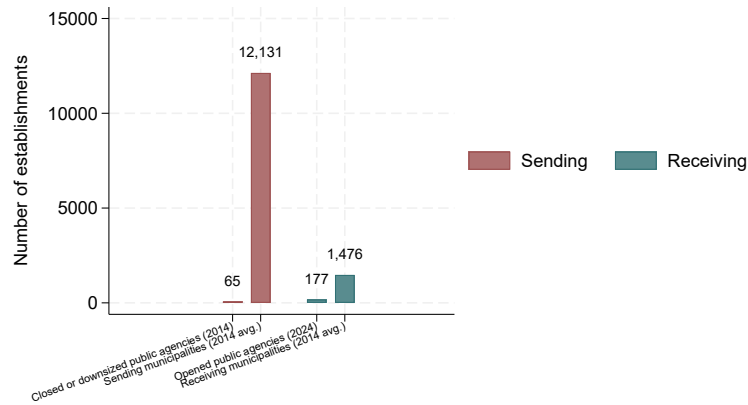
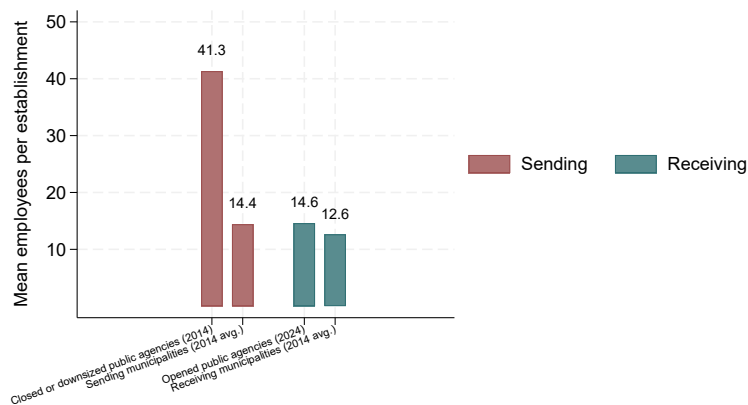


Figure 24. Share of women in the closing and opening groups and their municipal benchmarks. Closing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.



Robustness: downsized agencies remain open in 2024 and have a 2014-2024 employment change ≤ -1 pre-policy SD.
Pre-policy SD = SD of 2010-2013 employment changes among public agencies in sending postcodes.

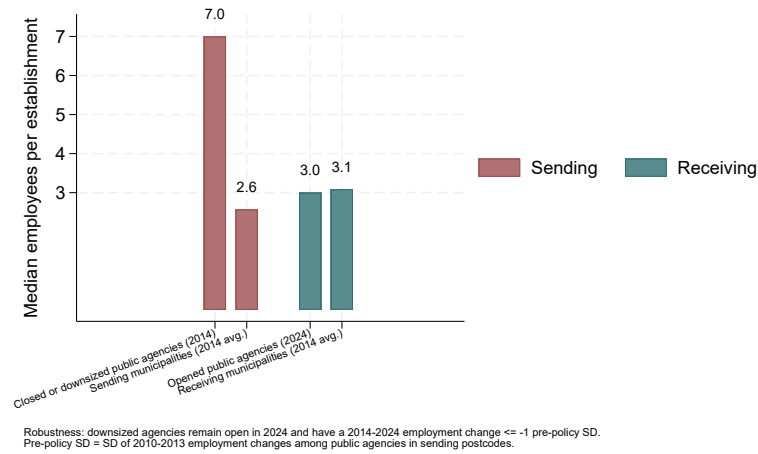
(a) Establishments



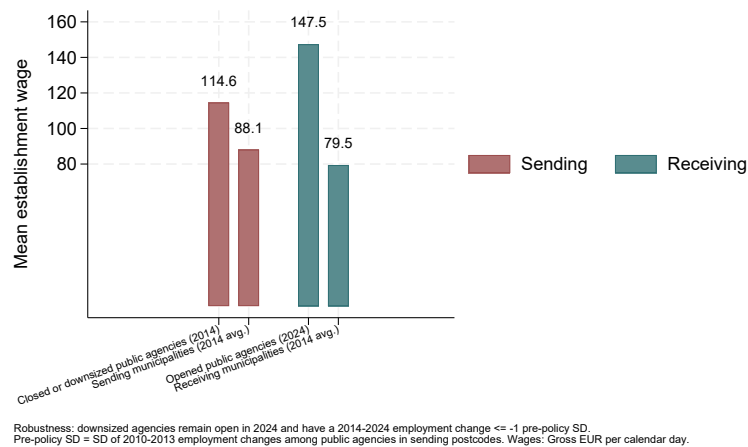
Robustness: downsized agencies remain open in 2024 and have a 2014-2024 employment change ≤ -1 pre-policy SD.
Pre-policy SD = SD of 2010-2013 employment changes among public agencies in sending postcodes.

(b) Mean employment

Figure 25. Sensitivity to including downsizing establishments among the sending candidates. The receiving comparison remains the opening group. Closing or downsizing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.



(a) Median employment



(b) Mean establishment wage

Figure 26. Employment size and establishment wages when the sending group includes both closures and downsizing establishments. Closing or downsizing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.

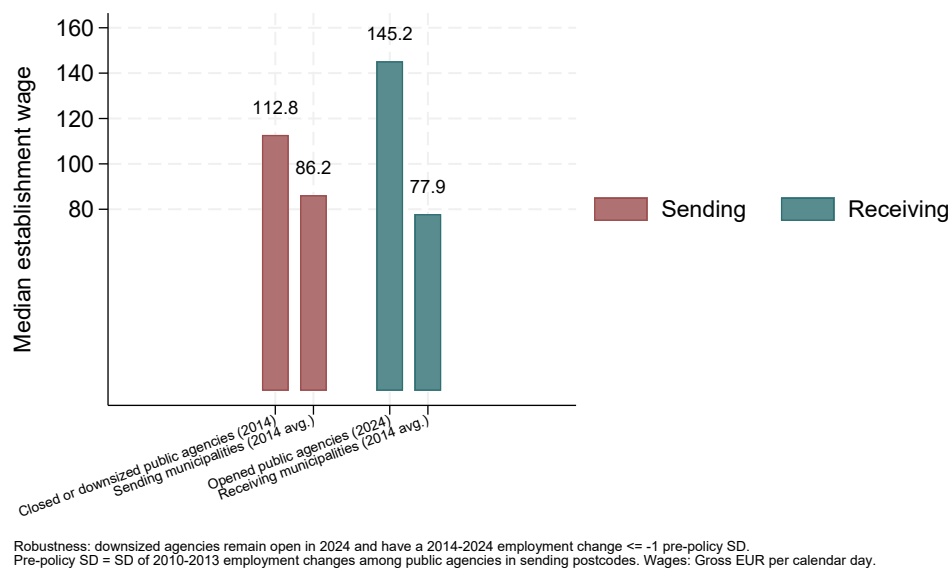


Figure 27. Median establishment wages when the sending group includes both closures and downsizing establishments. Closing or downsizing establishments are measured in 2014, opening establishments in 2024, and the municipal benchmarks in 2014.

Published Census counts

Table 2 uses the published county cells from tables 2000X-2041 and 2000X-2045. For each table, I select the total commuting category and either the total industry category or WZ 2008 section O. Each selection yields one observation per county. Workplace totals include both in-commuters and people working in their municipality of residence; residence totals include both out-commuters and residents working locally. I use the reported total cells directly, rather than adding overlapping industry aggregates or repeated commuting categories. Means and standard deviations describe the 412 counties in the source files and the 96 with Bavarian county codes. The Census's broader employment coverage and reference date distinguish these counts from the IAB panel.

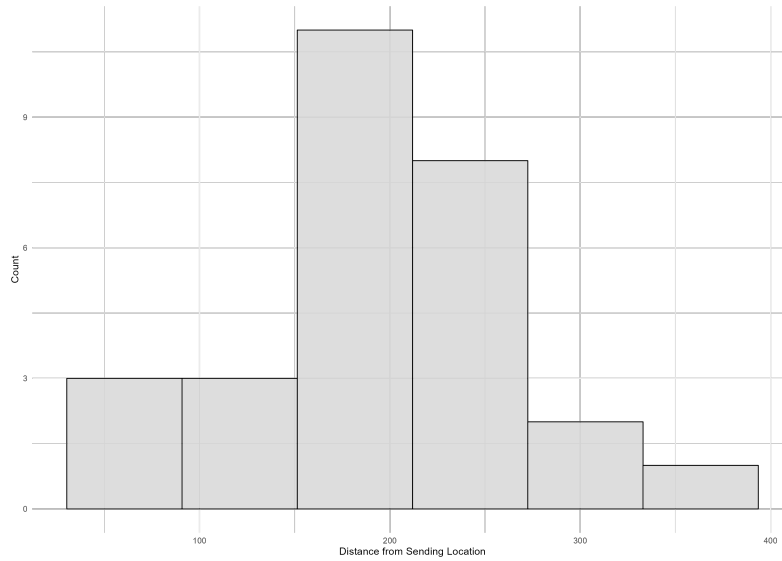


Figure 28. Distance between sending and receiving municipalities.

Appendix C: Empirical Methodology

Matching and comparison groups

I construct the receiving and sending matches separately, using the same matched pairs across the IAB and other municipal analyses. Let X_i contain the five 2014 levels and five pre-policy log changes described in Section 4. The squared Mahalanobis distance between a treated municipality and a potential control is

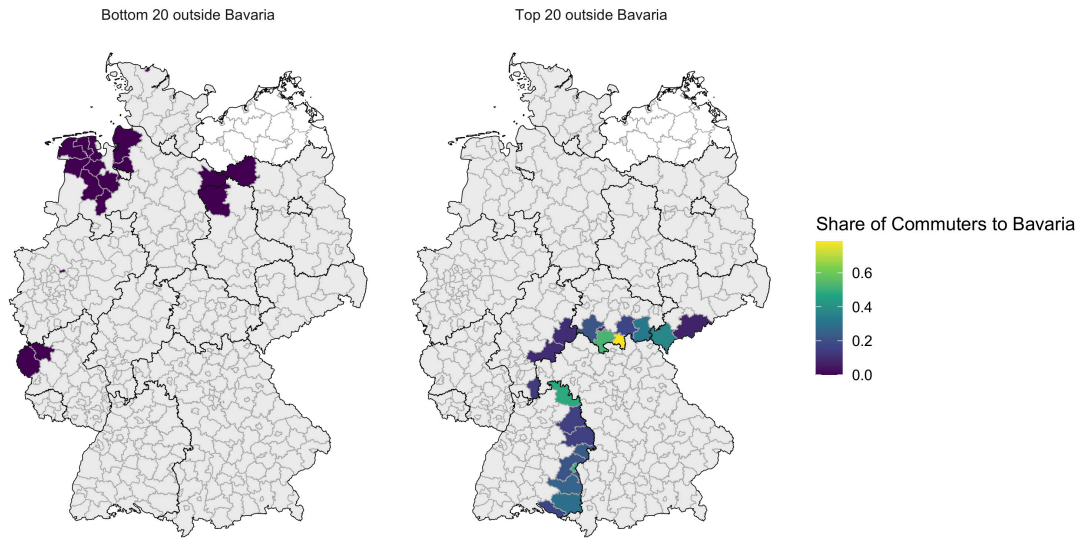
$$d_{ij}^2 = (X_i - X_j)'S^{-1}(X_i - X_j), \quad (22)$$

where S is the covariance matrix of the matching variables. Nearest-neighbor matching without replacement selects one control for each treated municipality. The covariance adjustment accounts for the different scales and correlations of the matching variables.

The main comparison pool excludes Bavarian municipalities to limit exposure to policy spillovers within the state. It also excludes municipalities in 20 non-Bavarian counties selected for their commuting links to Bavaria. Figure 29 shows the top and bottom 20 non-Bavarian counties by their share of commuters to Bavaria in 2013. The top-20 group defines the additional exclusion; the bottom-20 group is shown for comparison and does not define the donor pool. For the within-Bavaria robustness check, I exclude other municipalities named as origins or destinations in the relocation records from the control pool.

Selected counties (outside Bavaria): commuting to Bavaria (2013)

Top 20 and bottom 20 non-Bavarian origin counties by share of commuters to Bavaria

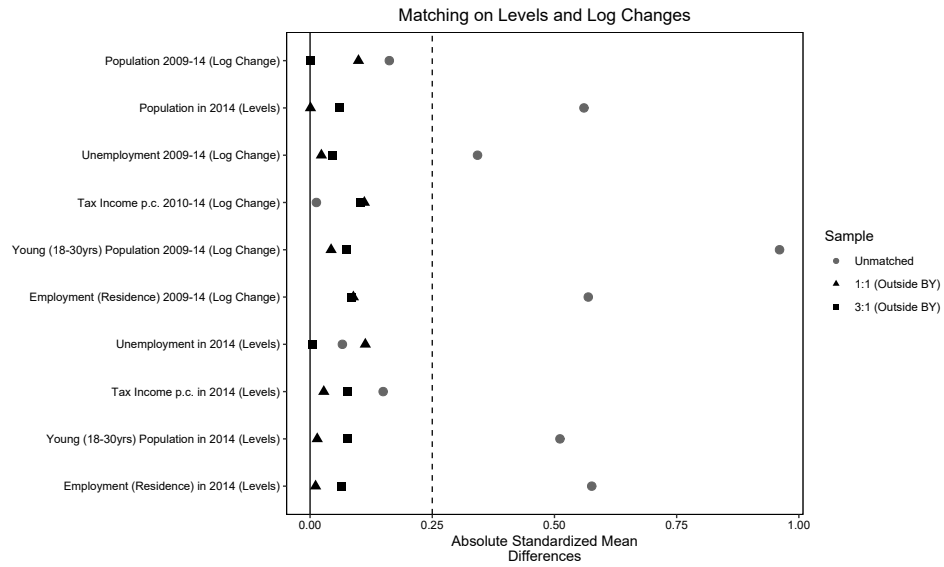


Source: Federal Employment Agency; County-Level Commuting Flows

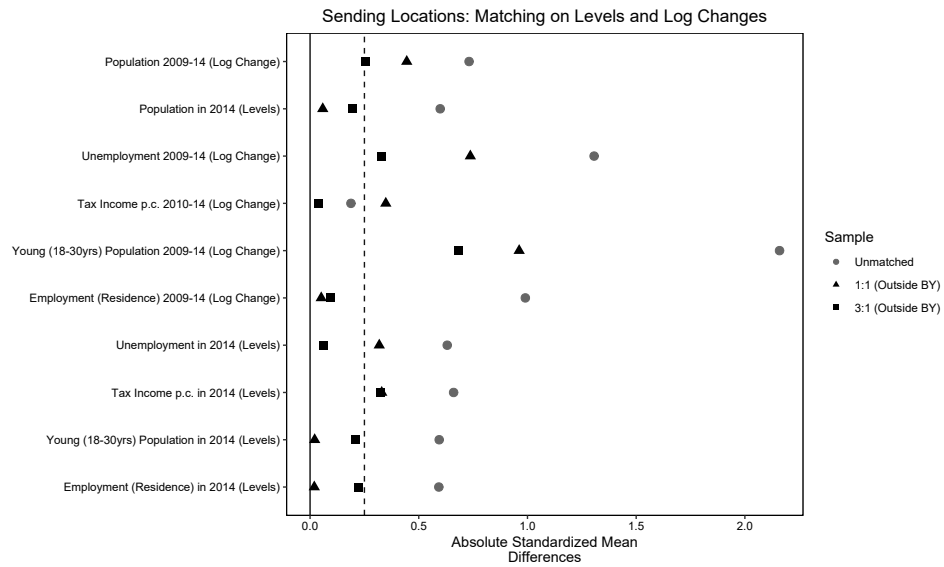
Figure 29. Commuting links used to restrict the comparison pool. Counties outside Bavaria with the lowest and highest shares of commuters to Bavaria in 2013. Municipalities in the top-20 counties are excluded from the main control pool, in addition to municipalities within Bavaria. Source: Federal Employment Agency county commuting flows.

Table 15 reports absolute standardized mean differences after matching with one and three outside-Bavaria controls. The receiving and sending columns use their respective treated-group standard deviations. Figure 30 also shows the unmatched comparison. These statistics describe balance on the matching variables; the event-study leads provide separate evidence on pre-treatment outcome dynamics.

I examine alternative comparison groups within Bavaria and matching ratios of three controls per treated municipality. These checks vary the available comparisons while retaining the distinction between receiving and sending locations. The within-state comparison is informative about Bavaria-specific shocks, but it may include municipalities indirectly affected by the program. I also examine sending-location estimates without Munich, whose scale differs sharply from that of the other origins.



(a) Receiving municipalities



(b) Sending municipalities

Figure 30. Covariate balance for Mahalanobis matching on 2014 levels and pre-policy log changes. The figures compare the unmatched pool with one-to-one and three-to-one outside-Bavaria matches. Symbols distinguish the three comparisons. The horizontal axis reports absolute standardized mean differences.

Covariate	Receiving		Sending	
	1:1	3:1	1:1	3:1
Population, 2014	0.001	0.060	0.058	0.197
Residence employment, 2014	0.011	0.064	0.019	0.221
Unemployment, 2014	0.113	0.005	0.318	0.060
Income per taxpayer, 2014	0.028	0.076	0.330	0.322
Population aged 18–30, 2014	0.015	0.077	0.021	0.210
Population change	0.099	0.001	0.445	0.256
Residence-employment change	0.089	0.085	0.051	0.096
Unemployment change	0.023	0.047	0.737	0.330
Income change	0.111	0.102	0.348	0.040
Population aged 18–30 change	0.043	0.074	0.962	0.681

Table 15. Balance after Mahalanobis matching with outside-Bavaria controls. Entries are absolute standardized mean differences for one and three controls per treated municipality, corresponding to Figure 30. Levels refer to 2014; changes are log differences from 2009 to 2014, except income per taxpayer, which uses 2010–2014.

Treatment cohorts and repeated relocations

The baseline treatment year is the first recorded relocation for a municipality among the projects included in the analysis. The last-treatment year is the latest such date. Both use the official treatment year when available and otherwise the recorded completion year. Repeated dates at a location therefore describe further exposure to the policy, rather than additional independent treated municipalities. Table 16 gives the first- and last-treatment distributions underlying Figure 2.

Year	Receiving		Sending	
	First	Last	First	Last
2015	3	1	2	1
2016	15	9	2	2
2017	7	8	1	1
2018	3	5	1	1
2019	0	1	0	0
2020	3	4	0	0
2021	2	2	2	2
2022	1	3	0	0
2023	1	2	0	1
2024	0	0	0	0
Municipalities	35	35	8	8

Table 16. First and last treatment years in the common matching roster. Each municipality appears once in each timing distribution. First and last refer to the earliest and latest recorded treatment among included projects, using the official year when available and the completion year otherwise.

For the last-treatment robustness specification, let L_i denote the final treatment year. I retain treated observations before G_i and from L_i onward, omitting the transition years

between them. All available control observations remain in the panel. The specification is

$$Y_{it} = \alpha_i + \delta_{p(i),t} + \tau^L D_i \mathbf{1}\{t \geq L_i\} + v_{it}. \quad (23)$$

It includes municipality and matched-pair-by-year fixed effects, with standard errors clustered by municipality. This is a difference-in-differences timing check alongside the baseline Sun–Abraham event study. The last recorded relocation does not necessarily mark the date at which an agency reaches its intended final staffing.

Table 17 compares the two timing definitions for private employment measured at workers’ residences. Both columns within each group use the same fixed effects and clustering. Point estimates are similar across the two definitions; estimates for sending municipalities remain imprecise.

Treatment date	Receiving		Sending	
	First	Last	First	Last
Private residence employment	223.4 (94.6)	249.1 (95.2)	3,786.2 (2,977.0)	3,525.0 (3,300.7)
Municipality–year observations	1,260	1,194	288	272

Table 17. Sensitivity to first and last treatment dates. Difference-in-differences estimates for private residence employment, measured in employees. All four specifications include municipality and matched-pair-by-year fixed effects; municipality-clustered standard errors appear in parentheses. The first-treatment specification uses observations before and after the first relocation. The last-treatment specification removes treated observations from the first treatment year through the year before the last, retaining observations before first treatment and from last treatment onward.

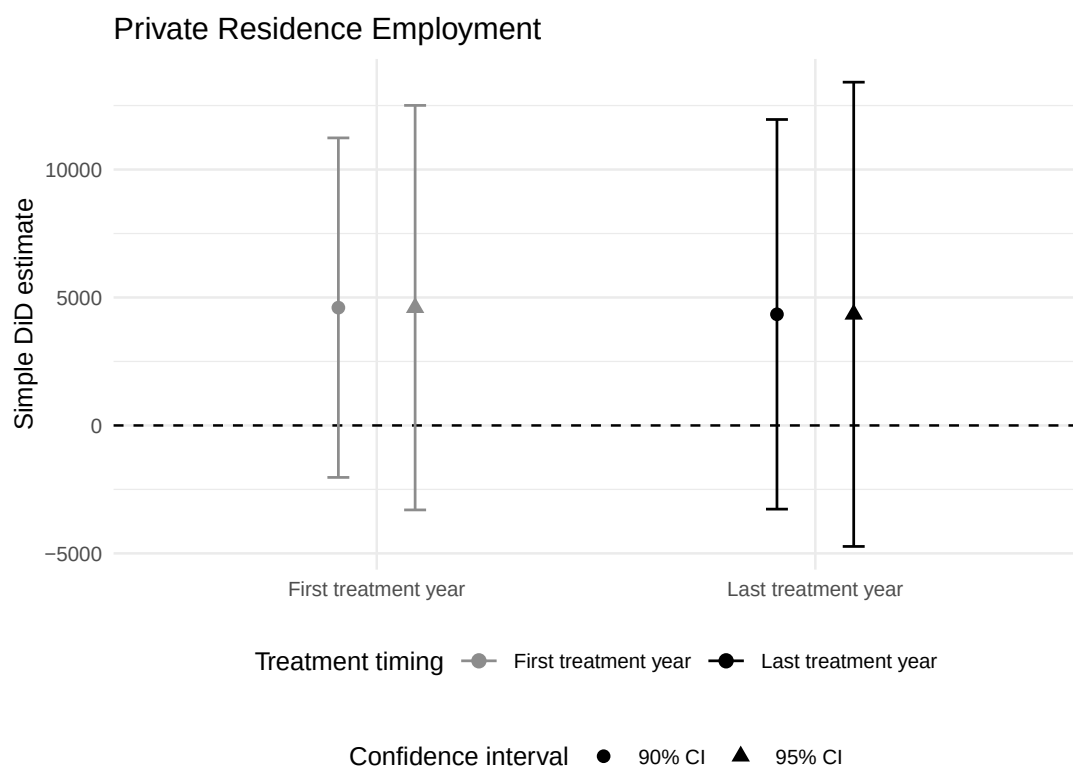
Event-time aggregation and inference

In the Sun–Abraham specification, each cohort’s event-time effects are measured relative to the year before first treatment. I aggregate cohort-specific coefficients at common event times using the estimator’s cohort weights. Cohorts contribute only where their outcomes are observed, so the composition of the aggregate effect can vary across event times. The municipal panel is allowed to be unbalanced. Municipality-clustered standard errors allow serial correlation within municipalities; the small number of sending locations remains relevant when assessing precision.

The Callaway–Sant’Anna specification uses never-treated municipalities as controls. For cohort g , the group-time effect compares its change from $g - 1$ to t with the corresponding change in the never-treated group:

$$ATT(g, t) = E[Y_{it} - Y_{i,g-1} \mid G_i = g] - E[Y_{it} - Y_{i,g-1} \mid G_i = \infty]. \quad (24)$$

The implementation uses a universal pre-treatment base period, permits an unbalanced panel, and aggregates group-time effects into dynamic effects. Bootstrap inference provides simultaneous confidence bands. This specification is estimated within the



(a) Sending: private residence employment

Figure 31. Sending-location sensitivity to first and last treatment dates. Difference-in-differences estimates for private residence employment, with 90% and 95% confidence intervals.

common matching design; it does not impose the matched-pair-by-year fixed effects of the Sun–Abraham regression.

Synthetic difference-in-differences for sending municipalities

Because only five municipalities are net senders, I complement the matched design with synthetic difference-in-differences (Arkhangelsky et al. 2021). I estimate one model per sending municipality. Donors are never-treated municipalities outside Bavaria with at least 10,000 inhabitants in 2014. As in the matched design, I exclude districts with strong commuting ties to Bavaria, every municipality named in the relocation records, and the municipalities of the other relocation programs listed in Appendix A. The pre-treatment window is 2008–2014 for all sending municipalities; years between 2015 and a later first relocation are not used. Unit weights reproduce each sending municipality’s pre-treatment outcome path up to a constant. Time weights select the pre-treatment years that best predict the donors’ post-treatment outcomes; in practice they fall almost entirely on 2014, so effects are measured relative to the last year before the program. Standard errors come from placebo replications. In each replication, every sending municipality’s timing is assigned to one of the 100 donors closest to it in 2014 population, and the models are re-estimated without the drawn donors. Placebo units of similar size matter because small municipalities have much noisier outcomes than large cities.

The donor weights match pre-treatment outcomes but not the characteristics used in the matching. To condition on these characteristics, I follow the two-step procedure of Kranz (2021). The covariates Z_i are the matching variables: the logs of 2014 population, unemployment, residence employment, population aged 18–30 and income per taxpayer; the 2009–2014 log changes in population, unemployment, residence employment and population aged 18–30; and the 2010–2014 log change in income per taxpayer. Because Z_i is fixed over time, it can only affect outcomes through year-specific coefficients. In the first step, I estimate

$$Y_{it} = \alpha_i + \delta_t + \sum_{s \neq 2014} \mathbb{1}[t = s] Z_i' \gamma_s + \varepsilon_{it} \quad (25)$$

by OLS on all untreated municipality-years: donor municipalities in every year and sending municipalities before their first relocation. Using only untreated observations keeps the relocations themselves out of the estimated covariate effects. In the second step, I remove the fitted covariate component,

$$\tilde{Y}_{it} = Y_{it} - \sum_{s \neq 2014} \mathbb{1}[t = s] Z_i' \hat{\gamma}_s, \quad (26)$$

and apply synthetic difference-in-differences to $\tilde{Y}_{it}$ with the same donors, time windows and placebo procedure. The adjustment allows, for example, larger municipalities to follow different trends than smaller ones. The placebo standard errors take $\hat{\gamma}_s$ as given and do not reflect estimation error from the first step (ibid.).

As a falsification test, I backdate treatment to 2012 for every sending municipality and estimate the same models with 2008–2011 as pre-treatment years and 2012–2014

as placebo post-treatment years. Because the analyzed relocations start in 2015, these estimates should be close to zero. The dynamic version reports the treated-minus-synthetic gap in each year relative to the time-weighted pre-treatment baseline, with pointwise placebo bands.

Appendix D: Long-Difference Results with Bavarian Control Group

Comparisons within Bavaria help assess whether the estimated responses reflect changes common to the policy state. I apply the matching approach described in Appendix C separately to receiving and sending municipalities, using untreated Bavarian municipalities as controls. These comparisons share state-level conditions with treated places, but may also share spillovers from the relocation program. They therefore complement the outside-Bavaria comparison rather than eliminate the identifying assumptions in Section 4.

Table 18 shows that the choice of comparison pool matters for private employment. With one Bavarian control per receiving municipality, the 2014–2024 private-workplace estimate is about -2.1% relative to 2014 employment, compared with 2.3% using outside-Bavaria controls; both estimates are imprecise. The corresponding residence-employment estimate falls from about 3.5% to 0.6% . The receiving-location estimates are therefore sensitive to the choice of control pool.

Municipalities	Control pool	Workplace	Residence
Receiving	Outside Bavaria, 1:1	0.0231 (0.0271)	0.0354 (0.0100)
Receiving	Bavaria, 1:1	-0.0213 (0.0281)	0.0056 (0.0110)
Receiving	Bavaria, 3:1	-0.0395 (0.0206)	-0.0025 (0.0092)
Sending	Outside Bavaria, 1:1	0.0204 (0.0574)	0.0604 (0.0155)
Sending	Bavaria, 1:1	-0.0372 (0.0529)	-0.0110 (0.0189)
Sending	Bavaria, 3:1	-0.0264 (0.0446)	-0.0026 (0.0143)

Table 18. Private-employment long differences, 2014–2024. Entries are treatment coefficients with HC1 standard errors in parentheses. Outcomes are log changes in IAB-covered employment outside public administration, measured at workplaces or residences. All specifications use pre-policy levels-and-trends matching and matched-set fixed effects. The control ratio is controls per treated municipality. The outside-Bavaria rows provide the reference comparison.

Additional municipal outcomes

Table 19 reports employment among people who both live and work locally in the supplementary sample of 34 receiving municipalities and 34 Bavarian controls. The outcome is expressed as a share of working-age population. These long differences use matched-pair fixed effects and pair-clustered standard errors; their sample and inference differ from the IAB specifications above.

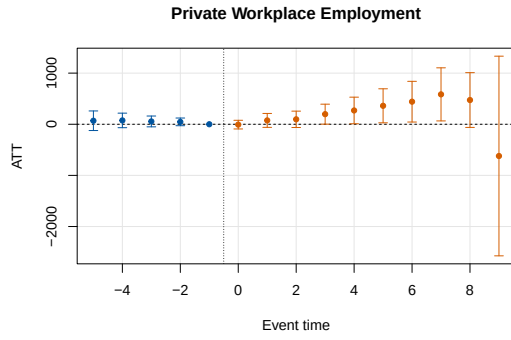
End year	Local-employment share
2015	-0.026 (–)
2016	-0.402 (0.232)
2017	-0.130 (0.216)
2018	0.222 (0.274)
2019	0.499 (0.350)
2020	0.400 (0.430)
2021	0.223 (0.428)
2022	0.238 (0.464)
2023	–

Table 19. Local-employment long differences with Bavarian controls. Employment comprises employees living and working in the same municipality, divided by the population aged 15–64. Entries are percentage-point changes since 2014, with pair-clustered standard errors in parentheses. Matching uses levels and pre-policy trends. The sample has 34 treated municipalities and 34 controls before endpoint restrictions. A dash indicates unavailable information.

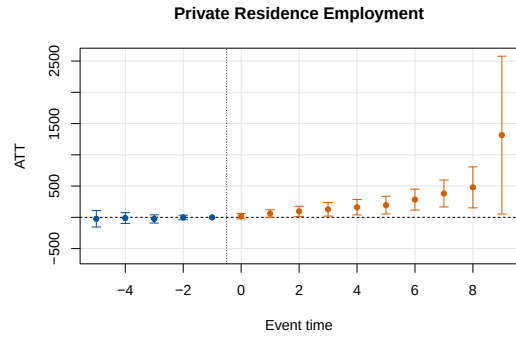
Appendix E: Event-Study Results with Bavarian Control Group

The event studies compare outcome paths around the first recorded relocation. Figure 32 holds the Sun–Abraham estimator and employment definition fixed while varying the control pool for receiving municipalities. The IAB specifications use municipality and matched-set-by-year fixed effects, never-treated controls, and municipality-clustered standard errors. Appendix C explains treatment cohorts, the year-before-treatment reference, and changing cohort support.

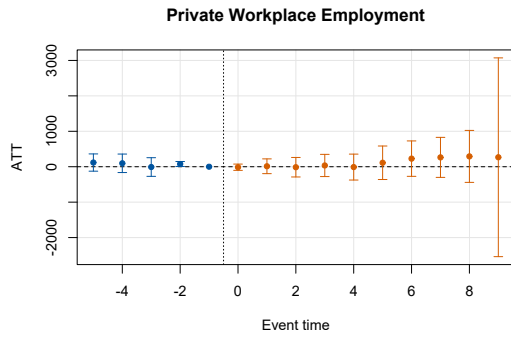
For receiving municipalities, Bavarian controls produce smaller and less precise post-treatment private-employment responses. At event year five, the workplace estimate is 1.3% of 2014 private workplace employment (standard error 2.8 percentage points), compared with 4.3% (2.0 percentage points) using outside-Bavaria controls; residence employment is 0.1% of its 2014 level (1.7 percentage points), compared with 3.3% (1.2 percentage points). The Bavarian-control workplace coefficient two years before treatment is 0.9% of its 2014 level (0.4 percentage points), so the leads should not be described as uniformly flat. The figure therefore reports receiving-location comparisons only; it does not establish how sending results respond to Bavarian controls.



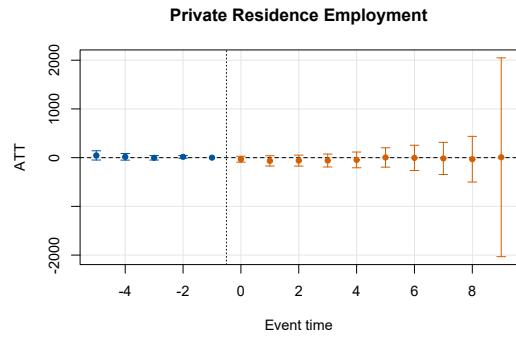
(a) Outside Bavaria: workplace



(b) Outside Bavaria: residence



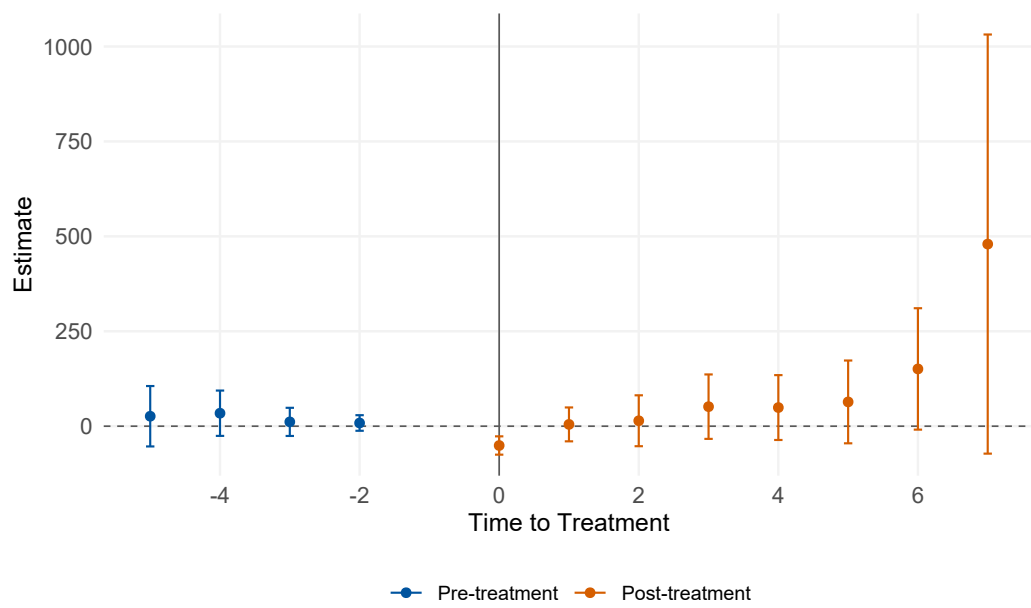
(c) Bavaria: workplace



(d) Bavaria: residence

Figure 32. Receiving private-employment dynamics under alternative control pools. Sun–Abraham estimates with one-to-one levels-and-trends matching. The upper row uses controls outside Bavaria and the lower row uses Bavarian controls. Employment is measured in employee counts, with year -1 as the reference.

Figure 33 reports the local-employment response in the supplementary municipal sample. It measures employees who live and work in the same municipality. These Sun–Abraham regressions use municipality and calendar-year fixed effects.



Bars are estimate +/- 1.96 standard errors.

Figure 33. Local employment in receiving municipalities with Bavarian controls. Sun-Abraham estimates in employee counts for one-to-one levels-and-trends matches. Intervals are estimate ± 1.96 standard errors.

The comparison-pool exercises show that the estimates are sensitive to using controls within the policy state. Appendix H examines the matching ratio, the estimator, and the exclusion of Munich; Appendix C reports the treatment-timing checks.

Appendix F: Quantitative Model

This appendix documents the data inputs, the derivation of the choice probabilities, the inversion of the baseline, and the construction of the counterfactual for the county model in Section 7. The implementation adapts the toolkit of Ahlfeldt, Redding, et al. (2015) and extends it to two worker types, public office space, and a government budget.

Data preparation. The model is quantified for the 412 German counties in their 2011 boundaries. The Census baseline requires total employment by workplace and residence, not only workers crossing a municipal boundary. The published 2011 files 2000X-2041 and 2000X-2045 contain separate commuting breakdowns and overlapping industry aggregates. I select the total-commuting cell and either the all-industry total or industry O, without summing totals with their components. Industry O covers public administration, defense, and compulsory social security and defines public employment; its complement defines private employment. Neither category is an employer-ownership classification.

The reduced-form targets come from the same matched-pair design as the municipal analysis, applied to counties. Relocated positions count jobs and paid training positions. A county that both sends and receives positions is classified by its net position. Private workplace employment nets out the relocated jobs held by employees subject to social security, and standard errors are clustered by matched pair. The model shock uses the same treated counties and positions as the reduced form.

Travel times. Road travel times τ_{ij} enter commuting as $d_{ij} = \exp(\kappa\tau_{ij})$, measured in minutes. Residence is the first index and workplace the second in the choice equations, while the productivity and amenity kernels index the recipient location first and the source of density second. Because the model covers all of Germany, travel links across the boundary of the policy state are retained.

Commuting parameters. Commuting gravity identifies the composite decay $\nu = \epsilon\kappa$. I estimate it from 2013 county-to-county commuting flows of the Federal Employment Agency by Poisson pseudo-maximum likelihood with origin and destination fixed effects and a control function for travel time. The estimate is $\hat{\nu} = 0.0429$ per minute (standard error 0.0038; 37,624 county pairs). Private conditional workplace choice is

$$\pi_{j|i}^P = \frac{D_{ij}E_j^P w_j^\epsilon}{\sum_s D_{is}E_s^P w_s^\epsilon}, \quad H_{Mj}^P = \sum_i H_{Ri}^P \pi_{j|i}^P. \quad (27)$$

Given observed private workplace and resident employment, this system identifies the transformed wages $\omega_j^P = E_j^P w_j^\epsilon$ up to scale. Following Ahlfeldt, Redding, et al. (ibid.), I set $E_j^P = 1$, so that $w_j = (\omega_j^P)^{1/\epsilon}$, and choose ϵ so that the variance of log model

wages across counties equals the variance of log observed gross wages in 2011. Because each county is its own wage group, this condition has the closed-form solution $\epsilon^2 = \text{Var}_j(\log \omega_j^P) / \text{Var}_j(\log w_j^{\text{data}})$, which gives $\hat{\epsilon} = 2.486$ and $\hat{\kappa} = \hat{\nu} / \hat{\epsilon} = 0.0172$.

Choice probabilities. Choice probabilities follow from the distribution of the best alternative. For private workers, independence gives $\Pr(\max_{i,j} u_{ijo}^P \leq u) = \exp(-\Phi^P u^{-\epsilon})$, which yields equations (4) and (6). For a public worker assigned to j , the analogous distribution is $\exp(-[(1-t)w_j]^\epsilon \Psi_j^G u^{-\epsilon})$, which yields equations (5) and (6). The expectations exist for $\epsilon > 1$. The flows satisfy the adding-up conditions

$$\sum_{i,j} N_{ij}^P = \bar{H}^P, \quad \sum_i N_{ij}^G = \bar{H}_{Mj}^G, \quad \sum_i H_{Ri}^m = \sum_j H_{Mj}^m = \bar{H}^m. \quad (28)$$

Private probabilities sum to one over residence–workplace pairs, whereas public residence probabilities sum to one separately within each assigned workplace. This keeps public workplaces exogenous.

Recovery of fundamentals. Residential inversion identifies effective amenities, which combine physical amenities and residential tastes. Define $\mathcal{W}_i^P = \sum_j D_{ij} E_j^P w_j^\epsilon$. The private residence margin gives

$$Z_i^{P,0} = \frac{H_{Ri}^{P,0} \Phi^{P,0}}{\bar{H}^P (1-t^0)^\epsilon \mathcal{W}_i^{P,0}}, \quad B_i^0 (T_i^P)^{1/\epsilon} = (Q_i^{R,0})^{1-\beta} (Z_i^{P,0})^{1/\epsilon}. \quad (29)$$

Public residences follow from equation (5) given the assigned workplaces. The model-implied public residences enter income, amenity density, and housing demand, so the baseline remains internally consistent.

Given wages and rents, zero profits recover private productivity, and factor payments recover private floor space:

$$A_j^{P,0} = \left(\frac{w_j^0}{\alpha} \right)^\alpha \left(\frac{q_j^0}{1-\alpha} \right)^{1-\alpha}, \quad L_j^{P,0} = \frac{1-\alpha}{\alpha} \frac{w_j^0 H_{Mj}^{P,0}}{q_j^0}. \quad (30)$$

The office rule in equation (13) then gives $\bar{L}^{G,0}$. Baseline payroll and office rents determine t^0 , baseline flows allocate gross earnings to residences, and disposable income determines $L_i^{R,0} = (1-\beta)v_i^0/Q_i^{R,0}$. The floor-space stock is $L_i = L_i^{P,0} + \bar{L}_i^{G,0} + L_i^{R,0}$, measured in effective model units.

For each candidate public spillover elasticity λ , the fundamentals remove the baseline density components:

$$a_j(\lambda) = \frac{A_j^{P,0}}{(\Upsilon_j^{P,0})^{\lambda_{PP}} (\Upsilon_j^{G,0})^\lambda}, \quad b_i = \frac{B_i^0}{(\Omega_i^0)^\eta}. \quad (31)$$

Baseline productivity is therefore the same for every candidate elasticity, and the fundamentals are held fixed within each comparison of the baseline with the policy equilibrium.

Land use. Because both rents are observed, the land-use wedge is the observed rent ratio, $\hat{\xi}_i = q_i^{\text{data}} / Q_i^{R, \text{data}}$, with a median of 1.145 and a range of 0.76 to 2.05. I treat every county as mixed use, so the no-arbitrage condition $q_i = \xi_i Q_i^R$ holds with equality and the corner cases of Ahlfeldt, Redding, et al. (2015) do not arise. The wedge applies to the floor space that remains after public offices and is held fixed in the counterfactuals. As a check, I compare $\hat{\xi}_i$ with the wedge implied by model wage bills, resident income, and the commercial land share in CORINE land-cover data; the two are positively correlated.

Baseline fiscal identity. Summing income across residences gives $\sum_i I_i = W^P + W^G$, and the public budget gives $\sum_i v_i = W^P - C^G$. At baseline, the office rule implies

$$C^{G,0} = \gamma_G \frac{1-\alpha}{\alpha} W^{G,0}, \quad t^0 = \left(1 + \gamma_G \frac{1-\alpha}{\alpha}\right) \frac{W^{G,0}}{W^{P,0} + W^{G,0}}. \quad (32)$$

After a relocation, office requirements keep their origin values while wages and rents change, so the counterfactual tax follows from equation (15) evaluated at counterfactual prices and occupancy.

Counterfactual shock. Gross departures cannot exceed baseline public employment, $\sum_r n_{rs} \leq \bar{H}_{Ms}^{G,0}$. Positions without an identifiable origin county are drawn from the national public workforce in proportion to baseline public employment, $o_j = n^{\text{pool}} \bar{H}_{Mj}^{G,0} / \bar{H}^G$, and carry the national average office intensity $\bar{\ell}^G = \sum_j \bar{H}_{Mj}^{G,0} \ell_j^{G,0} / \bar{H}^G$. Moves within a county cancel at the county level. For relocations into state-built offices completed by 2024, the new floor space equals the office space these positions bring,

$$B_r = \sum_s n_{rs}^b \ell_s^{G,0} + n_r^{b, \text{pool}} \bar{\ell}^G, \quad L_r^{\text{cf}} = L_r + B_r, \quad (33)$$

where n_{rs}^b and $n_r^{b, \text{pool}}$ are the corresponding positions. State-built offices raise $\bar{L}_r^G$ and L_r by the same amount, so only rented offices take space from private firms and housing.

Public payroll changes because transferred jobs move across wages and because all local wages adjust. The payroll change decomposes exactly into

$$W^{G, \text{cf}} - W^{G,0} = \sum_{r,s} n_{rs} (w_r^0 - w_s^0) + \sum_j (w_j^{\text{cf}} - w_j^0) \bar{H}_{Mj}^{G, \text{cf}}. \quad (34)$$

The first term is the €65 million reduction reported in Section 7.4, and the second term is the €14 million reduction from wage adjustment.

Welfare measures. For private workers, the change in expected utility is $\Delta \log \bar{U}^P = \epsilon^{-1} \log(\Phi^{P, \text{cf}} / \Phi^{P,0})$, which includes the tax factors inside Φ^P . For public jobs transferred from s to r , the assignment-conditional comparison is

$$R_{rs}^G = \frac{\bar{U}^{G|r, \text{cf}}}{\bar{U}^{G|s,0}} = \frac{1 - t^{\text{cf}}}{1 - t^0} \frac{w_r^{\text{cf}}}{w_s^0} \left(\frac{\Psi_r^{G, \text{cf}}}{\Psi_s^{G,0}} \right)^{1/\epsilon}. \quad (35)$$

This ratio compares a destination after the move with the origin before it. For the relocated cohorts, it is dominated by the baseline wage and access gap between Munich and the destinations, which is almost unchanged when all policy channels are switched off. I therefore do not interpret it as a welfare effect of the policy and do not pool it with private expected utility.

Solution and numerical checks. For each candidate elasticity, I invert the baseline, solve the model with a zero shock, solve the policy counterfactual, and run the auxiliary regressions. The zero-shock solution reproduces the baseline, with a maximum absolute log deviation below 10^{-10} . At the solution, I check the national masses of both worker types, public assignments, factor payments, housing expenditure, the floor-space stock, resident income, and the public budget.

Additional results. Figure 34 compares untargeted baseline outcomes with the data. Figure 35 reports the holdout test, Figure 36 the model-implied effects on treated and control counties, and Figure 37 the counterfactual with each channel switched off.

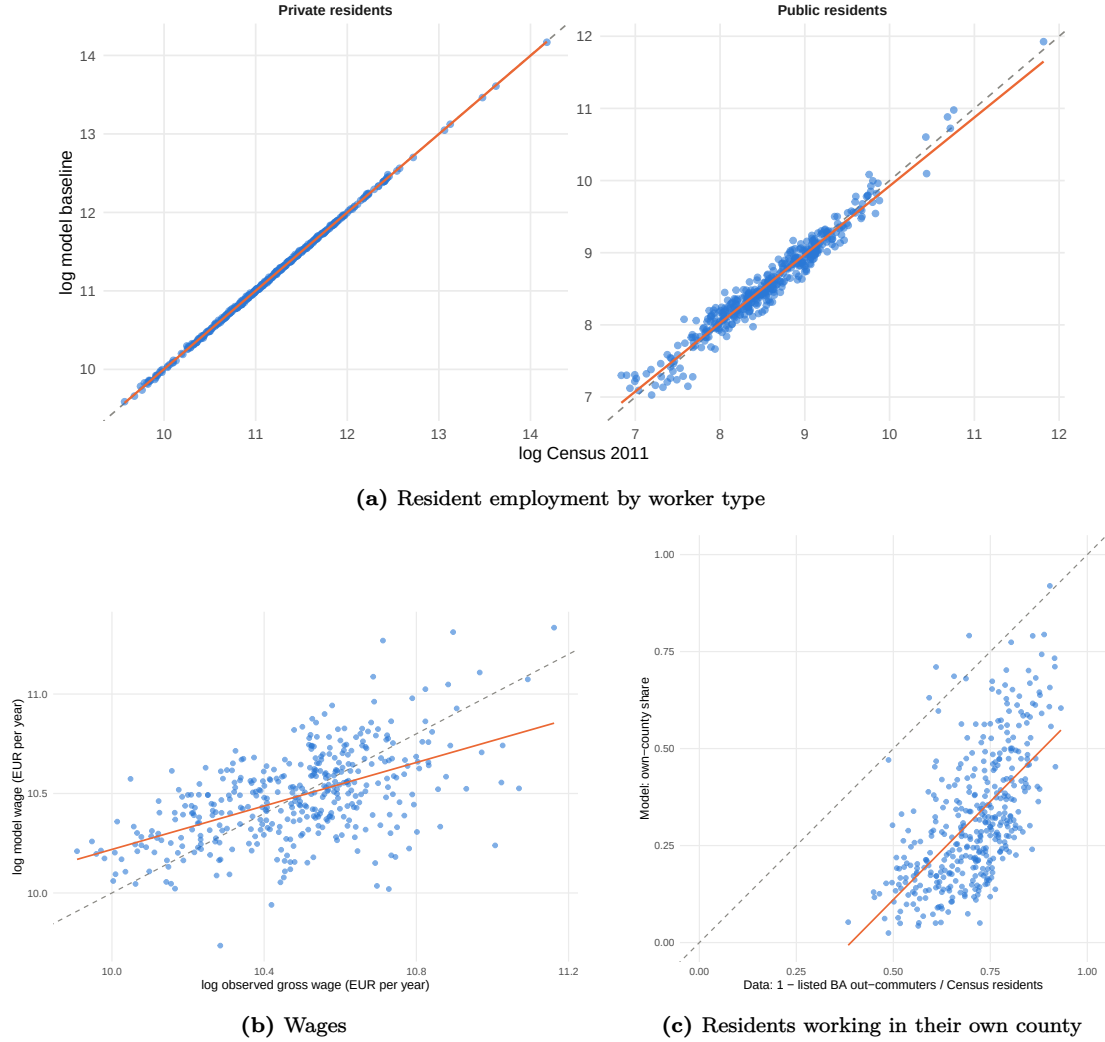


Figure 34. Baseline fit of the county model. Each point is a county. Panel (a) compares model and Census 2011 resident employment for private and public workers; private residents are targeted. Panel (b) compares model wages with observed gross wages in 2011. Panel (c) compares the model share of residents who work in their own county with the share implied by Federal Employment Agency out-commuting data and Census residents. Dashed lines mark the 45-degree line; solid lines are linear fits.

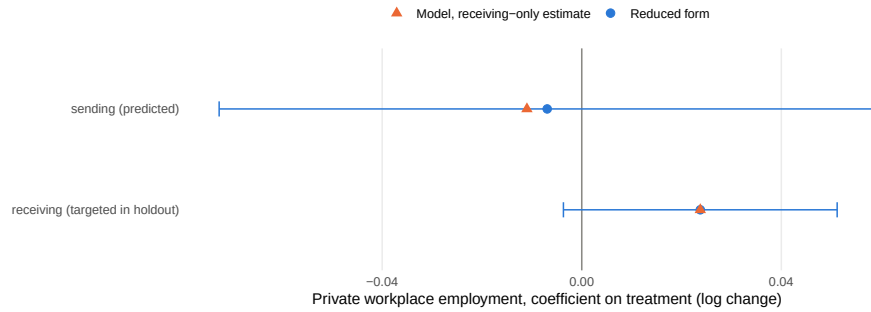


Figure 35. Holdout test. Estimating λ_{PG} from the receiving moment alone gives 0.369, compared with 0.364 when both moments are used. Triangles show model coefficients at this value and circles reduced-form coefficients with 95% confidence intervals. The sending coefficient is a prediction.

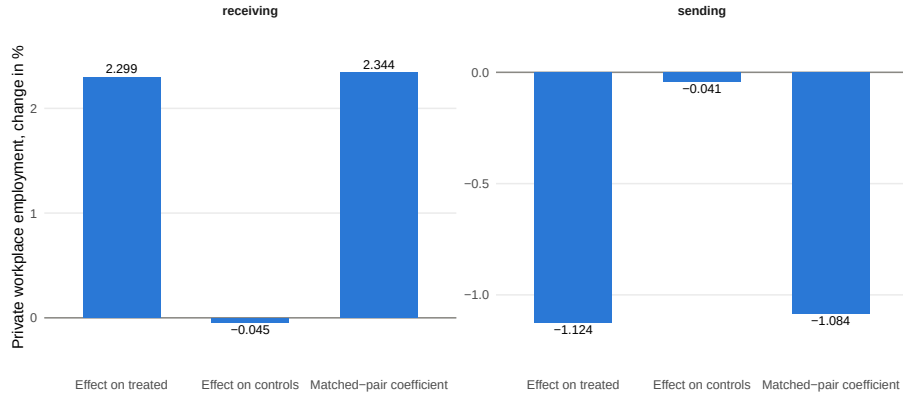


Figure 36. Model-implied effects on treated and matched control counties. Percentage changes in private workplace employment in treated counties, in their matched controls outside the policy state, and the resulting matched-pair coefficient. Effects on the controls shift the coefficient by 2.0% (receiving) and -3.6% (sending) of the effect on the treated counties.

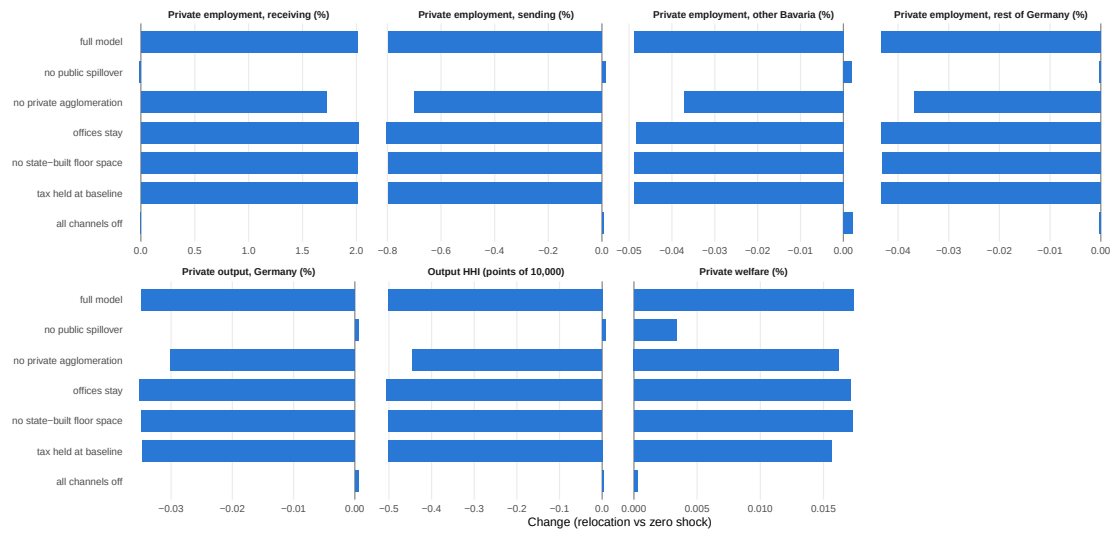


Figure 37. Counterfactual outcomes with one channel switched off at a time. No public spillover sets $\lambda_{PG} = 0$; no private agglomeration sets $\lambda_{PP} = 0$; offices stay keeps public office space at its origin; no state-built floor space takes all arriving offices from the existing stock; tax held at baseline fixes the tax rate. Output concentration is measured in HHI points on a scale from 0 to 10,000.

Appendix G: Year-by-Year Long-Difference Results

These tables report annual long-difference estimates. All changes are measured from 2014, and employment changes are log differences.

Private Employment

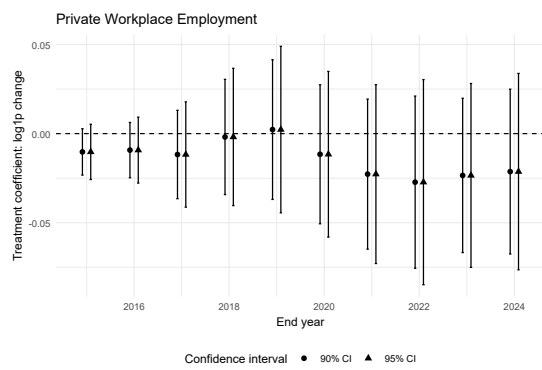
End year	Workplace			Residence		
	Estimate	SE	<i>p</i>	Estimate	SE	<i>p</i>
2015	-0.0139	0.0062	0.155	0.0082	0.0065	0.335
2016	0.0056	0.0115	0.632	0.0097	0.0060	0.123
2017	-0.0028	0.0191	0.886	0.0134	0.0059	0.032
2018	0.0113	0.0198	0.573	0.0155	0.0063	0.020
2019	0.0287	0.0188	0.137	0.0165	0.0084	0.061
2020	0.0197	0.0198	0.327	0.0190	0.0077	0.019
2021	0.0171	0.0224	0.452	0.0167	0.0087	0.063
2022	0.0137	0.0238	0.569	0.0275	0.0093	0.006
2023	0.0168	0.0249	0.504	0.0300	0.0097	0.004
2024	0.0231	0.0271	0.401	0.0354	0.0100	0.001

Table 20. Private employment in receiving municipalities, by endpoint year. Each coefficient compares the change since 2014 with the matched comparison. Outcomes are log changes. Matched-pair fixed effects and HC1 standard errors.

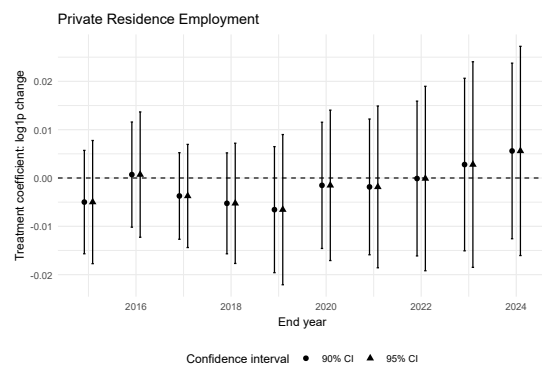
End year	Workplace			Residence		
	Estimate	SE	<i>p</i>	Estimate	SE	<i>p</i>
2018	0.0109	0.0199	0.607	0.0379	0.0124	0.028
2019	0.0242	0.0234	0.348	0.0450	0.0141	0.024
2020	0.0112	0.0213	0.622	0.0379	0.0123	0.027
2021	0.0329	0.0525	0.550	0.0335	0.0143	0.052
2022	0.0155	0.0551	0.787	0.0487	0.0151	0.015
2023	0.0248	0.0612	0.698	0.0580	0.0157	0.008
2024	0.0204	0.0574	0.733	0.0604	0.0155	0.006

Table 21. Private employment in sending municipalities, by endpoint year. Each coefficient compares the change since 2014 with the matched comparison. Outcomes are log changes. Matched-pair fixed effects and HC1 standard errors.

Comparison-Group Sensitivity



(a) Receiving: workplace



(b) Receiving: residence

Figure 38. Receiving private-employment long differences with Bavarian controls, across all available endpoints. One-to-one levels-and-trends matching. The outside-Bavaria comparison appears in Figure 5. The figures show 90% and 95% confidence intervals.

End year	Workplace			Residence		
	Estimate	SE	<i>p</i>	Estimate	SE	<i>p</i>
2015	-0.0103	0.0079	0.324	-0.0050	0.0065	0.523
2016	-0.0093	0.0094	0.340	0.0007	0.0066	0.917
2017	-0.0117	0.0151	0.444	-0.0037	0.0054	0.501
2018	-0.0019	0.0196	0.924	-0.0052	0.0063	0.416
2019	0.0023	0.0238	0.925	-0.0066	0.0079	0.416
2020	-0.0116	0.0237	0.628	-0.0015	0.0079	0.849
2021	-0.0227	0.0256	0.381	-0.0018	0.0085	0.830
2022	-0.0272	0.0294	0.360	-0.0001	0.0097	0.991
2023	-0.0235	0.0263	0.379	0.0028	0.0109	0.799
2024	-0.0213	0.0281	0.453	0.0056	0.0110	0.615

Table 22. Private employment with Bavarian controls in receiving municipalities, by endpoint year. Each coefficient compares the change since 2014 with the matched comparison. Outcomes are log changes. Matched-pair fixed effects and HC1 standard errors.

End year	Workplace			Residence		
	Estimate	SE	<i>p</i>	Estimate	SE	<i>p</i>
2018	-0.0049	0.0218	0.834	0.0391	0.0158	0.069
2019	0.0057	0.0278	0.848	0.0457	0.0182	0.066
2020	-0.0099	0.0256	0.718	0.0386	0.0167	0.082
2021	0.0221	0.0615	0.732	0.0416	0.0127	0.017
2022	0.0004	0.0636	0.996	0.0555	0.0152	0.011
2023	0.0111	0.0712	0.881	0.0635	0.0170	0.010
2024	0.0070	0.0669	0.920	0.0657	0.0173	0.009

Table 23. Private employment with Munich excluded before matching in sending municipalities, by endpoint year. Each coefficient compares the change since 2014 with the matched comparison. Outcomes are log changes. Matched-pair fixed effects and HC1 standard errors.

Appendix H: Additional Robustness Checks

These checks change one comparison at a time and retain the distinction between employment at workplaces and at residences. They supplement the Bavarian-control exercises in Appendices D and E. Table 17 in Appendix C reports the comparison of first and last treatment dates.

Matching ratio and sending-location leverage

Specification	Workplace	Residence
Receiving: 1:1	0.0231 (0.0271)	0.0354 (0.0100)
Receiving: 3:1	0.0069 (0.0294)	0.0350 (0.0083)
Sending: 1:1	0.0204 (0.0574)	0.0604 (0.0155)
Sending: 3:1	-0.0099 (0.0333)	0.0362 (0.0155)
Sending: without Munich	0.0070 (0.0669)	0.0657 (0.0173)

Table 24. Outside-Bavaria matching-ratio and Munich-exclusion comparisons. Log changes from 2014 to 2024 for private employment. Matching uses pre-policy levels and trends; matched-set fixed effects and HC1 standard errors apply. Entries are coefficients (standard errors). The without-Munich specification retains the one-to-one outside-Bavaria comparison for the remaining origins.

The receiving residence-employment long difference is similar with one and three outside-Bavaria controls, whereas workplace estimates remain imprecise. Excluding Munich, the sending residence estimate is 0.066, slightly above the full-sample estimate, and the workplace estimate remains imprecise. Figure 39 reports event studies for sending locations without Munich using total employment from municipal statistics, which includes public and private employees.

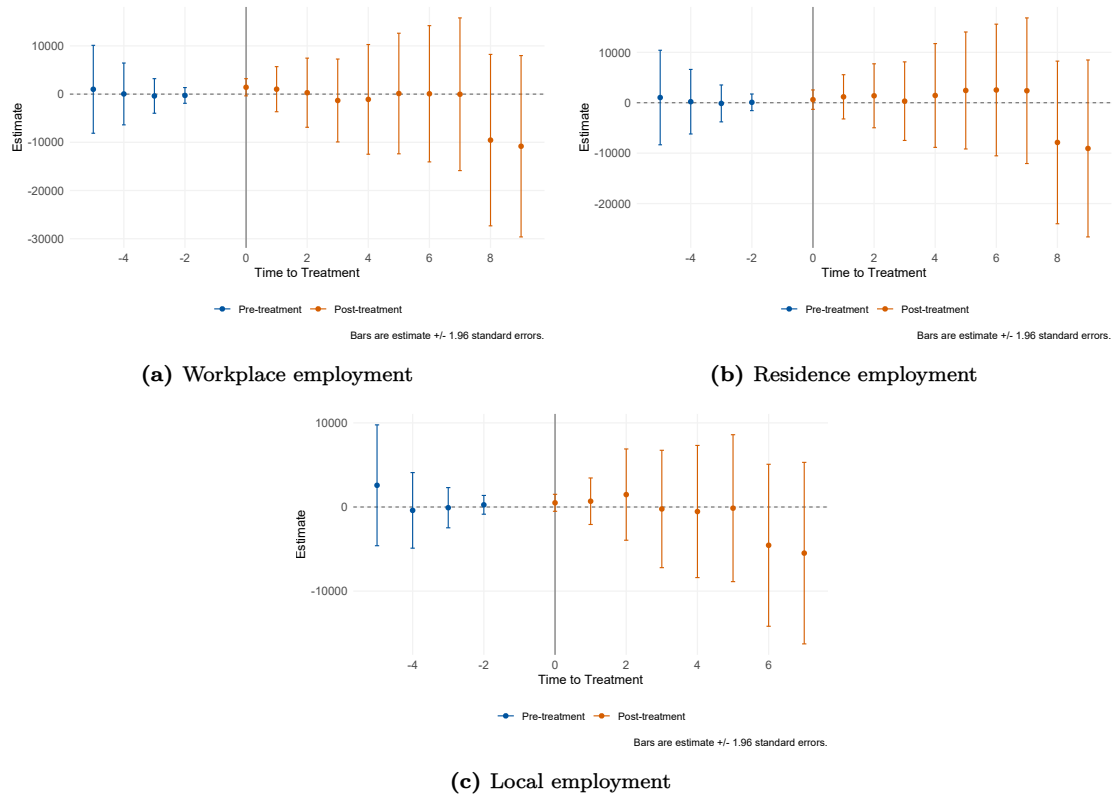
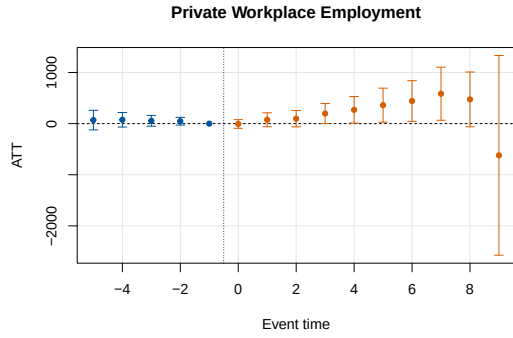


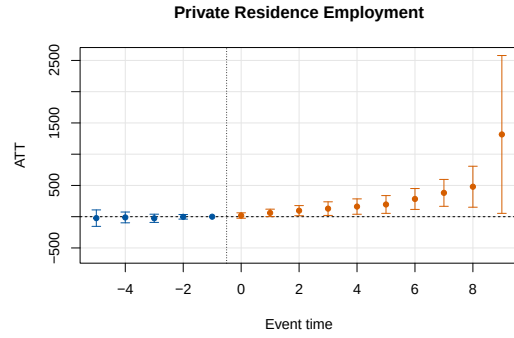
Figure 39. Sending-location employment responses excluding Munich. Sun–Abraham estimates in employee counts using one-to-one outside-Bavaria matches on levels and pre-policy trends. Employment includes public and private employees. Intervals are estimate ± 1.96 standard errors.

Estimator and event-time support

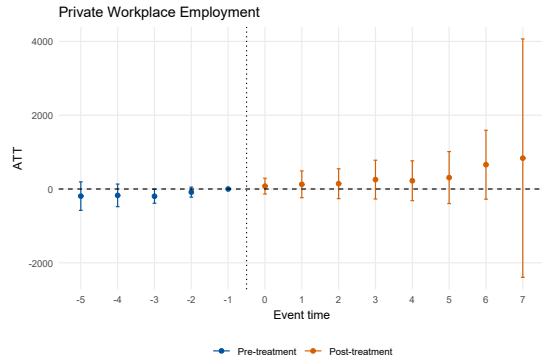
Figures 40 and 41 compare Sun–Abraham, Callaway–Sant’Anna, and two-way fixed-effect event studies for receiving and sending municipalities. The estimators aggregate treatment effects differently, and the Callaway–Sant’Anna implementation does not impose the matched-pair-by-year fixed effects of the other two.



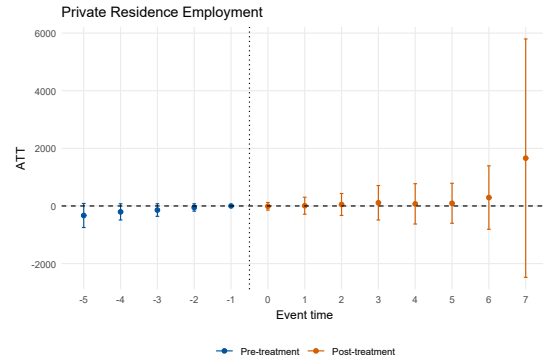
(a) Sun-Abraham: workplace



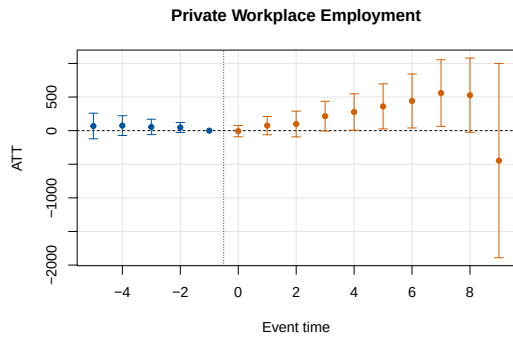
(b) Sun-Abraham: residence



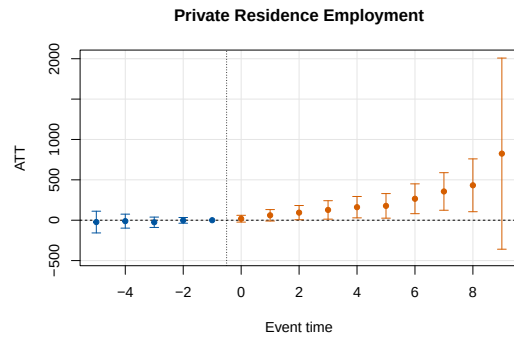
(c) Callaway-Sant'Anna: workplace



(d) Callaway-Sant'Anna: residence

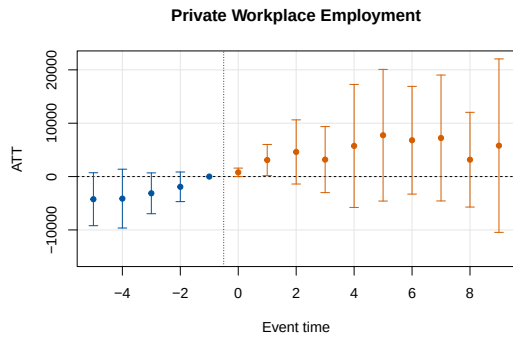


(e) TWFE: workplace

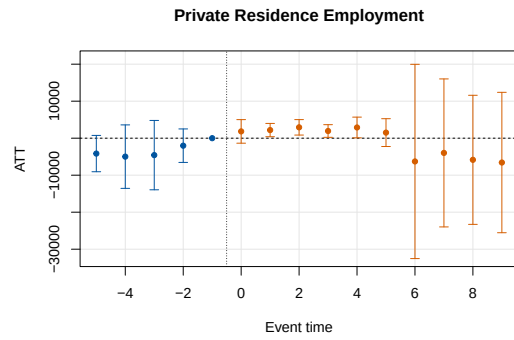


(f) TWFE: residence

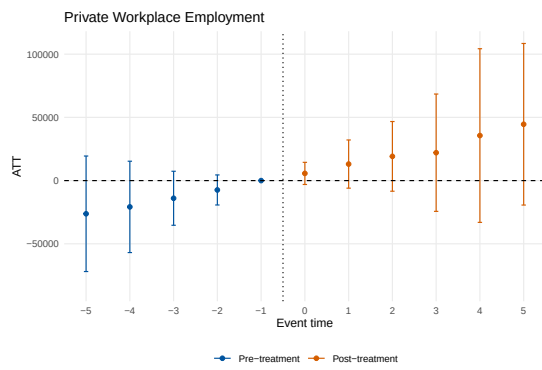
Figure 40. Receiving private-employment dynamics under alternative estimators, using the outside-Bavaria matching design. Table 25 reports estimates and interval conventions at event year five.



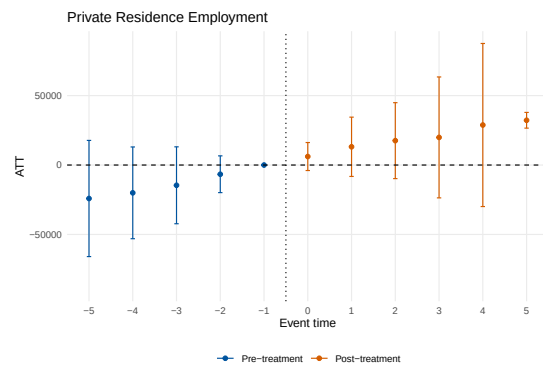
(a) Sun-Abraham: workplace



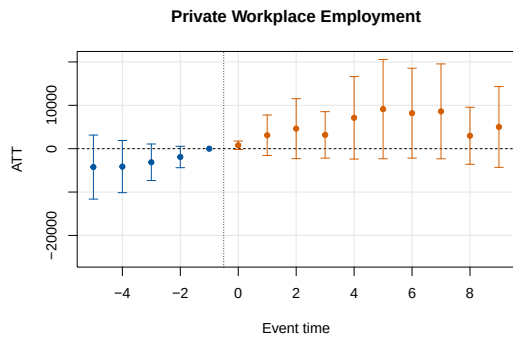
(b) Sun-Abraham: residence



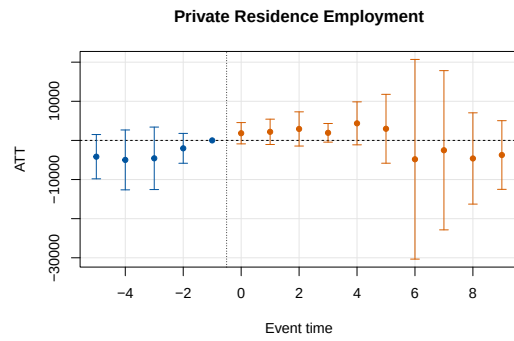
(c) Callaway-Sant'Anna: workplace



(d) Callaway-Sant'Anna: residence



(e) TWFE: workplace



(f) TWFE: residence

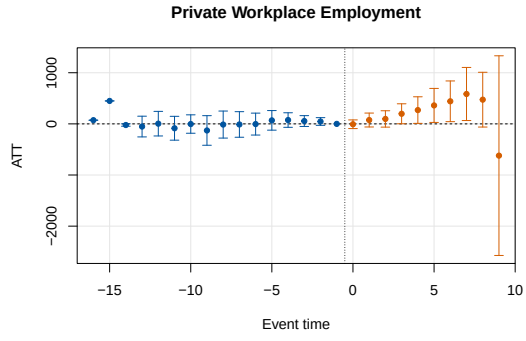
Figure 41. Sending private-employment dynamics under alternative estimators, using the outside-Bavaria matching design. Workplace and residence outcomes occupy the left and right columns. Panel scales differ.

Municipalities	Measure	Estimator	Estimate (SE)	Interval
Receiving	Workplace	SA	0.36 (0.17)	[0.03, 0.69]
Receiving	Workplace	CS	0.31 (0.15)	[-0.39, 1.01]
Receiving	Workplace	TWFE	0.36 (0.17)	[0.03, 0.69]
Receiving	Residence	SA	0.20 (0.07)	[0.06, 0.34]
Receiving	Residence	CS	0.09 (0.11)	[-0.60, 0.79]
Receiving	Residence	TWFE	0.18 (0.08)	[0.03, 0.33]
Sending	Workplace	SA	4.97 (3.52)	[-1.93, 11.88]
Sending	Workplace	CS	44.55 (32.61)	[-1,749.00, 1,838.11]
Sending	Workplace	TWFE	5.79 (3.80)	[-1.67, 13.24]
Sending	Residence	SA	1.34 (1.08)	[-0.77, 3.45]
Sending	Residence	CS	32.24 (25.16)	[-280.51, 344.99]
Sending	Residence	TWFE	2.15 (2.47)	[-2.69, 7.00]

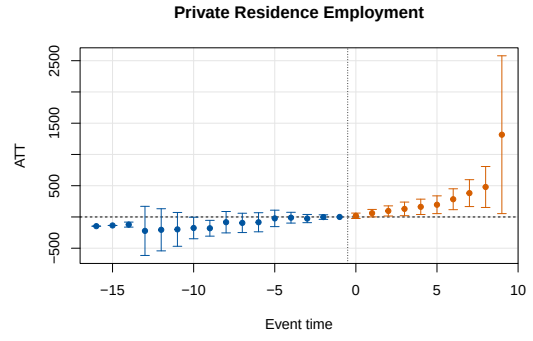
Table 25. Estimator comparison five years after first treatment. All entries are in thousands of employees. SA and TWFE intervals are pointwise 95% intervals based on municipality-clustered standard errors. CS intervals are bootstrap simultaneous bands, so interval widths are not directly comparable across estimators. Samples and specifications correspond to Figure 40.

For receiving workplace employment at event year five, the Sun–Abraham estimate is about 4.3% of 2014 private workplace employment and the Callaway–Sant’Anna estimate about 3.7%, but the latter simultaneous band covers both substantial losses and gains.

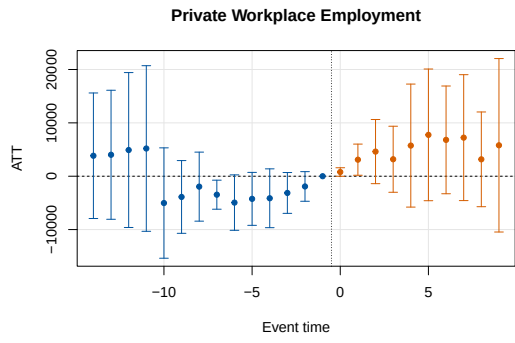
Figure 42 reports the full Sun–Abraham event-time ranges, which extend the windows shown above. Cohorts enter each event time only where their outcomes are observed, so remote leads and lags rest on changing cohort support.



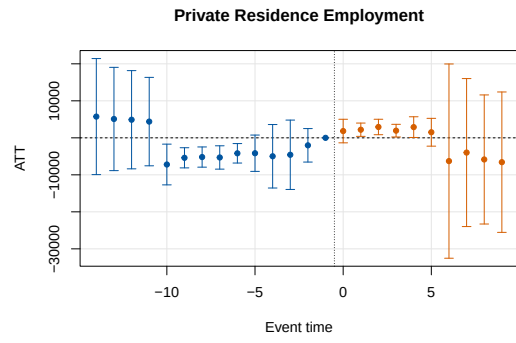
(a) Receiving: workplace



(b) Receiving: residence



(c) Sending: workplace



(d) Sending: residence

Figure 42. Full event-time support for private employment. Sun–Abraham estimates for receiving municipalities in the top row and sending municipalities in the bottom row. The contributing cohorts change across horizons.